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Aims and Scope

The Journal of Electromagnetic Engineering and Science (**JEES**) is an official English-language journal of the Korean Institute of Electromagnetic and Science (**KIEES**). This journal was launched in 2001 and has been published quarterly since 2003. It is currently registered with the National Research Foundation of Korea and also indexed in Scopus, CrossRef, EBSCO, DOI/Crossref, Google Scholar, and Web of Science Core Collection as Emerging Sources Citation Index (ESCI) journal. The objective of **JEES** is to publish academic as well as industrial research results and discoveries in electromagnetic engineering and science. The particular scope of the journal includes electromagnetic field theory and its applications; high frequency components, circuits, and systems; antennas; electromagnetic wave environments; and relevant industrial developments. In addition to the regular stream of contributed papers, the journal occasionally publishes special issues on specific topics of current interest, to promote a focused coverage of the selected topics. In addition, **JEES** regularly posts invited review papers by leading researchers in the relevant fields, which will provide an overview of various subjects from their own perspectives. The editors of **JEES** are aware of the importance of rapid publication for the delivery of new research results to its readers with minimum delay, and fully recognize the need for a short turnaround time. Manuscripts submitted to **JEES** take about 3–5 months from the initial submission to publication. This rapid process applies to all three types of manuscripts that **JEES** publishes: regular papers, letters, and invited papers. The publication of submitted manuscripts is subject to peer review, and the final decision by the Editor-in-Chief is regularly made based on three reviews from experts in the relevant area. This journal actively promotes research in electromagnetic wave-related engineering and sciences, and its ultimate aim is to benefit the global electromagnetic engineering and science community.

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Resonance Frequency and Bandwidth of the Negative/Positive n^{th} Mode of a Composite Right-/Left-Handed Transmission Line

Seong-Jung Kim · Jeong-Hae Lee*

Abstract

In this study, the analytic expression for the positive/negative n^{th}-mode resonance frequency of an N unit cell composite right-/left-handed (CRLH) transmission line is derived. To explain the resonance mechanism of the nth mode, especially for the negative mode, the current distribution of the N unit cell CRLH transmission line is investigated with a circuit simulation. Results show that both positive and negative n^{th} resonance modes have n times current variations, but their phase difference is 180° as expected. Moreover, the positive n^{th} resonance occurs at a high frequency, whereas the negative n^{th} resonance transpires at a low frequency, thus indicating that the negative resonance mode can be utilized for a small resonator. The correlation between the slope of the dispersion curve and the bandwidth is also observed. In sum, the balanced condition of the CRLH transmission line provides a broader bandwidth than the unbalanced condition.

Key Words: Bandwidth, Composite Right-/Left-Handed Transmission Line, Field Distribution, $\pm n^{th}$-Mode Resonance Frequency.

I. INTRODUCTION

Electromagnetic metamaterials (MTMs) [1] are broadly defined as artificial and effectively homogeneous electromagnetic structures with unusual properties not readily available in nature. MTMs enable the generation of left-handed waves through effective negative permittivity and permeability. MTMs have various applications, such as lens, absorber, and cloak, among others [2–4].

However, MTMs have a limitation when applied to radio frequency (RF) devices because they are lossy and have a narrow bandwidth (BW) in the microwave frequency range. To overcome these problems, the composite right-/left-handed (CRLH) transmission line (TL) was introduced [5]. Many miniaturized antennas, power dividers, resonators, filters, and couplers [6–12] can be designed and applied for many RF applications using CRLH TLs. However, until now, no detailed investigation on the positive/negative n^{th}-mode resonance frequency and its BW of N unit cell CRLH TL has been conducted as far as we know.

This study defines the negative mode and provides the guideline for designing a resonance antenna using the CRLH TL. The physical resonance mechanism can be explained by using the circuit model. Specifically, the difference between the negative and positive modes can be clearly seen, and the physical meaning of the negative mode is explained. Moreover, by deriving the analytic expression of the negative/positive n^{th} mode resonance frequency to show the correlation between

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the BW and the dispersion curve slope, this study provides the guideline for easily designing a resonance antenna.

II. Resonance Mechanism of the Negative/Positive nth Mode of the N Unit Cell CRLH TL

Fig. 1 shows the current distribution of the positive/negative nth mode for N cells with an open boundary condition at both ends. For the nth mode, because the total electrical length is a multiple of π with an open-ended boundary, the $n\pi/N$ phase delay is expected to occur per unit cell when $n = 0, \pm 1, \cdots, \pm N-1$, and the current distribution has n time variations of a standing wave pattern for both the positive and negative modes. The length of one variation corresponds to a half wavelength of each mode. Therefore, both the nth positive and negative modes have the same wavelength (or the same magnitude of a propagation constant) even though the corresponding frequencies are different because of different dispersion branches. However, the phase difference between the positive and the negative mode is 180° because of the opposite direction of the propagation direction between the two modes. The phase difference of 180° is easy to understand from the current standing wave pattern with an open boundary condition at both ends given by

$$I = I_0^+(e^{-j\beta z} - e^{+j\beta z}) = -2jI_0^+ \sin\beta z\,, \quad (1)$$

$$I = I_0^+(e^{+j\beta z} - e^{-j\beta z}) = +2jI_0^+ \sin\beta z\,, \quad (2)$$

where (1) and (2) are the positive mode and the negative mode, respectively.

To confirm the illustration in Fig. 1, the current distribution of the N unit cell CRLH TL with an open boundary condition is simulated with the Advanced Design System (Fig. 2). If a lossless structure with no resistance exists, only the magnitude of the reactance will distribute the amplitude of the current and the voltage. For $N = 3$, the voltages and the currents of each node in Fig. 2 are measured with probes labeled $P_{0,1,2,} \cdots, N$ and compared (Fig. 3).

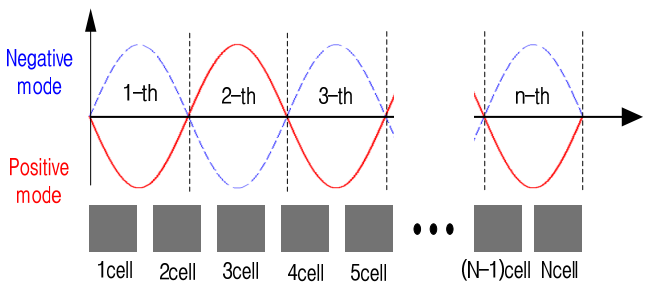
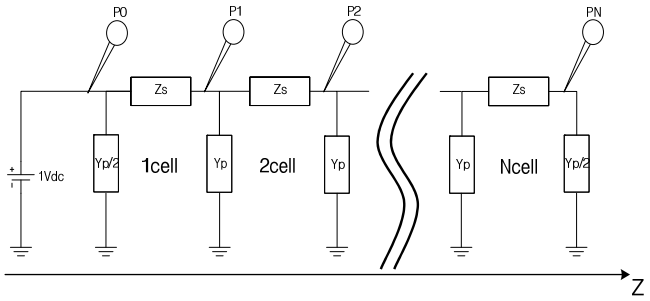


Fig. 1. Current distribution of the positive/negative nth mode for N cells with an open boundary condition.



$$\left(Z_S = \omega L_R - \frac{1}{\omega C_L},\ Y_P = \omega C_R - \frac{1}{\omega L_L},\ L_R = 13.7nH,\right.$$
$$\left. C_L = 0.18pF, L_L = 1.58nH,\ C_R = 1.97pF\right)$$

Fig. 2. Equivalent CRLH circuit with an open boundary condition at both ends.

The magnitude of the voltage and the current is normalized to 1, and the current flowing in the -z direction is a (+) sign. The left column of Fig. 3 shows the positive modes and the right column shows the negative modes. The result indicates that the voltage and the current have a cosine and sine distribution, respectively, except for the 0th mode.

For the 0th mode, no current flows and the voltage operates without a phase delay because of the zero value of Y_p. The 0th-order resonance with an open-ended boundary condition occurs at the shunt resonance ($Y_p = 0$). Moreover, the number of half wavelength variations increases as the mode number of n increases. When the distributions of the positive and negative modes are compared, note that the current distribution changes to 180° in the phase as discussed previously, and the voltage distribution is the same as that in the open boundary condition.

In terms of the circuit, the P_0 probe is fixed at +1 Vdc at the input node. Therefore, its amplitude is +1 V regardless of the mode. For the positive mode, $Y_p/2$ has a positive value because C_R is predominant over L_L; for the negative mode, $Y_p/2$ has a negative value because L_L is predominant over C_R. Therefore, the direction of the current changes in the negative mode. The reason for the existence of negative modes is that they are another combination that makes the total input reactance zero. In other words, Z_s and Y_p of the negative mode have the same ratio as Z_s and Y_p of the positive mode. The only difference is that the signs of $\mathrm{Im}(Z_s)$ and $\mathrm{Im}(Y_p)$ of the negative mode are negative and those of $\mathrm{Im}(Z_s)$ and $\mathrm{Im}(Y_p)$ of the positive mode are positive. Fig. 3 satisfies the condition of $1/\sqrt{L_L C_R} > 1/\sqrt{L_R C_L}$ with an open boundary condition. In this case, it has a positive mode when the angular frequency is larger than the shunt angular resonance frequency of $1/\sqrt{L_L C_R}$. Conversely, it has a negative mode when it is smaller than $1/\sqrt{L_R C_L}$. Similarly, if it is a short boundary condition, the 0th-order mode angular frequency is given by the series resonance of $1/\sqrt{L_R C_L}$. In addition, the vol-

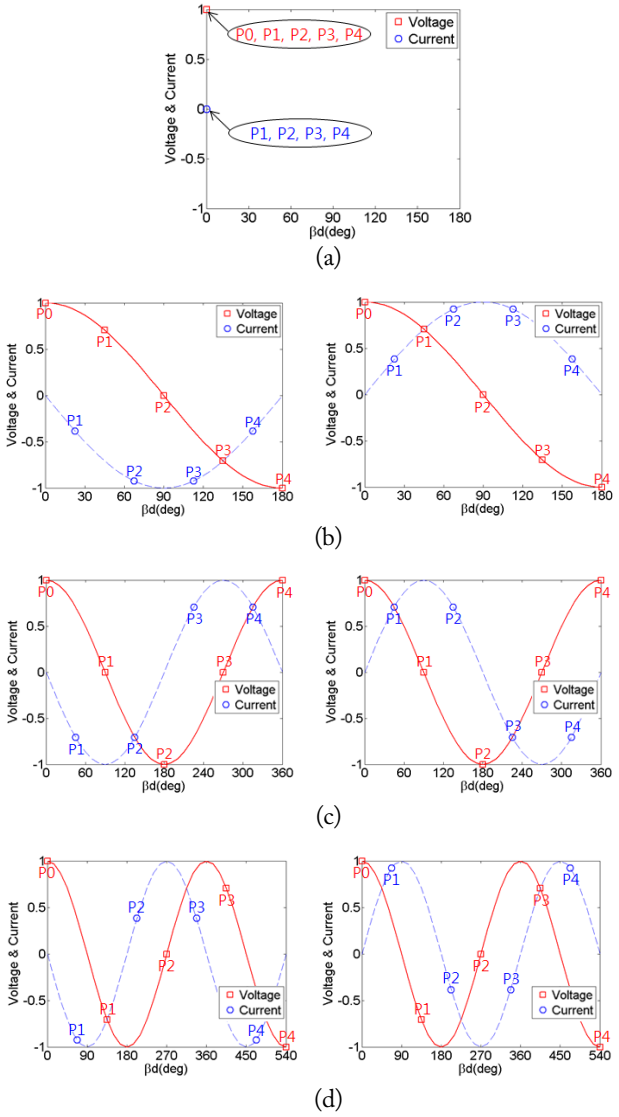


Fig. 3. Results of the voltages and the currents of each node. (a) 0[th] mode, (b) +1[th] mode and -1[th] mode, (c) +2[th] mode and -2[th] mode, (d) +3[th] mode and -3[th] mode.

tage follows the sine distribution, and the current follows the cosine distribution. The phases of the positive and negative mode currents are also 180° apart from each other.

III. Derivation of Resonance Frequency

The unit cell of CRLH TL can be drawn as an equivalent T circuit as shown in Fig. 4.

The following relation can be obtained by using the phase difference ($\theta = n\pi/N$) between two terminals:

$$V_n = V_{n+1} e^{j\theta}$$
$$I_n = I_{n+1} e^{j\theta} \quad , \tag{3}$$

where N is the number of total unit cells, and n is the number of

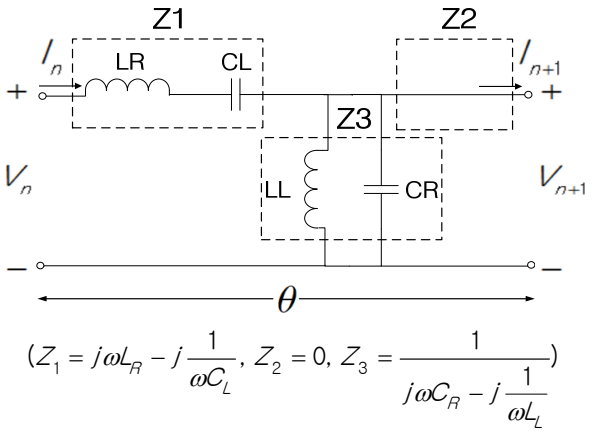


Fig. 4. Equivalent circuit of a CRLH unit cell.

arbitrary unit cells.

Eq. (3) is represented by a matrix. When it is compared with the ABCD matrix, the following equation is obtained:

$$\begin{bmatrix} A - e^{j\theta} & B \\ C & D - e^{j\theta} \end{bmatrix} \begin{bmatrix} V_{n+1} \\ I_{n+1} \end{bmatrix} = O \,. \tag{4}$$

As Eq. (4) has a non-trivial solution and the circuit in Fig. 4 is reciprocal ($AD - BC = 1$), we can summarize both as Eq. (5):

$$1 - e^{j\theta}(A + D) + e^{j2\theta} = O \,, \tag{5}$$

where

$$A = 1 + (j\omega L_R - j\frac{1}{\omega C_L})(j\omega C_R - j\frac{1}{\omega L_L}), \quad D = 1 \,.$$

Eq. (5) is a quartic equation for ω. By substitution into a quadratic equation, the following equation can be obtained:

$$\omega = \sqrt{\frac{\omega_E^2 + \omega_M^2 + 2\omega_R^2(1-\cos\theta) \pm \sqrt{(\omega_E^2 + \omega_M^2 + 2\omega_R^2(1-\cos\theta))^2 - 4\omega_L^2\omega_R^2}}{2}} \tag{6}$$

where

$$\omega_R = \frac{1}{\sqrt{L_R C_R}}, \ \omega_L = \frac{1}{\sqrt{L_L C_L}}, \ \omega_E = \frac{1}{\sqrt{L_L C_R}}, \ \omega_M = \frac{1}{\sqrt{L_R C_L}}$$

To confirm the result of Eq. (6), the resonance frequencies of the three-unit cell CRLH TL are calculated using the circuit simulation of Ansoft Designer and the full wave simulation of HFSS. Fig. 5 is the electrically coupled three-unit cell CRLH equivalent circuit employed in the circuit simulation. Fig. 6 shows the mushroom structure, and Fig. 5 illustrates its parameters (C_R, L_R, C_L, and L_L) and their values. The structure in Fig. 6 is obtained from [13] and simulated by High Frequency Structure Simulator (HFSS). Table 1 compares the resonance frequencies of the three-unit cell CRLH TL obtained from Eq. (6),

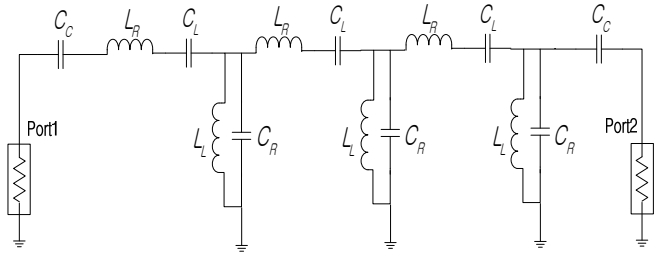


Fig. 5. Structure fed via electrical coupling (N=3 using the Ansoft Designer circuit tool). C_c =0.1 pF, L_R=13.7 nH, C_L =0.18 pF, L_L =1.58 nH, C_R=1.97 pF.

Table 1. Comparison of resonance frequencies of N=3 CRLH TL with an open boundary (unit: GHz)

	-2th mode	-1th mode	0th mode	$+1^{th}$ mode	$+2^{th}$ mode
Eq. (6)	2.29	2.55	2.85	3.58	4.00
Circuit	2.28	2.52	2.81	3.55	3.98
HFSS	2.71	2.88	3.16	3.59	3.94

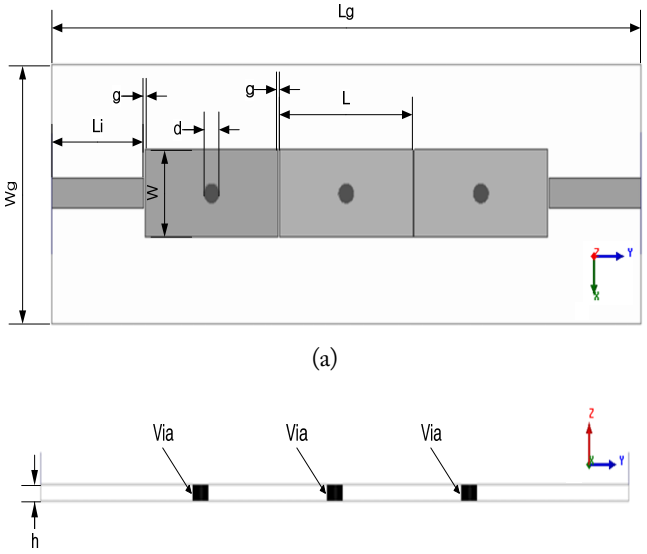


Fig. 6. Schematic of the proposed three-cell CRLH mushroom resonant antenna: (a) top view and (b) side view. L_g=90 mm, W_g=30 mm, L_i=13.8 mm, W_i=3.5 mm, L=20 mm, W=10 mm, d=2.4 mm, g=0.2 mm (Rogers RO-4003 substrate with thickness h=1.524 mm and dielectric constant of 3.55).

HFSS, and Ansoft Designer, respectively.

Table 1 shows that the results obtained by Eq. (6), and all the data in Table 1 are obtained with an unbalanced condition. The circuit simulation is almost the same as expected. However, the results from HFSS have some differences in the resonance frequencies caused by the errors of the extracted parameters (C_R, L_R, C_L, and L_L) from the real structure. The method of parameter extraction from the dispersion curve inevitably cannot produce perfect matched results in all modes. However, the overall results are in good agreement.

IV. Correlation between Slope and BW

The slope of the dispersion curve $df/d\theta$ (θ = electrical length per unit cell) is a measure of how much the frequency changes as the electrical length changes. The higher the slope is, the greater the frequency changes even with a change in electrical length. In other words, a large value of $\frac{1}{d\theta/df}$ means that the electrical length changes insensitively even when the frequency changes greatly. In a resonator, resonance depends on the electrical length because it occurs when the electrical length is an integer multiple of a half wavelength. Therefore, the slope of the dispersion curve correlates with the BW. The BW is defined as S_{11} <–10 dB, and the slope is calculated by differentiating the dispersion curve with the equation of $df/d\theta$. Tables 2 and 3 show the correlation between slope and BW for various cells and modes according to an unbalanced or a balanced condition [5]. The correlation coefficient is defined as

$$\rho = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \times \sum (y-\bar{y})^2}}$$
$$\rho : correlation\ coefficient$$
$$x : bandwidth\ of\ each\ modes$$
$$y : slope\ of\ dispersion\ curve \quad (7)$$

In Tables 2 and 3, the dimensions of slope and BW are in MHz/° and MHz, respectively. Except for the 0th mode of the unbalanced condition, Tables 2 and 3 show that the slope is always closely related to the BW. Fig. 7 illustrates that the BW, which is correlated with the dispersion curve slope, is significantly improved when the balanced condition is satisfied [14]. In other words, the overall BW improves because the average slope from 0° to 180° increases from 5.3 MHz/° to 25.3 MHz/°. This finding can be confirmed by comparing Tables 2 and 3.

V. Conclusion

This study has successfully derived the analytic expression for the positive/negative n^{th}-mode resonance frequency of an N unit cell CRLH transmission line. The resonance mechanism of the nth positive/negative mode is investigated by the current distribution of an N unit cell CRLH transmission line. Both the positive and the negative n^{th} resonance modes have n times current variation, but their phase difference is 180°. The slope of the CRLH dispersion curve is closely correlated with the BW. When the CRLH TL is designed with a balanced condition,

Table 2. Correlation coefficient between BW and slope in an unbalanced condition

# of cells		-4th mode	-3th mode	-2th mode	-1th mode	0th mode	$+1^{th}$ mode	$+2^{th}$ mode	$+3^{th}$ mode	$+4^{th}$ mode	Correlation coefficient
N=3	Slope (MHz/°)	-	-	3.01	5.73	-	8.04	5.27	-	-	0.990
	BW (MHz)	-	-	1.13	4.92	6.89	7.74	4.94	-	-	
N=4	Slope (MHz/°)	-	2.26	4.47	6.11	-	8.00	7.11	4.09	-	0.956
	BW (MHz)	-	0.46	2.02	4.90	5.16	5.18	5.61	2.32	-	
N=5	Slope (MHz/°)	1.81	3.60	5.27	6.14	-	7.72	7.80	6.09	3.33	0.932
	BW (MHz)	0.31	1.30	2.52	4.65	4.02	3.43	4.80	3.60	1.19	

C_c=0.1 pF, L_R=13.7 nH, C_L=0.18 pF, L_L=1.58 nH, C_R=1.97 pF.

Table 3. Correlation coefficient between BW and slope in a balanced condition

# of cells		-4th mode	-3th mode	-2th mode	-1th mode	0th mode	$+1^{th}$ mode	$+2^{th}$ mode	$+3^{th}$ mode	$+4^{th}$ mode	Correlation coefficient
N=3	Slope (MHz/°)	-	-	4.48	13.40	27.97	35.06	23.14	-	-	0.919
	BW (MHz)	-	-	35.30	172.0	407.4	426.9	156.5	-	-	
N=4	Slope (MHz/°)	-	3.44	8.36	16.61	27.97	35.07	31.20	17.97	-	0.928
	BW (MHz)	-	14.0	64.5	169.1	306.0	346.6	228.6	69.30	-	
N=5	Slope (MHz/°)	2.69	6.10	11.16	18.85	27.97	34.45	34.09	26.78	14.60	0.932
	BW (MHz)	6.90	30.70	78.70	156.8	245.0	282.5	232.2	129.1	36.20	

C_c=5 pF, L_R=13.7 nH, C_L=0.18 pF, L_L=13.7 nH, C_R=0.18 pF.

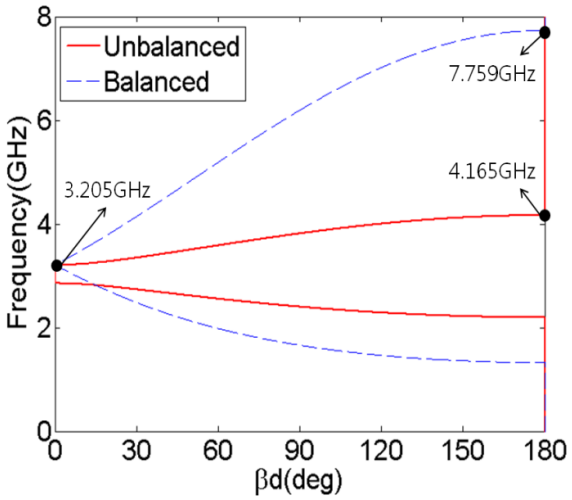


Fig. 7. Dispersion curves of slope and BW in Tables 2 and 3.

the BW increases from the slope.

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Design of a Small Radio Frequency Identification Tag Antenna Using a Corrugated Meander Line Applicable to a Drug Runout Sensor System

Dong-Ok Ko · Jong-Myung Woo*

Abstract

This article proposes an ultrahigh frequency band radio frequency identification (RFID) tag antenna for drug runout management that can be used in hospitals. The RFID tag antenna is designed to function as a sensor that alerts drug runout when a drug inside a drip chamber is completely consumed but does not work when a drug remains inside a drop chamber. A previously proposed 915 MHz dipole antenna, is too large to be attached to the drip chamber of a feeding bag. Moreover, the length of the dipole (L) should be increased for conjugate matching with an RFID chip. Therefore, the dipole antenna is downsized so that it can be attached to the drip chamber through a fine meander line structure coupling with a corrugate meander line. A transparent cover is added to enhance the grip force between the designed antenna and the drip chamber and to enable detachment. The dimensions of the completed antenna structure attachable to a drip chamber are 32.59 mm (height) and 13.5 mm (width). The gain reduction due to the decreased antenna length is enhanced. The fabricated antenna shows an average omni-directional read range of 10.65 m on a horizontal plane and has the function of sensing the presence of a drug.

Key Words: Meander Line, RFID, Tag Antenna.

I. Introduction

New types of diseases are continuously appearing, and the number of patients with chronic diseases is also increasing. Moreover, the human lifespan is increasing because of the extension of medical services. Therefore, the number of patients who are hospitalized for a long time for treatment is also increasing.

Consequently, hospitals need to efficiently manage their patients. Automatic medication management methods to handle drug administration information and medication positions are important, and there many studies have been conducted on these methods [1–4]. In accordance with these studies, a radio frequency identification (RFID) sensor-based system for managing the runout of a drug administered to a patient was proposed and studied [5, 6].

Previous studies on the wireless management of drug runout proposed a spherical three-axis polarized RFID tag antenna [7, 8]. In this management system, the RFID tag antenna is placed inside a feeding bag, where the antenna is exposed to the air when the drug is completely consumed. Impedance matching occurs between the tag antenna and an RFID chip to enable the

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wireless recognition of the information stored in the RFID and the transmission of the information to the manager. However, this management method has the drawback of antenna deterioration and drug degeneration due to the direct contact between the RFID tag antenna and the drug. Another study proposed a printed circuit board (PCB)-based tag antenna attached to the outside of a feeding bag [9]. However, the antenna is too large to be attached to the tip of the feeding bag structure, and thus information is transmitted even before the drug is completely consumed. Comparing these previously proposed systems, the drug runout management system including an RFID tag antenna attached to a drip chamber is more effective.

This article proposes an ultrahigh frequency (UHF) band (center frequency = 915 MHz) RFID tag antenna for drug runout detection that can be attached to a drip chamber. A dipole-type RFID tag antenna is downsized by using a corrugated meander line. The antenna is further downsized by using a fine meander line at the tip of the antenna where the electric current is low to make the antenna attachable to a drip chamber. A transparent cover is inserted between the antenna and the drip chamber to increase the grip force and to enable the detachment of the antenna and the viewing of the interior of the drip chamber.

II. Antenna Design

Fig. 1 shows the environment recognized by the RFID tag antenna for drug runout management through a reader. The RFID tag antenna is attached to the drip chamber of a feeding bag. When the drug remains inside the drip chamber, impedance matching does not occur between the tag antenna and the RFID chip because of the high dielectric constant of the drug, and the information in the chip is not read by the reader. Conversely, when the drug is completely consumed and the drip chamber becomes empty, the information is read by the reader as conjugate impedance matching occurs between the tag antenna and the RFID chip. Then, the information stored inside the RFID chip is recognized by the reader, enabling the RFID tag antenna to sensor the information on the runout of the drug administered to the patient. In this manner, the RFID tag antenna enables the recognition of the runout of the drug contained in the drip chamber so that the medication to the patient can be managed safely and conveniently.

Fig. 2 sequentially shows the simulation design procedures to develop the tag antenna. As shown in Fig. 2(a), the 915 MHz basic half-wavelength dipole designed for this purpose has a length of 149.5 mm. However, the dipole length (L) should be increased for conjugate matching with the impedance of the Higgs-4 RFID chip (Higgs) used for the antenna design (Z_{chip} = 18.42-j181.22 Ω). As the dipole antenna is spatially limited

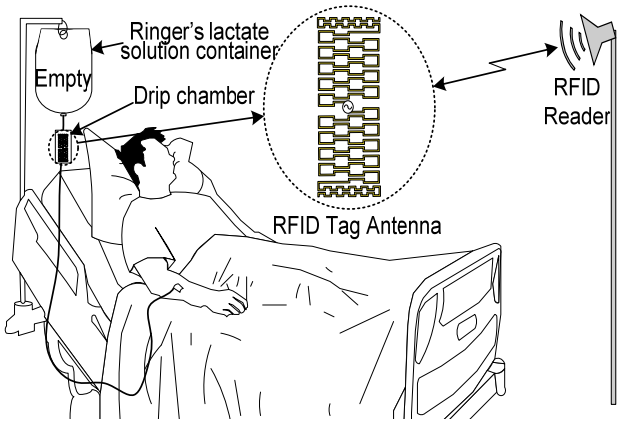


Fig. 1. Drug runout sensor system.

to be attached to a drip chamber (42 mm long), the radiation element of the antenna is downsized. Firstly, as shown in Fig. 2(b), the length of the RFID tag antenna is reduced to 37.2 mm by using a fine meander line. Then, to supplement the reduction of gain due to the electric current offset in the fine meander line antenna, the antenna is further downsized by using the corrugated meander line shown in Fig. 2(c), which is a magnified image of the line intervals at the feeding point. However, the antenna length is 45.39 mm, which is still inappropriate to be attached to the drip chamber. Therefore, a fine meander line is added at the tip of the antenna where the electric current is low to decrease the length of the antenna. As a result, the antenna length is decreased to 39.3 mm, as shown in Fig. 2(d).

Fig. 3 shows the final structure of the antenna designed in the present study. The antenna is attached to a cylindrical polyvinyl chloride (PVC, ε_r = 4) drip chamber 19 mm in diameter and 42 mm in height. To increase the grip force between the drip chamber and the antenna, a transparent cover, 1 mm thick and 37 mm long, is inserted between the antenna and the drip chamber. Owing to the dielectric constant of the drip chamber and the cover, the length of the antenna shown in Fig. 2(d) decreases by 14.5% from 39.3 mm to 32.59 mm in the absence of the drip chamber.

III. Simulation and Experimental Results

Fig. 4 shows the radiation pattern of the antennas shown in Fig. 2. Fig. 4(a) illustrates the radiation pattern of the basic half-wavelength dipole antenna shown in Fig. 2(a). This figure indicates that the basic half-wavelength dipole antenna has an omni-directional radiation pattern, which is a property of a dipole antenna, and a maximum gain of 1.86 dBi. Fig. 4(b) presents the radiation pattern of the fine meander line antenna shown in Fig. 2(b). The fine meander line antenna exhibits an omni-directional radiation pattern, which is a property of a basic half-wavelength dipole antenna, but the maximum gain is decreases

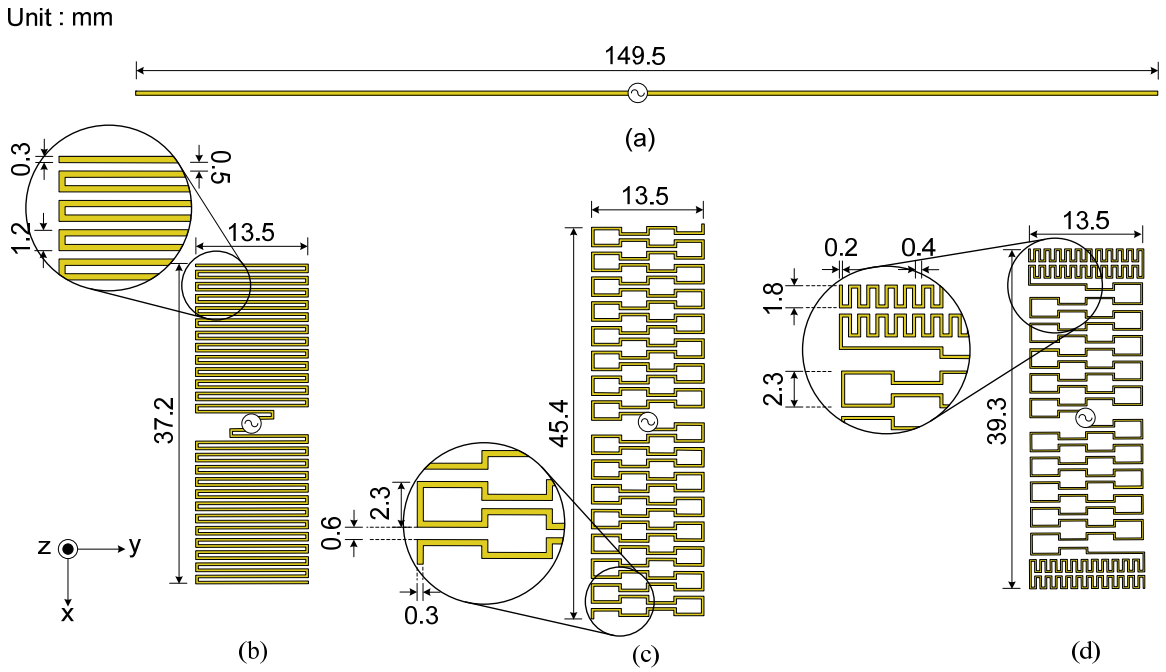


Fig. 2. Structure of the designed antenna. (a) Basic half-wavelength dipole antenna. (b) Fine meander line RFID tag antenna. (c) Corrugated meander line RFID tag antenna. (d) Composite meander line RFID tag antenna.

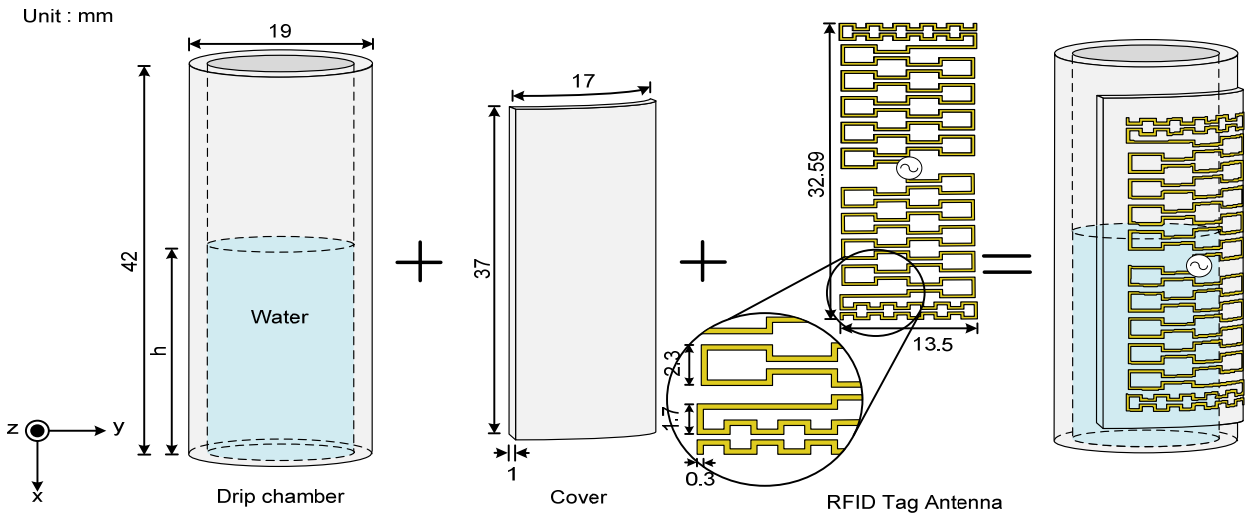


Fig. 3. Structure of the proposed antenna.

to –2.04 dBi because of the reduction of the antenna length and the electric current offset due to the meander line. Fig. 4(c) illustrates the radiation pattern of the corrugate meander line antenna shown in Fig. 2(c). A corrugate meander line is used to enhance the gain reduction caused by the fine meander line. The result shows that the gain increases by about 1 dBi to –1.03 dBi in comparison with that of the fine meander line antenna. Fig. 4(d) presents the radiation pattern of the composite meander line antenna shown in Fig. 2(d). The corrugated meander line and the fine meander line are combined to further downsize the antenna, and the resultant antenna exhibits an omni-directional radiation pattern and a maximum gain of –1.5 dBi.

Fig. 5 illustrates the properties of the final antenna structure (Fig. 3). Fig. 5(a) shows S_{11} of the antenna of the final design depending on the height (h) of the drug inside the drip chamber. As the drug height (h) decreases, the resonance frequency shifts toward the center frequency of 915 MHz. When the drug height become 0, impedance matching occurs at 915 MHz with S_{11} of –17.4 dB (bandwidth = 7.3 MHz, 0.79%), indicating that the RFID tag normally functions as a sensor only when the drug is completely consumed. Fig. 5(b) indicates that the antenna has an omni-directional radiation pattern with a gain of

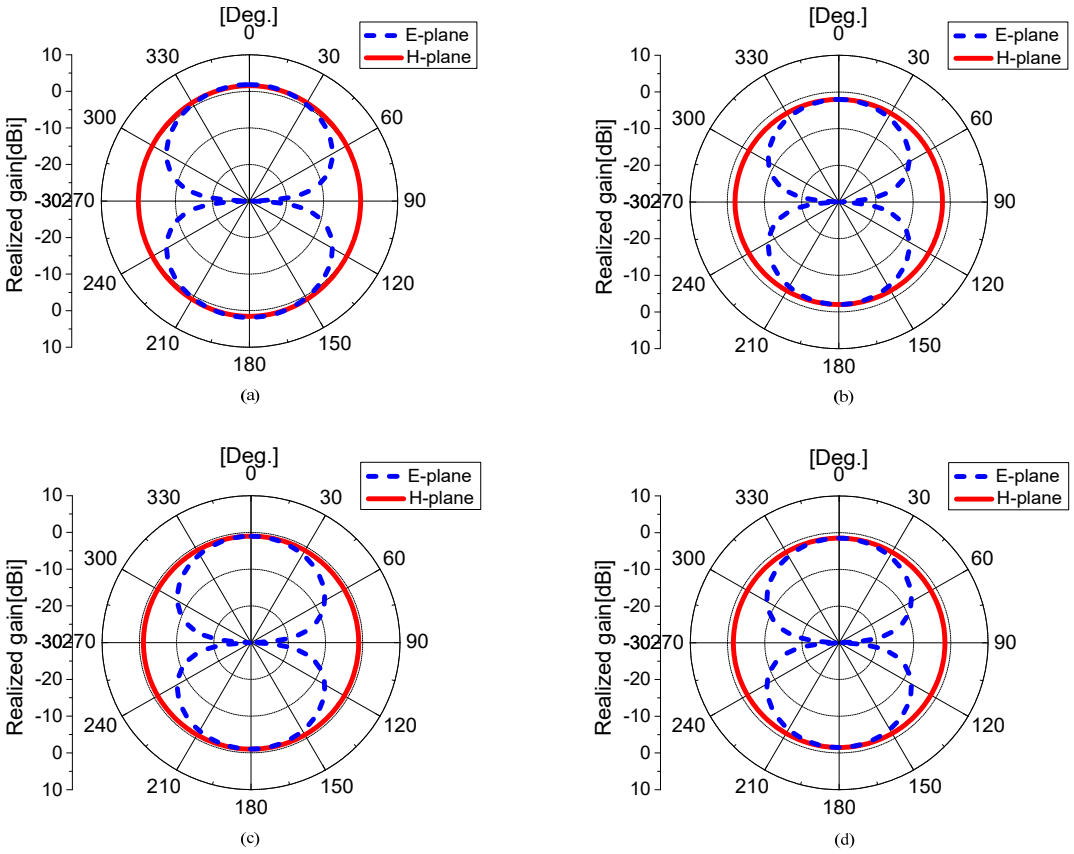


Fig. 4. Radiation pattern of the designed antenna. (a) Basic half-wavelength dipole antenna. (b) Fine meander line RFID tag antenna. (c) Corrugated meander line RFID tag antenna. (d) Composite meander line RFID tag antenna.

Table 1. Characteristics of the designed antennas (simulation)

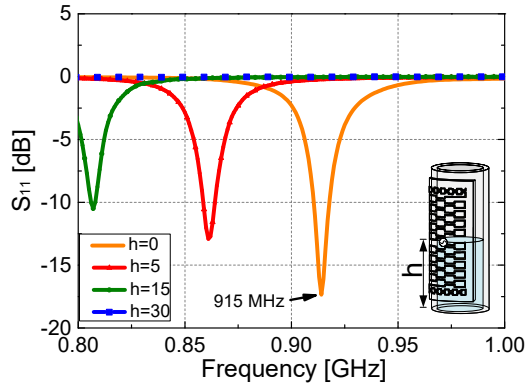
	Fig. 2(a) Basic half-wavelength dipole	Fig. 2(b) Fine meander line RFID tag antenna	Fig. 2(c) Corrugate meander line RFID tag antenna	Fig. 2(d) Composite meander line RFID tag antenna	Fig. 3 Proposed RFID tag antenna
Size (mm)	149.5 × 0.5	37.21 × 13.5	45.39 × 13.5	39.3 × 13.5	32.59 × 13.5
S_{11} (dB)	-14.86	-19.03	-18.18	-26.9	-17.4
Bandwidth (MHz)	89.5	11	12.94	11.98	7.3
Gain (dBi)	1.86	-2.04	-1.03	-1.5	-0.46

–0.46 dBi. As the properties of the tag antenna are not changed by the attachment of the RFID tag antenna to the drip chamber, a patient medication management system may be appropriately designed by using the RFID tag antenna developed in the present study.

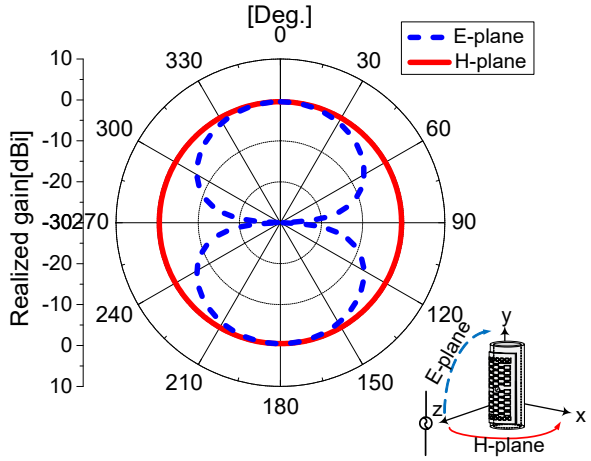
Table 1 shows the results of the simulation of the properties of the antennas designed in the study.

Fig. 6 illustrates the measurement setup for the property analysis of the completed antenna structure fabricated. The RFID reader is the Air-9000 Model of Alien with an intensity of 4 W. The sensing distance is measured with the RFID reader by attaching the antenna to the drip chamber and distancing it 1,500 mm from the ground. The antenna structure is fabricated by etching copper tape, and attached to a double-sided tape so that the structure can be detached and re-attached. The size of the fabricated antenna is 32.59 mm × 13.5 mm, which is equal to the simulated antenna.

Fig. 7 exhibits the measured properties of the fabricated antenna. Fig. 7(a) shows that S_{11} at 915 MHz is –37.79 dB (bandwidth = 12 MHz, 1.31%) when the drip chamber is empty. By contrast, when the water height (water is used instead of a drug) is 40 mm in the drip chamber, the reader is unable to recognize the RFID tag because no impedance matching occurs at 915 MHz. The E_ϕ component on the xz-plane shows an



(a)



(b)

Fig. 5. Return loss and radiation pattern of the proposed antenna (simulation). (a) S_{11}. (b) Radiation pattern.

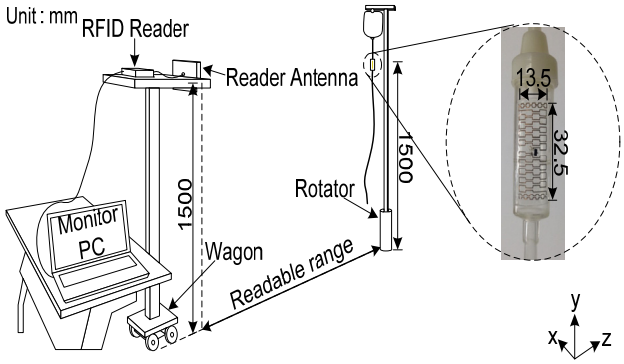
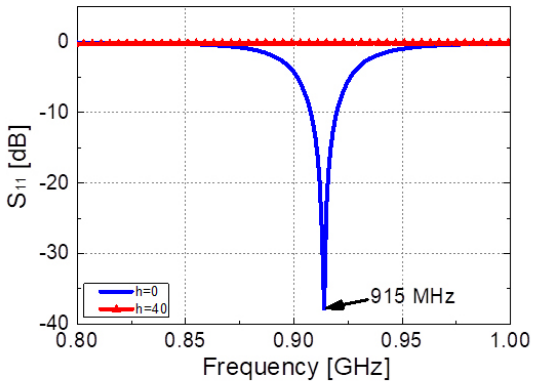


Fig. 6. Measurement setup.

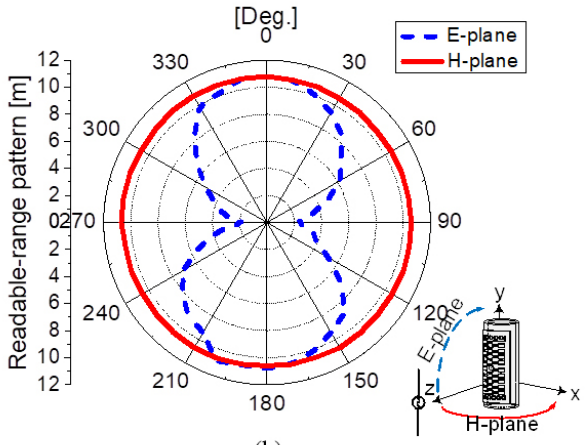
omni-directional read range of 10.65 m on average, and the E_θ component on the yz-plane presents an "8"-shaped read range, which indicates that the half-wavelength dipole property is maintained.

IV. Conclusion

This article proposes a UHF-band RFID tag antenna for drug runout detection that can be attached to a drip chamber.



(a)



(b)

Fig. 7. Return loss and readable-range pattern of the fabricated antenna (measurement). (a) S_{11}. (b) Readable-range pattern.

The function of the antenna is based on the change in the impedance matching depending on the presence of a drug inside a drip chamber. The antenna is downsized by using a meander line to enable impedance matching between an RFID chip (Z_{chip} = 18.43-j181.22 Ω) and the tag antenna and to shorten the antenna. In addition, the gain is increased by 0.54 dB through the combination of a fine meander line and a corrugate meander line. The dimensions of the completed designed antenna are 32.59 mm (height) and 13.5 mm (width). The measurement of the sensing distance of the designed antenna shows that the antenna has an average omni-directional read range of 10.65 m on a horizontal plane. This result verifies that the RFID tag antenna proposed in this article can be attached to a drip chamber to function as a drug runout sensor, which transmits information from a sickroom through an RFID reader.

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Analysis and Optimization of Wireless Power Transfer Efficiency Considering the Tilt Angle of a Coil

Wei Huang · Hyunchul Ku*

Abstract

Wireless power transfer (WPT) based on magnetic resonant coupling is a promising technology in many industrial applications. Efficiency of the WPT system usually depends on the tilt angle of the transmitter or the receiver coil. This work analyzes the effect of the tilt angle on the efficiency of the WPT system with horizontal misalignment. The mutual inductance between two coils located at arbitrary positions with tilt angles is calculated using a numerical analysis based on the Neumann formula. The efficiency of the WPT system with a tilted coil is extracted using an equivalent circuit model with extracted mutual inductance. By analyzing the results, we propose an optimal tilt angle to maximize the efficiency of the WPT system. The best angle to maximize the efficiency depends on the radii of the two coils and their relative position. The calculated efficiencies versus the tilt angle for various WPT cases, which change the radius of RX (r_2 =0.075 m, 0.1 m, 0.15 m) and the horizontal distance (y=0 m, 0.05 m, 0.1 m), are compared with the experimental results. The analytically extracted efficiencies and the extracted optimal tilt angles agree well with those of the experimental results.

Key Words: Coupling Coefficient, Mutual Inductance, Power Transfer Efficiency, Tilt Angle, Wireless Power Transfer.

I. INTRODUCTION

Wireless power transfer (WPT) based on the magnetic resonant coupling scheme has recently become widely used in many applications, such as electric vehicles, robotics, and medical devices [1, 2]. The WPT using magnetic resonant coupling can improve the power transfer efficiency (PTE) or extend the service range compared with inductively coupled systems. To improve the PTE, researchers have used various methods, such as impedance matching [3, 4], adjusting the coupling among multiple transmitters (TXs) or multiple receivers (RXs) [5], and applying the optimum condition of the values of phase difference and amplitude ratio between two adjacent TXs [6], among others.

Adding more TXs and RXs overcomplicates the circuit and renders it difficult to apply in practice. The most widely considered configuration of the WPT system is using two parallel coaxial circular coils: a single TX coil and a single RX coil. The PTE is known to decrease sharply when angular misalignment or lateral misalignment occurs [1, 7–9]. A WPT system with two parallel coaxial coils can achieve maximum PTE compared with other configurations [10]. However, the experimental result in [11] showed that the maximum transmission efficiency is reached when two coaxially located circular coils have an angle difference of 45°. The PTE can be improved by changing the tilt angle of coils for the given position and the radii of coils.

In this study, we analyze the effect of the tilt angle of the coil on the PTE of the WPT system. A mutual inductance between two coils with the tilt angle located at an arbitrary position in a 3D coordinate is extracted using numerical analysis through the

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Neumann equation. Using the extracted mutual inductance and the equivalent circuit model of the WPT system, the PTE depending on the tilt angle is extracted. We propose an optimal tilt angle to maximize the efficiency of the WPT system. The angle at which the efficiency is maximized depends on the radii of the two coils and their relative position.

This paper is organized as follows. Section II describes the equivalent circuit model of the WPT system based on magnetically resonant coupling, and the coupling coefficient analysis considering angular misalignment is introduced. Section III shows the simulated and experimental efficiencies of the WPT system for three different RX coil radii, and the efficiency of the WPT system is analyzed when angular misalignment or lateral misalignment occurs. We also discuss the relationship among the radii of RX, the tilt angle of the coil, and the PTE. Section IV presents the findings.

II. Efficiency Analysis of the WPT Tilted Coil

An equivalent circuit model of a magnetic resonant coupled WPT system with a single TX coil and a single RX coil is shown in Fig. 1.

In Fig. 1, k is the coupling coefficient between a TX coil and an RX coil; V is the phasor of the voltage source in the TX; L_{TX} and L_{RX} are the self-inductances of the two coils; and R_{P1} and R_{P2} are the parasitic resistances of the two coils. C_1 and C_2 are the capacitances of TX and RX, respectively, to satisfy $\omega_0 = 1/\sqrt{L_{TX}C_1} = 1/\sqrt{L_{RX}C_2}$, where ω_0 is the resonant frequency of the WPT system. R_S is the source resistance, and R_L is the load resistance.

The circuit model can be analyzed by Kirchhoff's voltage law equations for two coils as follows:

$$(R_{P1}+R_S+j\omega L_{TX}+\frac{1}{j\omega C_1})\boldsymbol{I_{TX}}+kj\omega\sqrt{L_{TX}L_{RX}}\boldsymbol{I_{RX}}=0\,, \tag{1}$$

$$kj\omega\sqrt{L_{TX}L_{RX}}\boldsymbol{I_{TX}}+(R_{P2}+R_L+j\omega L_{RX}+\frac{1}{j\omega C_2})\boldsymbol{I_{RX}}=0\,, \tag{2}$$

where $\boldsymbol{I_{TX}}$ and $\boldsymbol{I_{RX}}$ are the phasors of the currents in the TX and RX, respectively.

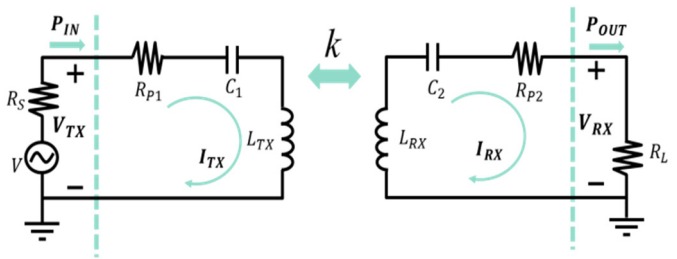


Fig. 1. Circuit modelling of the WPT system with a single TX and a single RX.

The PTE is defined as the ratio between the power delivered to a load and the power supplied by the source as follows:

$$\eta=\frac{P_{out}}{P_{in}}=\frac{\mathrm{Re}\left[V_{RX}I_{RX}^{*}\right]}{\mathrm{Re}\left[V_{TX}I_{TX}^{*}\right]}=\frac{k^2\omega^2L_{TX}L_{RX}R_L}{R_{P1}(R_{P2}+R_L)^2+k^2\omega^2L_{TX}L_{RX}(R_{P2}+R_L)} \tag{3}$$

Eq. (3) can be rewritten as $\eta = Ak^2/\left(B+Ck^2\right)$, where $A=\omega^2L_{TX}L_{RX}R_L$, $B=R_{P1}(R_{P2}+R_L)^2$, $C=\omega^2L_{TX}L_{RX}(R_{P2}+R_L)$, and $k\in[0,1]$. The derivative of η for k is as follows:

$$\frac{\partial\eta}{\partial k}=\frac{2ABk}{\left(B+Ck\right)^2}\,. \tag{4}$$

By setting Eq. (4) to 0, k will be 0. Therefore, $k=0$ is an extreme point of function η. When $k\in[0,1]$, η' is greater than zero, η is monotonically increasing in this range, and η increases when k increases.

To calculate the coupling coefficient between two coils, we consider the two coils located in the 3D space as shown in Fig. 2.

The coupling coefficient k is acquired using

$$k=\frac{M}{\sqrt{L_{TX}L_{RX}}}\,, \tag{5}$$

where M is the mutual inductance between the two coils.

The mutual inductance M can be calculated using the Neumann formula [12]:

$$M=\frac{N_{TX}N_{RX}\mu_0}{4\pi}\oint_{l_{TX}}\oint_{l_{RX}}\frac{d\boldsymbol{l_{TX}}\cdot d\boldsymbol{l_{RX}}}{R}\,, \tag{6}$$

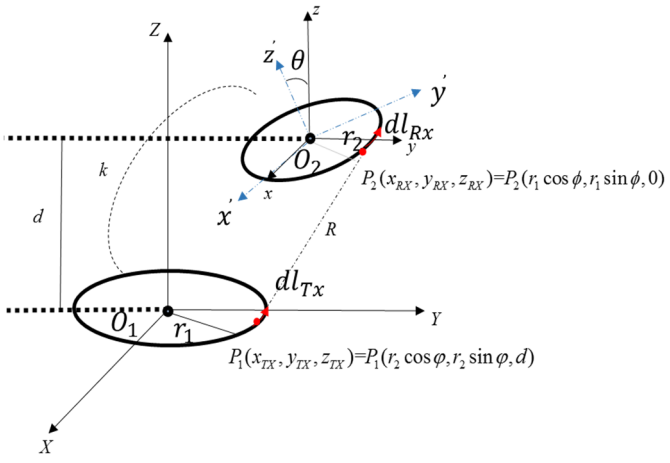


Fig. 2. Configuration of the WPT system with angle mis-alignment and horizontal misalignment.

where N_{TX} and N_{RX} are the number of TX and RX coil turns, respectively, and μ_0 is the magnetic permeability of free space. In Fig. 2, r_1 represents the radius of TX, and r_2 is the radius of RX. The centers of the TX coil and RX coil are $O_1(0, 0, 0)$ and $O_2(0, y, d)$, respectively. The horizontal distance is represented by y, and d is the vertical distance from the center of the TX coil to the center of the RX coil. The point P_1 on the TX coil is (x_{TX}, y_{TX}, z_{TX}), and the point P_2 on the RX coil is (x_{RX}, y_{RX}, z_{RX}). The dl_{TX} and dl_{RX} are counter-clockwise directional differential length change vectors for the TX and RX coils, respectively. R is the distance between P_1 and P_2 as follows:

$$R = \sqrt{(x_{TX} - x_{RX})^2 + (y_{TX} - y_{RX})^2 + (z_{TX} - z_{RX})^2} \cdot \tag{7}$$

The parametric equations of the TX coil and the RX coil are described using ϕ and φ $(0 \le \phi, \varphi \le 2\pi)$, respectively. When two coils have no angular misalignment, the parametric equations of the two coils can be represented as follows:

$$(x_{TX}, y_{TX}, z_{TX}) = (r_1 \cos\phi, r_1 \sin\phi, 0). \tag{8}$$

$$(x_{RX}, y_{RX}, z_{RX}) = (r_2 \cos\varphi, r_2 \sin\varphi, d). \tag{9}$$

In Fig. 2, two coils are non-coaxial if $y \neq 0$. The normal direction of the RX coil plane is z', and the normal direction of the TX coil plane is z. The angle θ represents the tilt angle between the two coils, and it is defined as the angle between the normal vector of plane z and the normal vector of plane z'. On the basis of the TX coil plane, we build a Cartesian coordinate system xyz. According to the xyz coordinate system, the $x'y'z'$ coordinate system can be established by drawing x' parallel to x in the RX coil plane and drawing y' perpendicular to $x'z'$. The rotation matrix, which performs the rotation relationship between xyz and $x'y'z'$, can be represented as follows:

$$T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}. \tag{10}$$

Therefore, the P_2 of the RX coil considering the tilt angle can be represented using Eqs. (9) and (10) as follows:

$$\begin{bmatrix} x_{RX} \\ y_{RX} \\ z_{RX} \end{bmatrix} = T \begin{bmatrix} r_2 \cos\varphi \\ r_2 \sin\varphi \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \\ d \end{bmatrix} = \begin{bmatrix} r_2 \cos\varphi \\ r_2 \cos\theta \sin\varphi + y \\ r_2 \sin\theta \sin\varphi + d \end{bmatrix}. \tag{11}$$

The dl_{TX} and dl_{RX} for P_1 in Eq. (8) and P_2 in Eq. (11), respectively, are as follows:

$$d\boldsymbol{l}_{TX} = (-\sin\phi \boldsymbol{x} + \cos\phi \boldsymbol{y}) r_1 d\varphi \tag{12}$$

And

$$d\boldsymbol{l}_{RX} = (-\sin\varphi \boldsymbol{x} + \cos\theta\cos\varphi \boldsymbol{y} + \sin\theta\cos\varphi \boldsymbol{z}) r_2 d\varphi., \tag{13}$$

where x, y, z are the base vectors in the xyz coordinate. Substituting Eqs. (7), (8), (11), (12), and (13) into Eq. (6), the coupling coefficient between the TX coil and the RX coil can be calculated. If the centers and the radii of the two coils are given, then the coupling coefficient is a function of the tilt angle as follows:

$$k(\theta) = \alpha_1 \cdot \oint_{l_{TX}} \oint_{l_{RX}} \frac{\alpha_2 + \alpha_3 \cos\theta}{\sqrt{(\alpha_4 + \alpha_5 \cos\theta + \alpha_6 \sin\theta)}} d\phi d\varphi \tag{14}$$

where $\alpha_1 = \frac{N_{TX} N_{RX} \mu_0}{4\pi\sqrt{L_{TX} L_{RX}}}$, $\alpha_2 = r_1 r_2 \sin\phi \sin\varphi$, $\alpha_3 = r_1 r_2 \cos\phi \cos\varphi$, $\alpha_4 = r_1^2 + r_2^2 + y^2 + d^2 - 2r_1 y \sin\phi - 2\alpha_3$, $\alpha_5 = 2r_2 \sin\varphi y - 2\alpha_2$, and $\alpha_6 = 2r_2 d \sin\varphi$.

In Eqs. (3) and (4), we show that η increases when k increases. To achieve an optimal tilt angle θ_{opt}, which gives the maximum PTE for the given center positions and radii of the two coils, we set θ_{opt} as a value that satisfies $\partial k(\theta)/\partial\theta = 0$ and $\partial^2 k(\theta)/\partial\theta^2 < 0$.

III. Simulated and Experimental Results

To verify the above analysis and to determine the maximum transmission efficiency, we implement a WPT system. The parameters of the implemented WPT system are listed in Table 1.

We execute simulations and experiments for various cases. Three different receivers are considered; cases I, II, and III have different radii of the RX coils. The results show that the inductances and resistances of these three cases differ. Different capacitors are used to ensure that the LC resonance for each case occurs at the same resonant frequency as described in Table 1.

For the given vertical distance d, we simulate the PTE using Eqs. (3) and (14) by changing the horizontal distance (y) and the tilt angle of the RX coil (θ). Fig. 3 shows the numerical extracted coupling coefficients for cases I, II, and III when $O_2(0,0.05,0.15)$, respectively. Fig. 3 clearly shows that the coupling coefficient is related to the tilt angle of the RX coil.

To examine the effect of the tilt angle on the coupling coefficient for the coaxially parallel circular TX and RX coils, we extract the coupling coefficients using Eq. (14) versus the radii of the RX coil, r_2. In this simulation, we change the tilt angle of the RX coil in the range of -90° to 90° and change r_2 from 0 to 0.15 m. The simulated results are shown in Fig. 4.

The results show that the optimal tilt angle is not always zero and depends on the radius of the RX coil. The simulation results indicate that the angle between the TX and the RX for the

Table 1. Parameters of the WPT system

Parameter	Sym-bol	Value
Resonant frequency	f	100 kHz
Transmitter coil inductance/radius	L_1 / r_1	217.5 μH / 0.15 m
Transmitter coil resistance	R_{P1}	4.5 Ω
Transmitter capacitance	C_1	11.65 nF
Load resistance	R_L	20 Ω
Source impedance	R_S	50 Ω
Number of turns	N_{TX}, N_{RX}	18,18
Receiver coil inductance/radius	L_2 / r_2	
Case I		80.3 μH / 0.075 m
Case II		120.64 μH / 0.1 m
Case III		217.5 μH / 0.15 m
Receiver capacitance	C_2	
Case I		31.54 nF
Case II		20.45 nF
Case III		11.65 nF
Receiver coil parasitic resistance	R_{P2}	
Case I		2.1 Ω
Case II		3.2 Ω
Case III		4.5 Ω

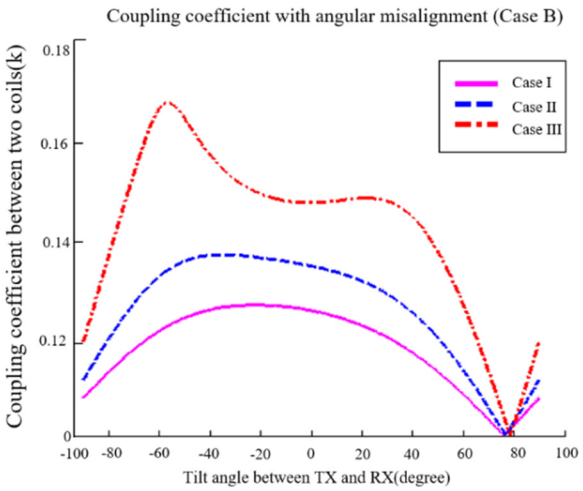


Fig. 3. Coupling coefficient with angular misalignment.

maximum efficiency is 0° when the radius of the RX coil is less than 0.1 m for the case with $d = 0.15$ m. When the radius is larger than 0.1 *m* for this case, the angle for the maximum efficiency is non-zero. We plot the change of the optimal tilt angle for the various radii of the RX coil for the coaxially located TX and RX coils in Fig. 5.

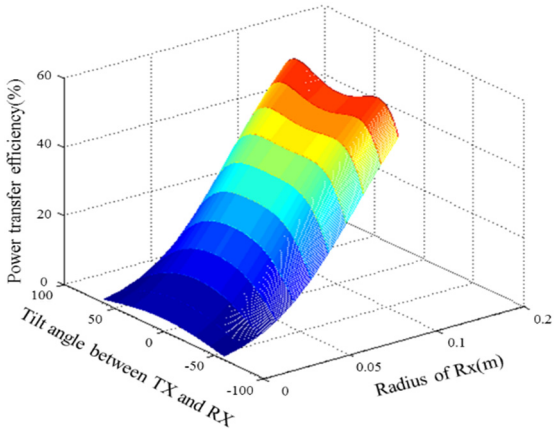


Fig. 4. Simulated coupling coefficient with various tilt angles and radii of RX.

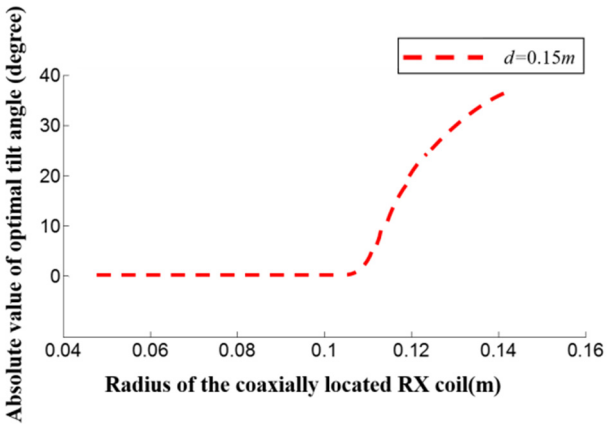


Fig. 5. Extracted optimal tilt angle versus radii of the coaxially located RX coil.

In addition, we analyze the PTE for various cases with the horizontal offset between the TX and RX coils. We extract the PTE by rotating the RX coil in the range of -90° to 90° and by changing the horizontal distance, *y*, from 0 to 0.15 m along the *y*-axis positive direction. The simulated results are shown in Figs. 6(a), 7(a), and 8(a) for cases I, II, and III, respectively. The results show that the PTE strongly depends on the tilt angle of the RX coil. The condition for the maximum PTE and its value for each case are summarized in Table 2.

The simulated maximum PTE is denoted by $\eta_{\max}$, which us obtained by utilizing the optimal values of the angle between the two coils and the horizontal distance. When the radius of the RX coil is small ($r_2 = 0.075$ m), such as in case I, the RX coil, which is coaxially aligned with the TX coil with no tilt angle, gives the maximum PTE. However, as the radius of the RX increases ($r_2 = 0.1$ m, $r_2 = 0.15$ m) in cases II and III, the maximum PTE is achieved by using the non-coaxially located RX coil with the non-zero tilt angle.

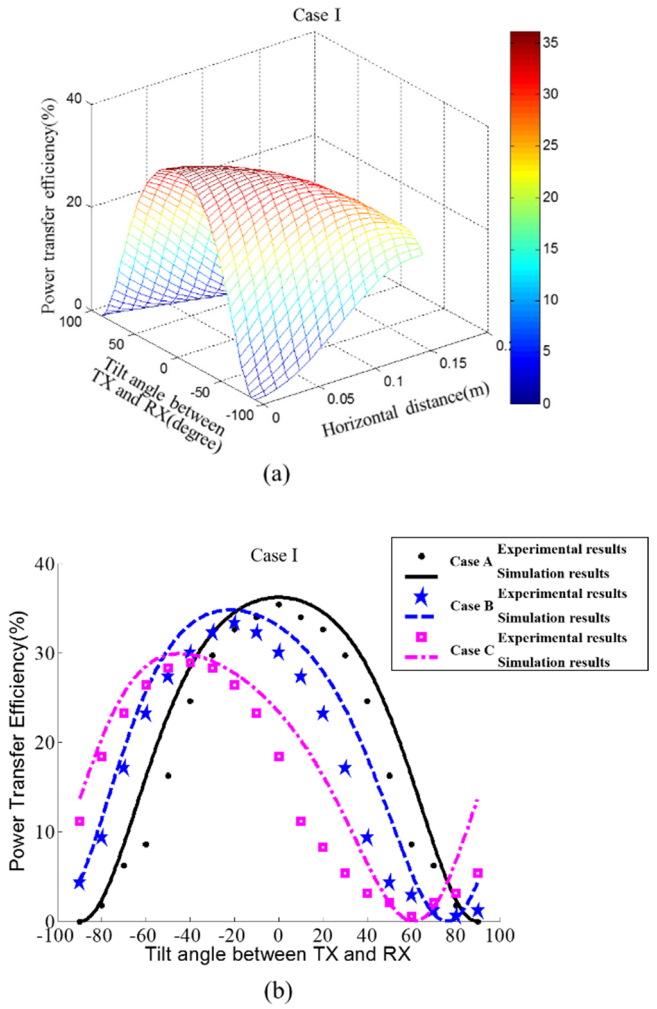


Fig. 6. (a) Simulated transmission efficiency for case I with various tilt angles and horizontal distances. (b) Transmission efficiency with various tilt angles and some special horizontal distance in Fig. 6(a) and the experimental results.

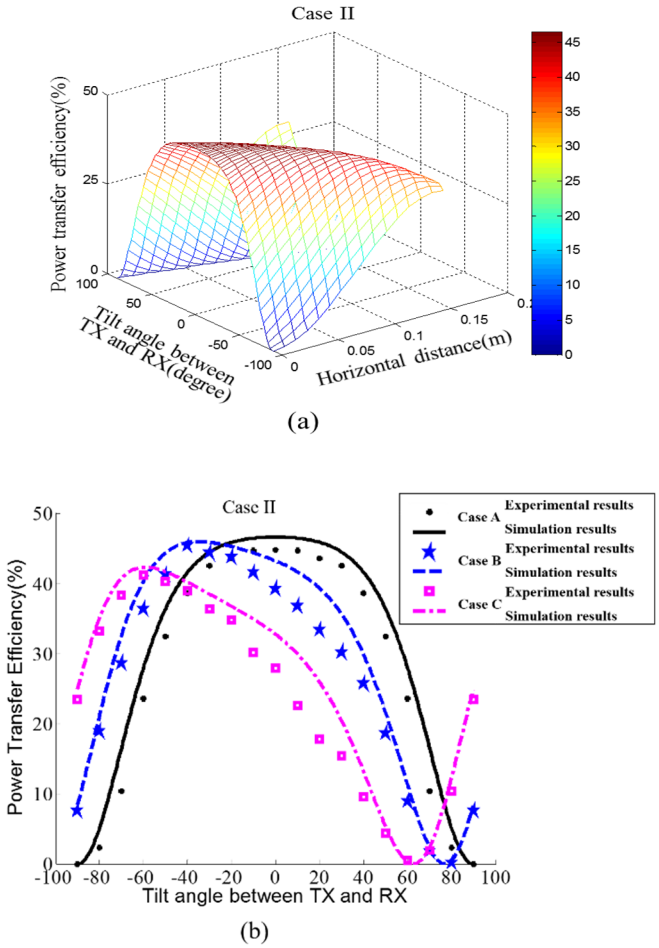


Fig. 7. (a) Simulated transmission efficiency for case II with various tilt angles and horizontal distances. (b) Transmission efficiency with various tilt angles and a special horizontal distance from Fig. 7(a) and the experimental results.

To verify the simulated results, we measure the PTE for the WPT system listed in Table 1 by changing the tilt angle for the three different horizontal offset positions (case A, $y = 0$ m; case B, $y = 0.05$ m; and case C, $y = 0.1$ m). The measured PTE is acquired using the following equation:

$$\eta_P = \frac{P_{RX}}{P_{TX}} = \frac{V_L I_L \cos\gamma_L}{V_{TX} I_{TX} \cos\gamma_{TX}}, \tag{15}$$

where η_P is the ratio of the real power dissipated in the load, P_{RX}, to the power provided by the source driving the TX coil, P_{RX}; V_{TX}, I_{TX}, V_L, and I_L are the voltage and current amplitudes of the source and the load; and γ_{TX} and γ_L are the phase difference between the voltage and the current signals in the source and the load. The measured results are plotted in Figs. 6(b), 7(b), and 8(b) for cases I, II, and III, respectively. Fig. 6(b) shows the simulated transmission efficiency with various tilt angles and a number of special horizontal distances from Fig. 6(a). The experimental results are presented in Fig. 6(b). Fig. 7 shows the simulated and experimental results when the radius of RX is 0.1 m. In Fig. 7(b), when the position of the RX coil is 0.05 m in the positive y-axis and the angle between the RX coil and the TX coil is −34.7667°, the maximum transmission efficiency is obtained. The simulation and experimental results are given in Fig. 8, where the radius of RX is the same as that of TX. In Fig. 8(b), the bifurcation phenomenon occurs when the RX coil is 0.1 m or 0.15 m in the positive y-axis. In this study, the bifurcation phenomenon is not considered. Fig. 8(b) shows that when the radii of the two coils are equal, a special angle and a special horizontal distance are given to ensure the maximum transmission efficiency. In this experiment, the maximum transmission efficiency is reached when the angle between TX and RX is about −56.4247° and the RX coil is located at 0.05 m in the positive y-axis.

The measured results are compared with those of the simulated ones. As the horizontal offset increases, the tilt angle of the RX coil for the maximum PTE also increases. The simula-

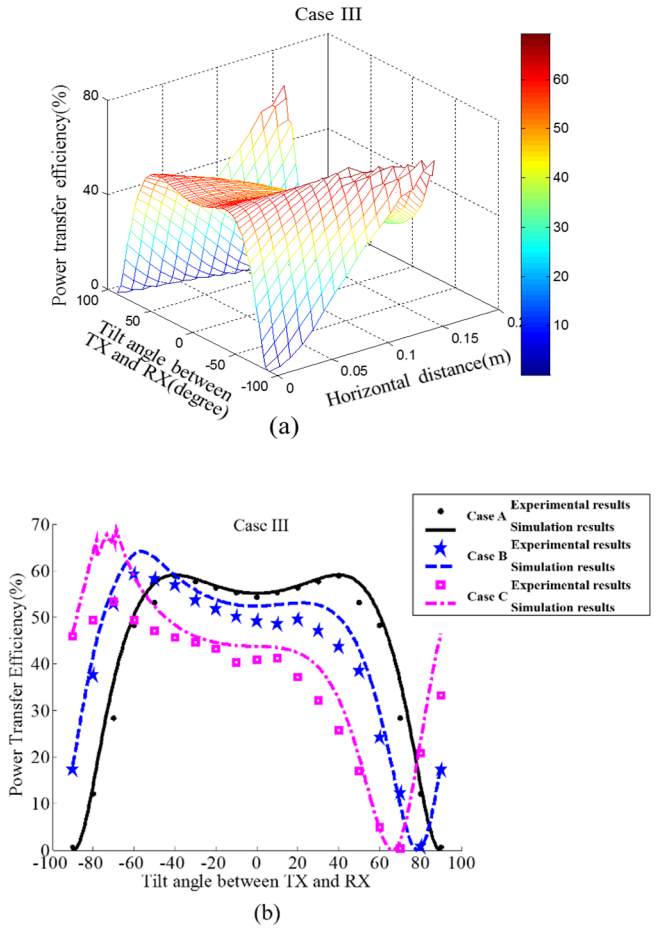


Fig. 8. (a) Simulated transmission efficiency for case III with various tilt angles and horizontal distances. (b) Transmission efficiency with various tilt angles and some special horizontal distance in Fig. 8(a) and the experimental results.

Table 2. Position of the coil and tilt angle for the maximum PTE

	r_2 (m)	O_2 for η_{max}	Optimum θ (°)	η_{max} (%)
Case I	0.075	O_2(0,0,0.15)	0	36.2213
Case II	0.1	O_2(0,0.05,0.15)	-34.7669	45.9719
Case III	0.15	O_2(0,0.05,0.15)	-56.4247	64.1415

Table 4. Comparison of accuracy between the simulated results and the experimental results

	Case A (y=0 m)	Case B (y=0.05 m)	Case C (y=0.1 m)
Case I	3.8505	3.9162	4.1944
Case II	3.3371	3.3221	3.5365
Case III	3.1352	3.7635	6.7635

ted PTEs agree well with the measured results as shown in Figs. 6(b), 7(b), and 8(b). The results show that the predicted maximum tilt angle for each case is accurate. The simulated and measured values of PTE for each case are presented in Table 3. Table 4 shows the accuracy between the simulated results and the experimental results using root mean square errors for the WPT system with one TX and one RX. As shown in Table 4, the experimental results agree with the measured ones.

IV. Conclusion

We analyze the effect of the tilt angle of the coil on the PTE of the WPT system. Based on our analysis results, the PTE of a WPT with two circular coils can be optimized by changing the tilt angle, relative position, and radii of coils. To archive the optimized PTE, we propose a method to calculate the mutual inductance between the two coils with the tilt angle located at an arbitrary position in a 3D coordinate through the Neumann equation. Using the extracted mutual inductance and the equivalent circuit model of the WPT system, the coupling coefficient and the PTE depending on the tilt angle are extracted. We propose an analytical scheme to extract the optimal tilt angle that maximizes efficiency. To demonstrate our proposed theoretical method, we design experiments using a WPT with two circular coils. By changing the position, radius, and the tilt angle of the RX coils, we compare the transmission efficiency of the different cases. The suggested analysis effectively predicts the optimum tilt angle of the RX coil and the PTE for the various cases.

Table 3. Simulated and measured values of PTE for each case

	Case A (y=0 m)				Case B (y=0.05 m)				Case C (y=0.1 m)			
	Simulation		Experiment		Simulation		Experiment		Simulation		Experiment	
	θ	η_{max}	θ	η_{max}	θ	η_{max}	θ	η_{max}	θ	η_{max}	θ	η_{max}
Case I	-0.05	33.5	0	35.4	-21.8	32.2	-20	33.4	-44.7	27.6	-40	28.9
Case II	-0.04	46.6	0	44.8	-35.0	46.0	-40	45.4	-59.1	42.3	-60	41.2
Case III	±40.2	59.1	40	58.9	-56.2	64.1	-60	59.2	-69.4	66.6	-70	53.3

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Study on Timing Characteristics of High-Voltage Pulse Generation with Different Charging Voltages

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Abstract

The time synchronization of each sub-unit of a pulsed generator is important to generate an output high-power radio frequency (RF) signal. To obtain the time synchronization between an input RF signal fed by an external source and an electron beam produced by an electric pulse generator, the influence of different charging voltages on a delay and a rise time of the output pulse waveform in the electric pulse generator should be carefully considered. This paper aims to study the timing characteristics of the delay and the rise time as a function of different charging voltages with a peak value of less than –35 kV in the high-voltage pulse generator, including a trigger generator (TG) and a pulse-forming line (PFL). The simulation has been carried out to estimate characteristics in the time domain, in addition to their output high-voltage amplitude. Experimental results compared with those obtained by simulation indicate that the delay of the output pulses of the TG and PFL, which are made by controlling the external triggering signal with respect to different charging voltages, is getting longer as the charging voltage is increasing, and their rise times are inversely proportional to the amplitude of the charging voltage.

Key Words: Charging Voltage, High-Voltage, Rise Time, Time Delay

I. INTRODUCTION

Pulsed power systems with a high-voltage pulse have been developed to produce a high-power electromagnetic pulse using a switch that can handle the high-voltage capacity and that can be triggered by an external triggering pulse to synchronize with the system. For example, the high-power electromagnetic pulse can be produced by sending the external triggering signal into a spark gap switch installed before a broad impulse antenna, after charging the high-voltage spark gap switch regulated by changing the insulating gas pressure and the gap distance between the two main electrodes of the spark gap switch [1]. Another way to produce a high-power electromagnetic pulse is using a vacuum amplifier that emits an intense electron beam from the cathode that interacts with a microwave provided by an external driver in an input cavity. For an efficient interaction between an intense electron beam and the input cavity, the voltage for the cold cathode emission needs to reach above a threshold level of 140 kV [2]. Therefore, key technical points to get the high-power electromagnetic pulse are essentially dependent on whether and how much electrical pulse energy with more than a couple hundred kilovolts is gained. This is because it is related directly to an electron beam acting as a key factor when it interacts with the external radio frequency (RF) input signal at the input part in the vacuum tubes [3]. There are largely two steps to generate an electrical pulse with more than a couple hundred kilovolts in

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a pulsed generator. First, direct current (DC) volts will be charged up to several tens of kilovolts. Then, high-voltage stacking will be conducted using a trigger generator (TG) to produce a high-voltage electrical pulse by charging many capacitors in parallel. If there is no voltage loss in this circuit, the target amplitude of the voltage is equal to the result of multiplying the number of stacking capacitors by the amplitude of the voltage of the single capacitor. To get the optimum value of the target voltage amplitude, therefore, the characteristic of a high-voltage switch connecting one stage to another in the capacitor bank plays a key role [4]. Moreover, it can influence the general performance of a pulsed generator. This is why it is so important to research the characteristics of the high-voltage switch.

Another thing that must be considered to generate a good waveform at the output of a pulsed generator after achieving an acceptable high-voltage level at the voltage stacking structure is a pulse-forming line (PFL) as a sub-system of a pulsed generator. It can largely be categorized into a pulse-forming network [5–8] and a PFL [9–12]. The pulse-forming network can be applicable to creating a long pulse duration, but it has the disadvantage of having many parasitic components and low flexibility to the impedance of the load comparatively. However, the PFL can be useful for creating a short pulse duration, and it has good compatibility with the load impedance match compared to the pulse-forming network.

Fig. 1 shows the general timing diagram for generating a high-power RF signal in a pulsed power system including a vacuum tube. It indicates how the vacuum tube is operated. First, a driver can be ready by receiving the trigger by signaling a pulse modulator and the enable time signal sequentially, which results in its RF pulse. Second, a pulsed magnet for producing a sufficient magnetic field to guide the electron beam in the vacuum tube given by a pulsed generator during the enable time can be operated by the external trigger on the signal. Third, a pulsed generator can produce an electron beam when the triggering signal is given. Lastly, the RF pulse from the output port of the vacuum tube can be attained within the duration of the electron beam.

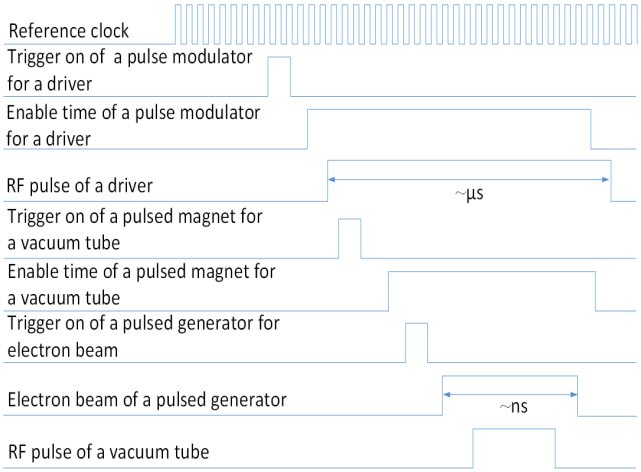


Fig. 1. General timing diagram for capturing an RF signal in a vacuum tube.

As explained above, an analysis in the time domain is important to determine whether all of the parameters related to the high-power RF generation through a vacuum tube are synchronized with each other. Particularly, time synchronization in a pulse generator should be analyzed because it has a time length in nanoseconds, and the characteristics of the output electric short pulse of a pulsed generator can have a significant impact on the generation of an electron beam as a source of the vacuum tube.

If the output RF pulse level of the pulsed generator is tuned by the variation in the electron beam intensity with the fixed input RF pulse level driven by a driver instead of the variation in the driver's RF pulse power level, the interaction timing in an input cavity of a vacuum tube between the input RF pulse and the electron beam driven by the electric pulse generated by high-voltage pulse generator can largely affect the output RF pulse power level of the pulsed generator.

However, conventional studies have relatively rare data on the time-varying characteristics of a high-voltage pulse with the amplitude of more than –500 kV. Therefore, a time analysis of the generation of an electric high-voltage pulse driving the electron beam of a pulsed generator is the focus of this paper. In particular, pulse characteristics over time produced by a TG and a PFL are researched in detail, because the output power level and efficiency of a vacuum tube depend largely on the time synchronization of the electron beam driven by the electric high-voltage pulse with the input RF signal in the vacuum tube.

II. Design

1. Design of the Capacitor-Charging Power Supply

A capacitor-charging power supply (CCPS) having an output voltage of more than 100 kV is designed to charge capacitors in the high-voltage stacking circuit (HVSC) and is simulated with PSIM [13].

It is designed to work from 60 Hz with three-phase and 380 volts in alternating current (VAC). Its AC input is full-wave rectified through the rectifier. After that, a filtered voltage at the capacitor attained on the secondary of the first transformer is delivered to the second transformer, called the main transformer, through the inverter module to get the target-filtered voltage by the full-bridged rectifier on the secondary of the second transformer, as shown in Fig. 2. It has a rise time of less than 2 ms to reach the desired output voltage of 100 kV, as shown in Fig. 3.

Fig. 4 indicates the primary current with the maximum value of 572 A of the main transformer having a turn ratio of 196 to 1,

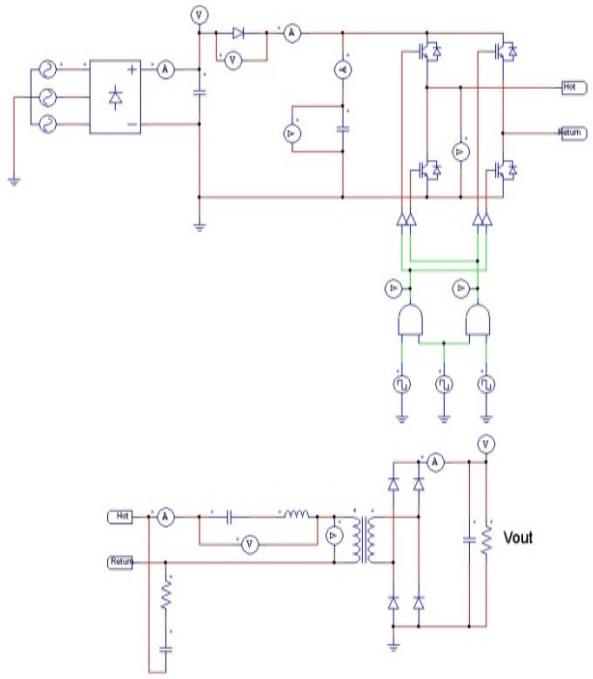


Fig. 2. Circuit diagram of the CCPS.

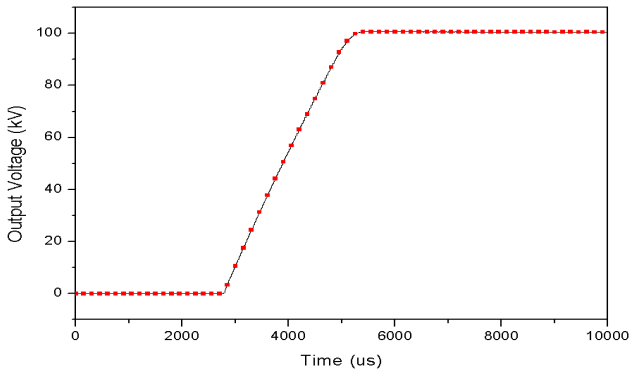


Fig. 3. The simulated maximum output voltage of the CCPS.

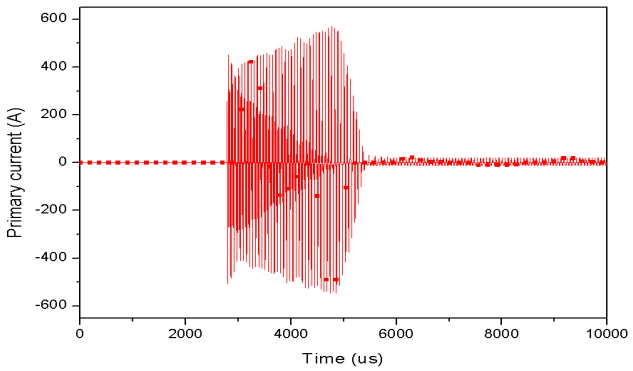


Fig. 4. The simulated primary current of the main transformer in the CCPS.

and Fig. 5 shows the secondary current with a maximum value of 2.91 A of that in the CCPS.

2. Design of the TG

Fig. 6 indicates the circuit diagram of the TG simulated by

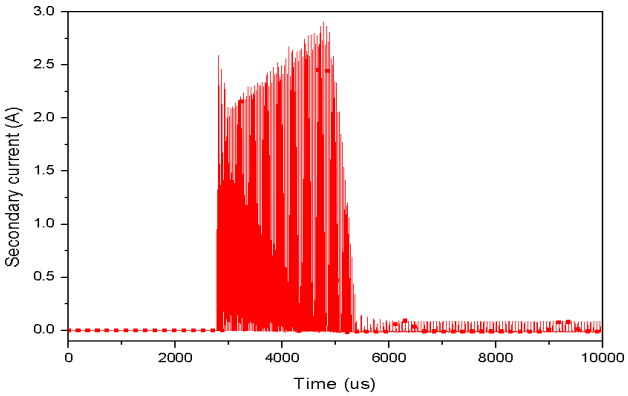


Fig. 5. The simulated secondary current of the main transformer in the CCPS.

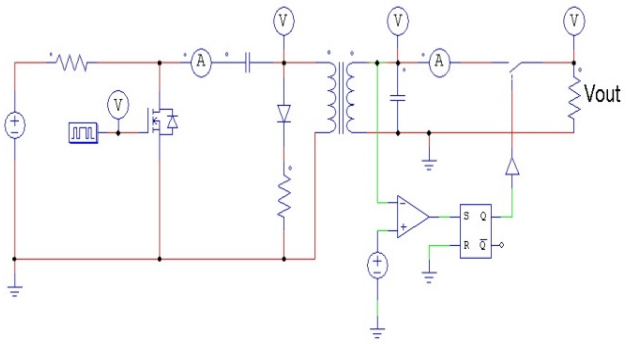


Fig. 6. Circuit diagram of the TG.

PSIM. Its applied input voltage is set to 300 VDC as a bus voltage by taking into account the full-wave rectification and power loss with the input voltage, frequency, and phase at 220 VAC, 60 Hz, and single, respectively. Detailed values in millimeters of the design parameter of the spark gap of the TG are shown in Fig. 7.

Fig. 8 indicates the primary current and voltage at the transformer having the turn ratio of 2 to 400. When the critical level of the output voltage in the circuit with the load resistance of 6 MΩ is set to −60 kV, the characteristic of the rise time is shown in Fig. 9.

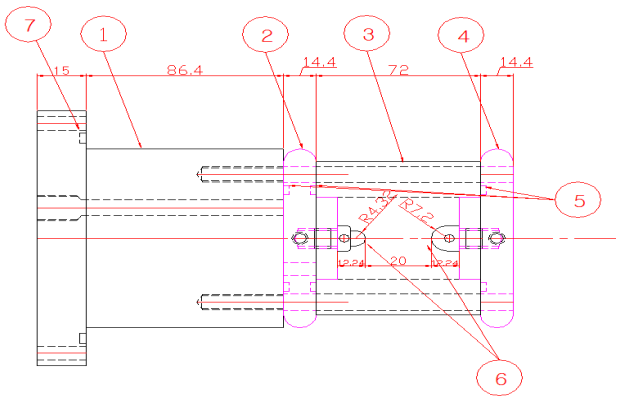


Fig. 7. Designed spark gap of the TG.

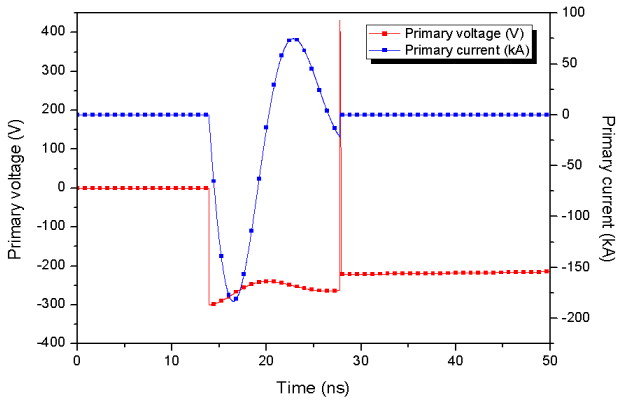


Fig. 8. The simulated primary current and voltage of the transformer in the TG.

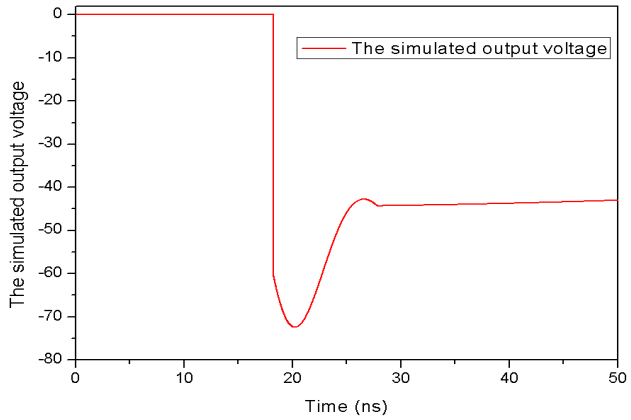


Fig. 9. The simulated output voltage of the TG.

The spark gap of the TG is set to remain at a high resistance to the circuit initially so that capacitor installed in front of the spark gap is allowed to charge until the breakdown voltage of the spark gap is reached.

The number in Table 1 describing parameters of the design of the TG indicates the index circular number, as shown in Fig. 7.

3. Design of the PFL

To produce a pulse duration of more than 100 ns and to keep the same amplitude between the charging voltage and the load voltage, a PFL, called the Blumlein PFL [14], with two transmission lines, as shown in Fig. 10, is designed if it is more sensitive to parasitic inductance.

Table 1. Design parameters of the TG in Fig. 7

Number	Description
1	Insulted frame #1 (Acetal)
2	Input frame (100 kV max.)
3	Insulated frame #2 (Acetal)
4	Output frame (100 kV max.)
5	O-ring (G-35)
6	Electrode (Cu-W, 70:30)
7	O-ring (G-75)

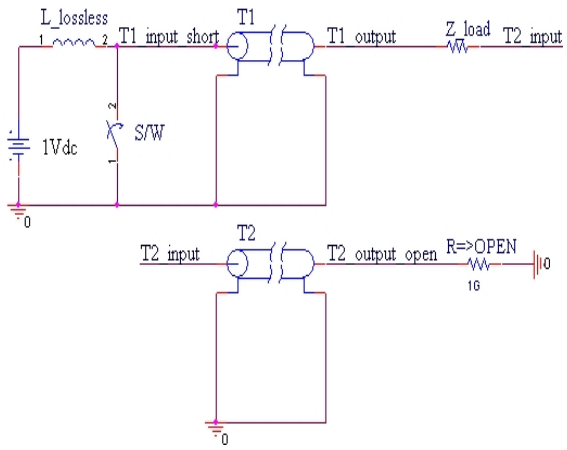


Fig. 10. Circuit diagram of a PFL with two transmission lines.

The load impedance (Z_L) of the PFL, comprised of two lossless transmission lines, is shown in Eq. (1).

$$Z_L = 2 \times \ Z_{o,PFL} = 120\sqrt{\frac{\mu_r}{\epsilon_r}}\ln\left(\frac{R_o}{R_i}\right)[\Omega] \quad (1)$$

where $Z_{o,PFL}$ is the characteristic impedance of the transmission line, ε_r is the relative permittivity of deionized water as a dielectric, μ_r is the relative permeability of deionized water as a dielectric, R_o is the radius of the PFL's outer conductor, and R_i is the radius of the PFL's inner conductor.

The maximum pulse duration on the condition of a full width at half maximum (FWHM) can be defined, as shown in Eq. (2)

$$\mathrm{PW_{FWHM}} = 2 \times \frac{l\sqrt{\varepsilon_r}}{c}[\mathrm{s}] \quad (2)$$

where l is the length of each transmission line and c is the speed of light in a vacuum.

If a pulsed generator with this PFL is one of the sub-systems in a transceiver system, the pulse duration, which is dependent on the PFL, plays a crucial role as a system parameter in terms of whether its performance is effective against any threat, because the pulse duration is related to the transit time in a vacuum tube with a cavity.

Fig. 10 shows a conceptual configuration of the PFL with two lossless transmission lines. When the voltage V_{input}=1 VDC is applied to the input terminal and then the switch is closed, an input voltage signal will flow toward the load Z_{load}, where it will be reflected with a reflection coefficient of 0.5.

The part where inner and middle conductors are connected has a reflection coefficient of –1, and the reflection coefficient at the end of the transmission line comprised of middle and outer conductors is 1.

If the switch is closed at the time of 60 ns after charging and the transmit time of two transmission lines of T1 and T2 is 60

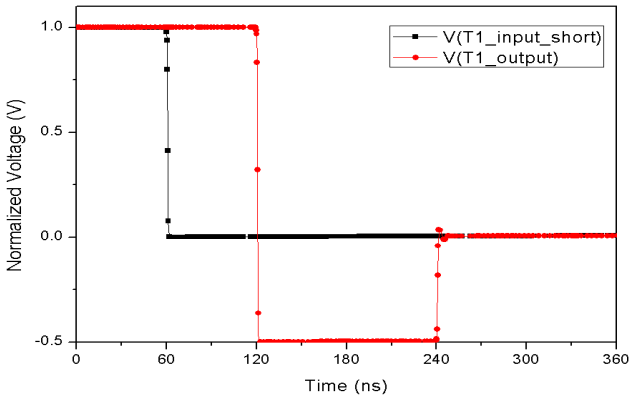


Fig. 11. Voltage distribution formed on the left and right terminals of T1 over time.

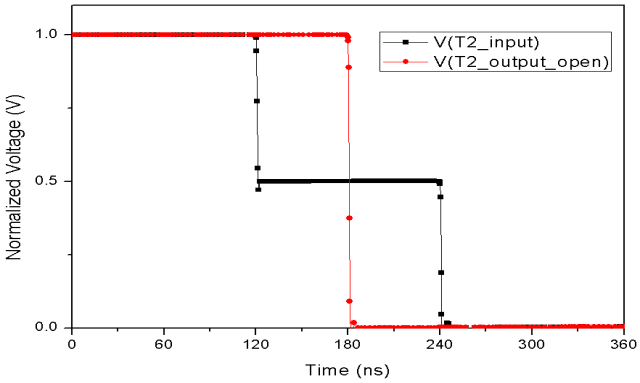


Fig. 12. Voltage distribution formed on the left and right terminals of T2 over time.

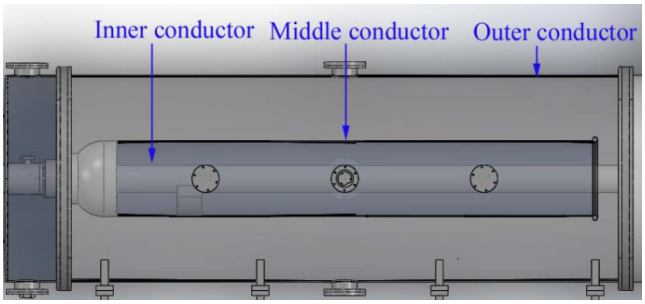


Fig. 13. Schematic of the PFL structure.

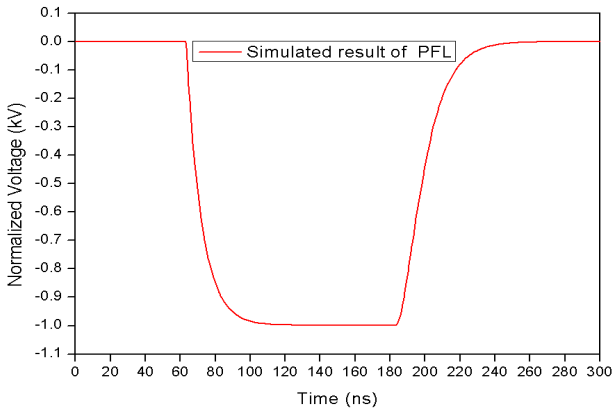


Fig. 14. The simulated pulse waveform of the PFL.

ns each, the lossless PFL can produce a pulse duration of 120 ns. Figs. 11 and 12 describe the voltage formation at both end terminals of T1 and T2 over time.

The structural diagram of the PFL is shown in Fig. 13. A simulated pulse duration of 120 ns on the condition of a FWHM with a voltage of –1 kV is shown in Fig. 14.

To withstand the applied high voltage of down to –500 kV without the electric breakdown and to lessen the physical length of the PFL, the deionized water was used as a dielectric of the PFL. Table 2 indicates the radius and physical length of each connector of the PFL. Two designed transmission conductors have been submerged in deionized water to calculate the characteristic impedance of the PFL as a coaxial cable structure.

All of conductors making up the PFL have been designed to be immersed in the deionized water with a relative permittivity of 78.3 to calculate the characteristic impedance of the PFL as that of a coaxial structure [15].

III. Experiment Results

The fabricated spark gap of the TG is shown in Fig. 15. The measured rise time of the TG is shown in Fig. 16, and it is 7 ns longer than its simulated rise time at the point where it reaches an output voltage of –60 kV.

This slight rise time difference between the measured and simulated results can be explained by the inference, which is caused by the difference in capacitance between the output port cable of the TG and the high-voltage probe of the measurement instrument. In addition, the time difference in reaching the peak output voltage would result from the effect of the capacitance and inductance in the circuit.

The experiment was carried out 10 times, as shown in Fig. 17, with a time step of one minute and two different N_2O_2 gas pressure levels of 0.3 MPa and 0.4 MPa. A bare cable composed of metal composite wire having a combination of tungsten com-

Table 2. Design parameters of the PFL

Parameter	Value
Radius of the inner conductor (mm)	55.0
Radius of the middle conductor (mm)	149.0
Radius of the outer conductor radius (mm)	406.0

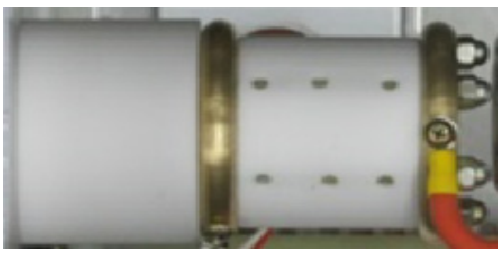


Fig. 15. Fabricated spark gap of the TG.

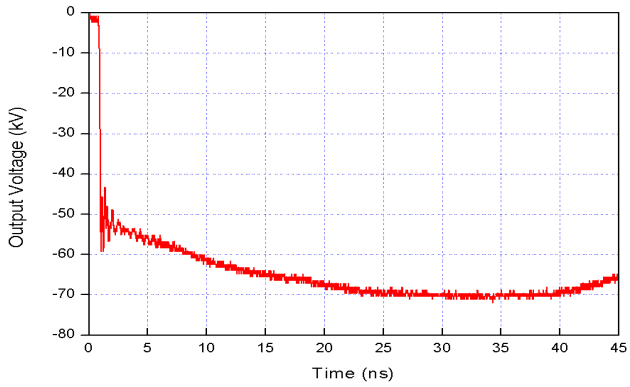


Fig. 16. Measured rise time of the output voltage of the TG.

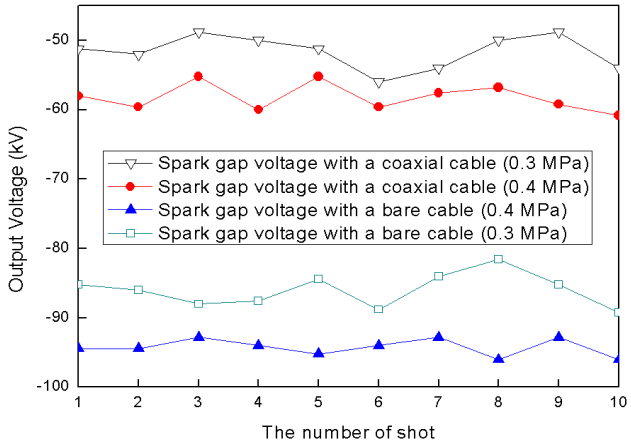


Fig. 17. Breakdown voltage with different cable types and pressure levels.

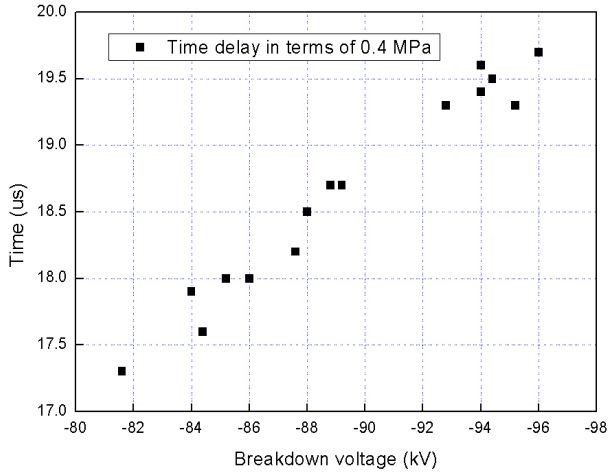


Fig. 18. Time delay in terms of 0.4 MPa.

bined with copper at a percentage of 30% to 70%, respectively, has been applied to achieve an output voltage of down to –100 kV of the TG.

The breakdown voltage is higher when a bare copper cable is applied compared to a coaxial cable and a higher pressure is applied. For each shot, it can be seen that it is constant typically if the experimental condition is the same.

The time delay between the applied charging voltage and the breakdown voltage is approximately linearly proportional to the amplitude of the breakdown voltage, as shown in Fig. 18.

We have conducted an experiment by applying the output signal of the TG to the HVSC comprised of eight high-voltage spark gap switches to analyze how the rise time will change when the applied voltage increases. The spark gap switch of Model 40265 made by L_3 is used as an active load instead of a resistive load in this experiment.

As the charging voltage increases, the rise time of the output voltage signal gets shorter generally, as shown in Fig. 19. Special care with the gas pressure across the electrodes within the spark gap switch must be taken to ensure the spark gap is closed without any effect of variations in pressure across two electrodes.

The measurement setup for analyzing the time delay of a pulsed electric generator is composed of the CCPS, the TG, the HVSC, and the PFL.

A fabricated PFL is shown in Fig. 20. Ground impedance was set to be less than 10 Ω to suppress the floating noises in which the high-frequency components are related to the very fast rising part of the transient, while the low-frequency components are related to the long tail part. High-voltage waveforms are well captured simultaneously over time from the TG, the HVSC, and the PFL.

Typical waveforms over time are formed, as shown in Fig. 21.

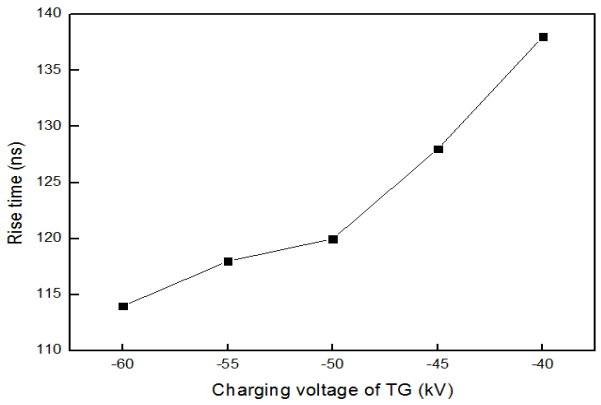


Fig. 19. Fitting curve of the rise time with respect to different charging voltages.



Fig. 20. Fabricated PFL.

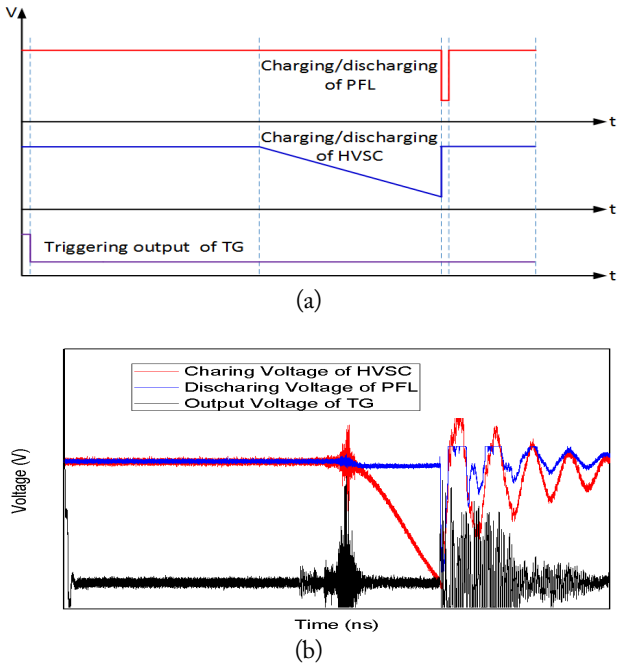


Fig. 21. Typical waveforms with respect to time. (a) Individual charging and discharging sequence. (b) The triggering output of the TG, the charging of the HVSC, and the discharging of the PFL.

The time sequence is largely divided into three stages. First, the output voltage of the TG is produced, and then this signal is applied to the HVSC to charge the high-voltage target to drive the PFL. Finally, charging and discharging occur in the PFL to generate a high-voltage short pulse with the designed pulse duration.

Once the control signal being actively high is optically delivered to the TG, the output pulse signal of the TG is generated after reaching the target charging voltage level of the TG in around 2–7 μs. Then, the HVSC is charged up to the breakdown level for around 3 μs, as shown in Fig. 22.

The PFL begins charging and discharging for a time designed by twice the electrical length of the transmission. Reflected pulse signals emerge after the main pulse is generated due to a stray impedance condition. For example, if the HVSC is charged down to −45 kV, the amplitude of the output pulse signal of the PFL reaches around −450 kV, as shown in Fig. 22.

The reason why the charging time is longer when the charging voltage is higher is discussed. As the target charging voltages of the spark gap switch installed in the TG are −35, −45, −50, and −55 kV, as shown in Fig. 22, the output signals of the TG appear around 2.5, 3.5, 3.7, and 3.8 μs, respectively. Each delay reflects how long it takes to get the target breakdown voltage amplitude, as the wanted breakdown level is higher. As soon as the output signal of the TG is generated, the high-voltage process in the HVSC is started. The process of charging the HVSC is completed from 5.0 to 6.5 μs, depending on the amplitude of the output voltage of the HVSC. A high-voltage pulse is produced after completion of charging the HVSC for

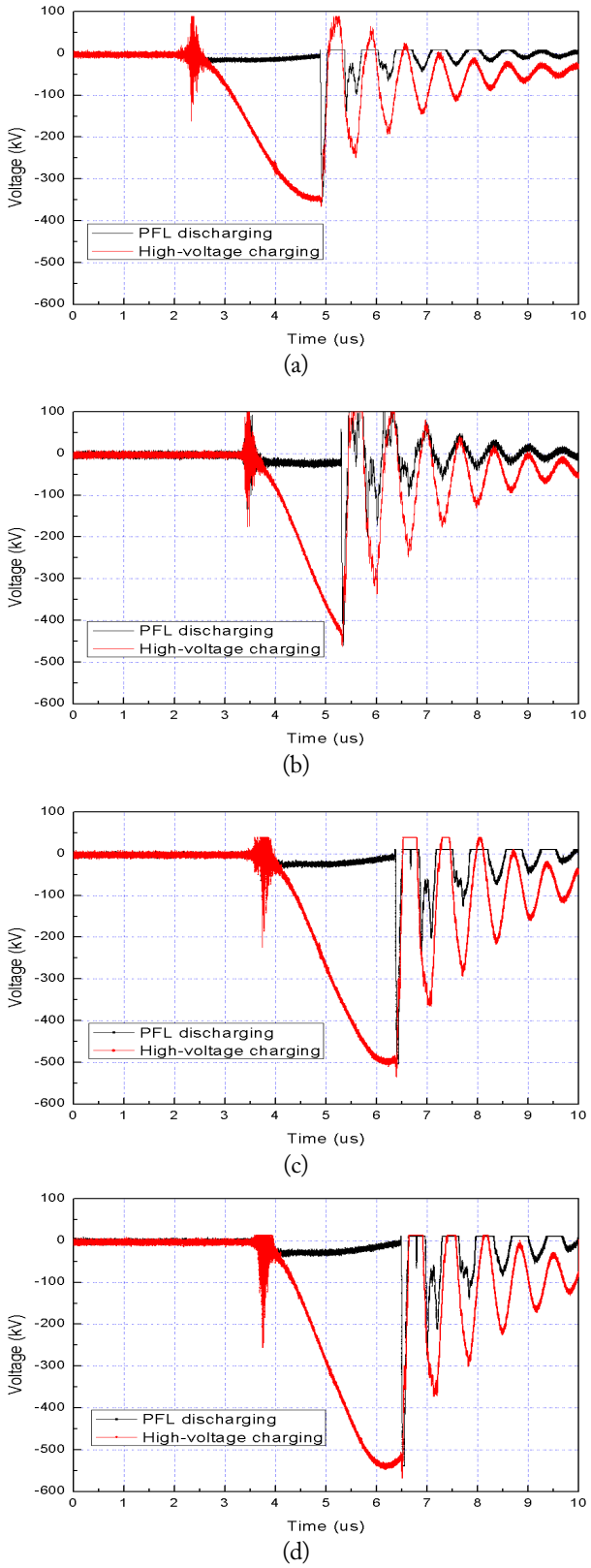


Fig. 22. Time characteristic of measured output waveforms with respect to the change in charging voltage: (a) −35 kV, (b) −45 kV, (c) −50 kV, and (d) −55 kV.

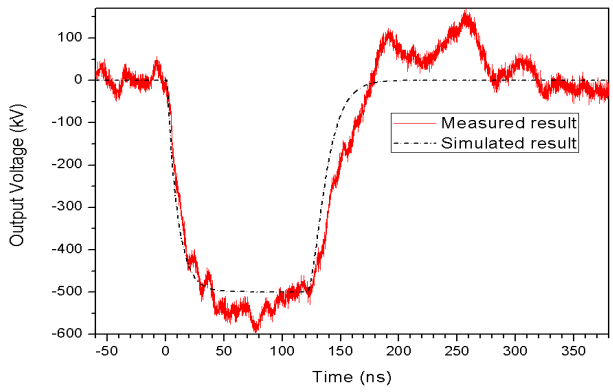


Fig. 23. The result of the output pulse waveform of PFL.

the period during which the pulse duration is linearly proportional to the electrical length of the PFL. This means the internal triggers in the pulsed generator must be matched with the delay and the rise time of the electric high-voltage pulse generated by the high-voltage pulse generator in a time domain when the amplitude of the charging voltage in the pulse generator is lowered or raised to get the changeable output RF pulse level of the pulsed power generator with the fixed input RF pulse level driven by a driver instead of the variation in the input RF pulse power level driven by it.

The agreement on the simulated and measured amplitudes of the output voltage with down to −500 kV at the peak point is shown in Fig. 23, and the measured rising trace is followed by the simulated rising trace. Furthermore, the measured pulse duration is around 120 ns at the FWHF compared to the simulated pulse duration. However, the measured voltage pulse contains a slight ripple due to a stray capacitance.

IV. Conclusions

In this paper, a time analysis with the output electric pulses of the TG and the PFL composing the pulsed generator was performed.

It was found that the delay in the output electric pulses of the TG and the PFL, which are made by controlling the external triggering signal through different charging voltages, is proportional to the amplitude of the charging voltage, and their rise time gets shorter as the charging voltage increases.

The simulation has been carried out using PSIM and PSPICE codes to estimate characteristics over time, such as the high-voltage amplitude, the rise time, and the pulse duration of its output pulse waveform. Data achieved through the simulation have been used for its fabrication. It has been tested to verify that the high-voltage pulse waveform has been generated without breakdown, and its delay and rise time attained repeatedly have a consistent relation with the change in the amplitude of the charging voltage. Results from simulations have been compared with those obtained experimentally.

The time analysis presented in this paper can be helpful for designing a vacuum tube to produce a relativistic output RF signal in terms of the time synchronization of an input RF signal with an intense electron beam.

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Design of a Wideband Antipodal Vivaldi Antenna with an Asymmetric Parasitic Patch

Jihoon Bang · Juneseok Lee · Jaehoon Choi*

Abstract

An antipodal Vivaldi antenna with a compact parasitic patch to overcome radiation performance degradations in the high-frequency band is proposed. For this purpose, a double asymmetric trapezoidal parasitic patch is designed and added to the aperture of an antipodal Vivaldi antenna. The patch is designed to efficiently focus the beam toward the end-fire direction at high frequencies by utilizing field coupling between the main radiating patch and the inserted parasitic patch. As a result, this technique considerably improves the gain and stability of radiation patterns at high frequencies. The proposed antenna has a peak gain greater than 9 dBi over the frequency range of 6–26.5 GHz.

Key Words: Antipodal Vivaldi Antenna, High Gain, Parasitic Antenna, Ultra-Wideband Antenna.

I. INTRODUCTION

The Vivaldi antenna has received a great deal of attention for many applications, such as satellite communications, electronic warfare systems, remote sensing, radio telescopes, radio astronomy, radar, and microwave imaging systems because of its wideband characteristic and high directivity. It has numerous advantages in terms of weight, cost, scan angle capability, ease of fabrication, and system integration. Recently, some applications have required a Vivaldi antenna with higher directivity or gain over a wider bandwidth to achieve the desired system performance such as range resolution [1, 2]. However, a conventional Vivaldi antenna has limited bandwidth because of performance degradation, including directivity or gain reduction and radiation pattern distortion at high frequencies. Numerous factors, including undesired radiation generated by unwanted currents traveling along the termination section and the phase reversal problem, contribute to these limitations. These factors stem primarily from the structural limitations of the conventional Vivaldi antenna. Therefore, a central issue in the further development of the Vivaldi antenna is to improve the directivity at higher frequencies and the stability of the radiation pattern over the frequency range. Several techniques, such as the use of dielectric lens [1, 3], directors [4, 5], and negative index metamaterial [6], have been proposed to overcome these issues. These materials are located in the aperture of the tapered slot to focus energy toward the end-fire direction. Although these methods are suitable for large antenna structures such as the horn antenna [7], they cannot be utilized in a low-profile Vivaldi antenna because of increasing complexity and a high manufacturing cost. In addressing this issue, one novel technique using an elliptical parasitic patch has improved the high-frequency radiation performance without additional complex structures such as lenses or directors [8]. However, performance improvements using this method occur over a limited frequency range, and radiation pattern distortion is present at high frequencies

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due to the increased off-axis radiation as the frequency increases, thus resulting in a poor side-lobe level and front-to-back ratio. This method also results in a reasonable enlargement of the antenna dimension.

In this letter, an antipodal Vivaldi antenna (AVA) with a compact parasitic patch is proposed to enhance antenna performance in areas such as the gain and stability of the radiation pattern at high frequencies. For this purpose, we design a double asymmetric trapezoidal parasitic patch that efficiently focuses the beam toward the end-fire direction and reduces the off-axis radiation at high frequencies. The proposed design is proven capable of enhancing the directivity and improving the stability of the radiation pattern at higher frequencies.

II. Antenna Design

1. Antenna Geometry

Fig. 1 shows the configuration of the designed antennas. The designed Vivaldi antenna is a dual-elliptically tapered antipodal slot antenna (DETASA) [9], which is a modified form of AVA [10]. The conventional DETASA, shown in Fig. 1(a), differs from the exponentially tapered antipodal Vivaldi antenna in that the inner and outer edges of the conducting arms are all elliptically tapered [11]. The conventional DETASA comprises two main parts: a tapered slot radiator and a feed transition. The tapered slot radiator is formed by two conducting arms that are symmetrically printed on opposite sides of the dielectric substrate. The inner and outer slot tapers of the conducting arms follow the outline of a quarter ellipse with two different radii as shown in Fig. 1(a). The feed transition transforms a 50-Ω microstrip line into a parallel stripline by linearly tapering the microstrip line and elliptically tapering the ground plate to feed the tapered slot radiator. The design parameters of the conventional DETASA are identical to those presented in [8]. Fig. 1(b) illustrates the structure of the proposed DETASA with the addition of a double asymmetric trapezoidal parasitic patch in the aperture of the conventional DETASA. The size of the proposed antenna is set to 124 mm × 66 mm, of which the length is L_a mm longer than that of the conventional DETASA. The dielectric substrate is Taconic/TLC-30 with a permittivity of 3.0, tangent loss of 0.0028 and a thickness of 62 mils (1.575 mm). The inserted parasitic patch with various design parameters (L_1, L_2, L_3, α, and β) is located a distance of d away from the center of the flared throat. The parasitic patch is designed to focus the energy toward the end-fire direction over a wide bandwidth, including higher frequencies.

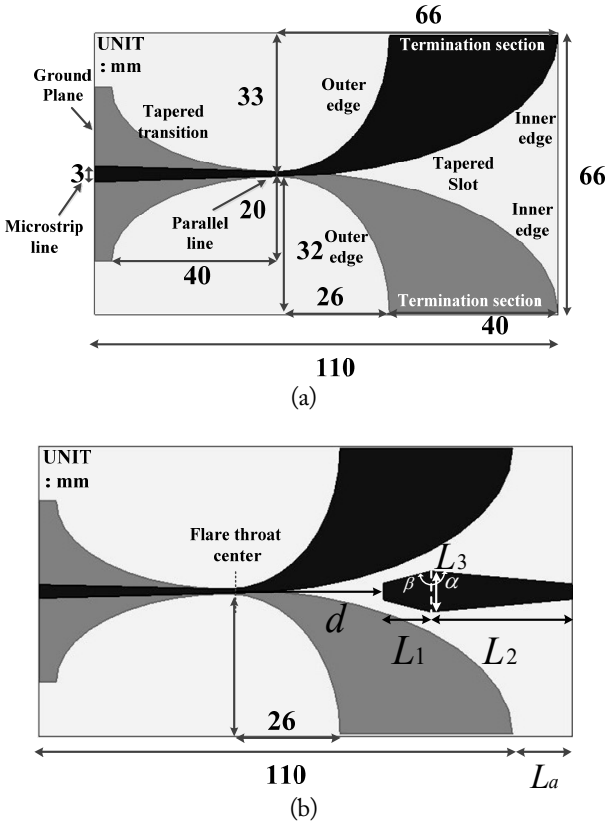


Fig. 1. Configurations of Vivaldi antennas: (a) conventional DETASA (without parasitic patch) and (b) proposed DETASA (with parasitic patch).

2. Parasitic Patch Design

Unlike a conventional DETASA, which produces fields between the inner edges of two conducting arms like a traveling wave antenna [12], the DETASA with a parasitic patch generates fields between the parasitic patch and the inner edges of two conducting arms because the fields couple to both sides of the parasitic patch. Therefore, the field distribution produced in the aperture of the Vivaldi antenna is dependent on the shape and the size of the inserted parasitic patch.

The reasons for designing the parasitic patch in a double asymmetric trapezoidal shape are as follows. (1) To ensure that the field is well coupled to both sides of the parasitic patch, the left trapezoid is arranged opposite to the end-fire direction so that the sides of the left trapezoid of the parasitic patch have an increasing profile similar to that of the inner edge of the conducting arm. (2) At the same time, to transform the E-field distribution produced at the aperture of the antenna into a plane-like wave form, the right trapezoid of the parasitic patch is designed such that its sides have a decreasing profile towards the end-fire direction. Unlike other shapes such as elliptical [8] and rectangular, the double asymmetric trapezoidal shape has a degree of freedom in determining the profile ratio. This degree of freedom enables the beam to focus while reducing the off-axis radiation more effectively over wider frequency ranges. However, the fields cannot be coupled suitably to the parasitic patch if the size of the parasitic patch is extremely larger than the wavelength. For this purpose, the size (L_1, L_2, and L_3), angle

Table 1. Design parameters of the proposed parasitic patch

Parameter	Value	Parameter	Value
L_1	8.8 mm	α	76°
L_2	35.2 mm	β	85°
L_3	8.08 mm	d	36 mm
L_a	14 mm		

(α and β), and position (d) of the parasitic patch are optimized through a parametric study and the final design parameters are listed in Table 1.

To investigate the effect of the inserted parasitic patch, simulated electric field distributions in the near-field region of three antennas (conventional DETASA without a patch, referenced DETASA with the parasitic ellipse presented in [8], and the proposed DETASA) at 15 GHz and 22 GHz are shown in Fig. 2. The simulated current distribution of the proposed Vivaldi antenna at a high frequency of 22 GHz is illustrated in Fig. 3. The simulation is performed using the CST Microwave Studio software (2017 version 15.0; CTS Computer Simulation Technology GmbH, Darmstadt, Germany). As shown in Figs. 2 and 3, the radiation of the proposed Vivaldi antenna is more focused toward the end-fire direction at high frequencies than those of the other two Vivaldi antennas because of the strong electromagnetic field coupling between the man radiating element and the inserted parasitic patch. Clearly, the conventional DETASA generates spherical-like waves at both frequencies, as shown in region A of Fig. 2, and the off-axis radiation results in low directivity. Conversely, the proposed DETASA generates plane-like waves at both frequencies, as shown in region D of Fig. 2. Most of the energy is radiated toward the end-fire (axis) direction with a little off-axis radiation, thus resulting in high directivity at those frequencies. The referenced DETASA generates plane-like waves relatively well at 15 GHz but not at 22 GHz, as shown in regions B and C of Fig. 2.

One can expect radiation performance degradation at high frequencies in the referenced DETASA.

III. Results and Analysis

A prototype of the antenna is fabricated and tested. Fig. 4 shows the fabricated prototype. A 50-Ω SMA connector is used to feed the antenna. The simulated and measured reflection coefficients of the conventional (without a parasitic patch) and the proposed (with a parasitic patch) Vivaldi antennas are shown in Fig. 5. The measured results are given up to 26.5 GHz only because of the frequency limit of the SMA connector. As illustrated in Fig. 5, inserting the parasitic patch affects the reflection coefficient, but the –10 dB reflection coefficient bandwidth is not significantly changed. On the other hand, the inserted parasitic patch strikingly enhances the gain as shown in

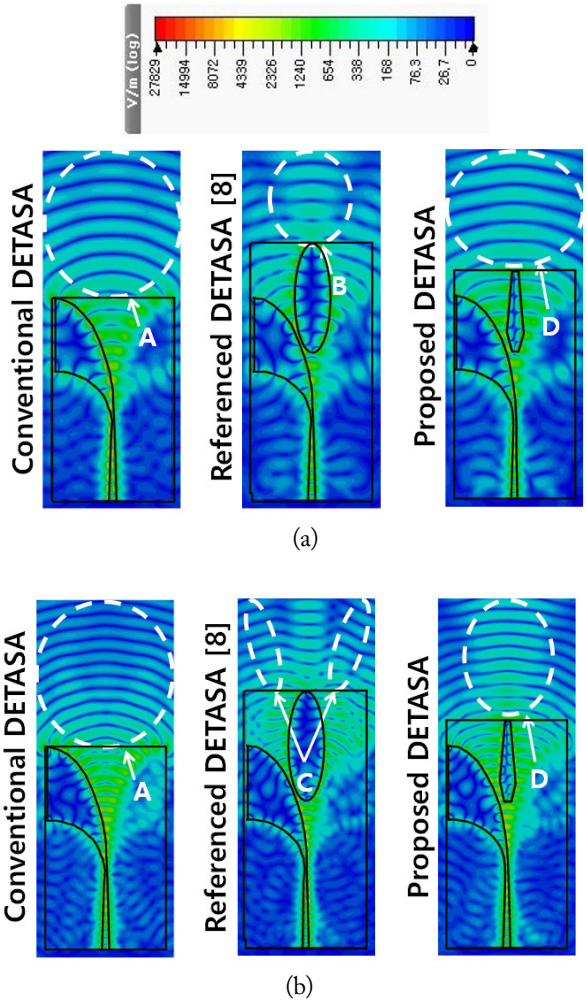


Fig. 2. Electric field distributions of three DETASA configurations: (a) 15 GHz and (b) 22 GHz.

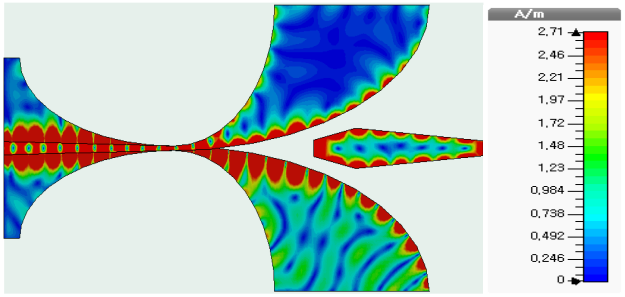
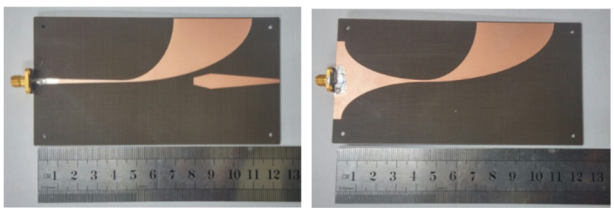


Fig. 3. Current distribution of the designed antennas at 22 GHz (phase: 0°).



(a) (b)

Fig. 4. Photograph of the fabricated prototype. (a) Top view and (b) bottom view.

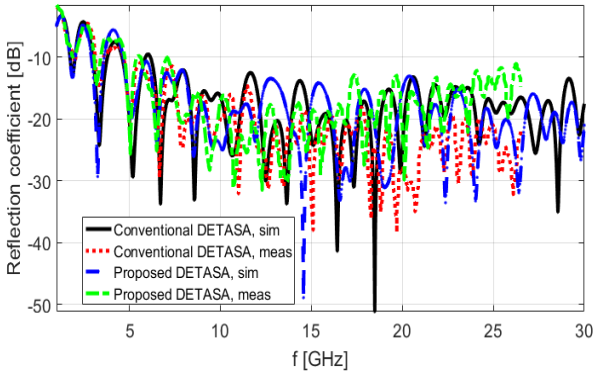
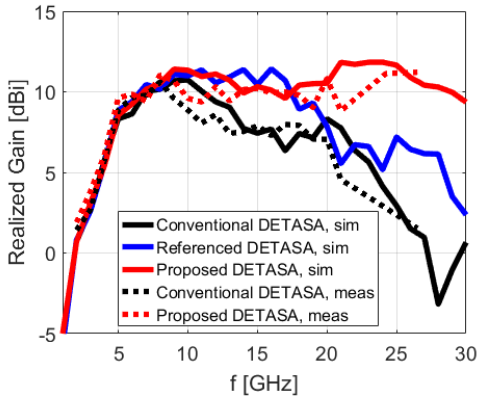
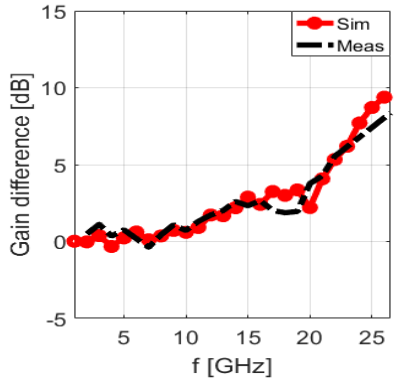


Fig. 5. Simulated and measured reflection coefficients of the conventional (without parasitic patch) and the proposed (with parasitic patch) DETASAs.

Fig. 6, which shows the simulated and measured realized gain variations with frequency for the three antennas and the gain differences between the conventional and proposed DETASAs. Some deviations exist between the simulated and the measured realized gain, which is possibly due to alignment errors between the antenna under test and the referenced antenna. The simulated and measured gain improvements indicate good agreement as shown in Fig. 6(b). Note that the gain of the proposed DETASA is significantly improved at higher frequencies above 20 GHz in contrast to those of the other two designed antennas. Even with the reduced antenna size compared with the referenced DETASA, the gain of the proposed DETASA shows similar improvement over the lower frequency range, with a significant improvement of above 20 GHz. As a result, the double trapezoidal parasitic patch enables gains greater than 9 dBi in the frequency range of 6–26.5 GHz. The simulated E-plane electric field patterns for three antennas at 22 GHz are shown in Fig. 7. The field pattern of the proposed antenna is well formed to generate end-fire radiation toward the axial direction of the slot aperture at 22 GHz, unlike the other two antennas. The measured E-plane radiation patterns of the referenced [8] and the proposed Vivaldi antennas are illustrated in Fig. 8. At low frequencies, No notable difference is observed between the measured radiation patterns of the two DETASAs. For frequencies of 18–21 GHz, the gain is quite improved (2–3 dB), and the directivity and gain are enhanced significantly for frequencies above 22 GHz.



(a)



(b)

Fig. 6. (a) Simulated and measured realized gain in the end-fire direction for the three designed antennas. (b) Simulated and measured gain difference between the conventional and the proposed DETASAs.

IV. Conclusion

An AVA with a compact parasitic patch to overcome the radiation performance degradations in the high frequency band is proposed. For this purpose, a double asymmetric trapezoidal parasitic patch is designed and added in the aperture of an AVA, which is intended to focus the beam efficiently toward the end-fire direction at higher frequencies by utilizing the electromagnetic coupling between the main radiation patch and the inserted parasitic patch. As a result, the proposed Vivaldi antenna shows a notably improved gain (or directivity) at frequencies above 20 GHz and a stable radiation pattern over the wide frequency range without the requirement of an additional complex structure.

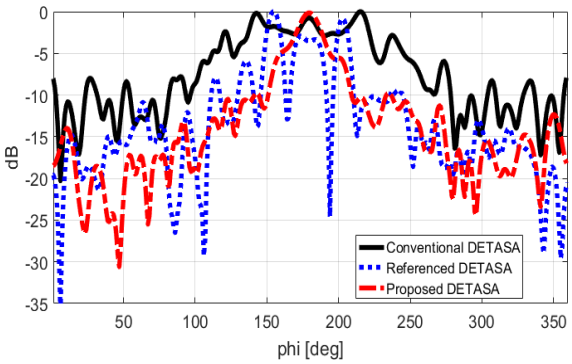


Fig. 7. Simulated E-plane electric field pattern for the three designed antennas at 22 GHz.

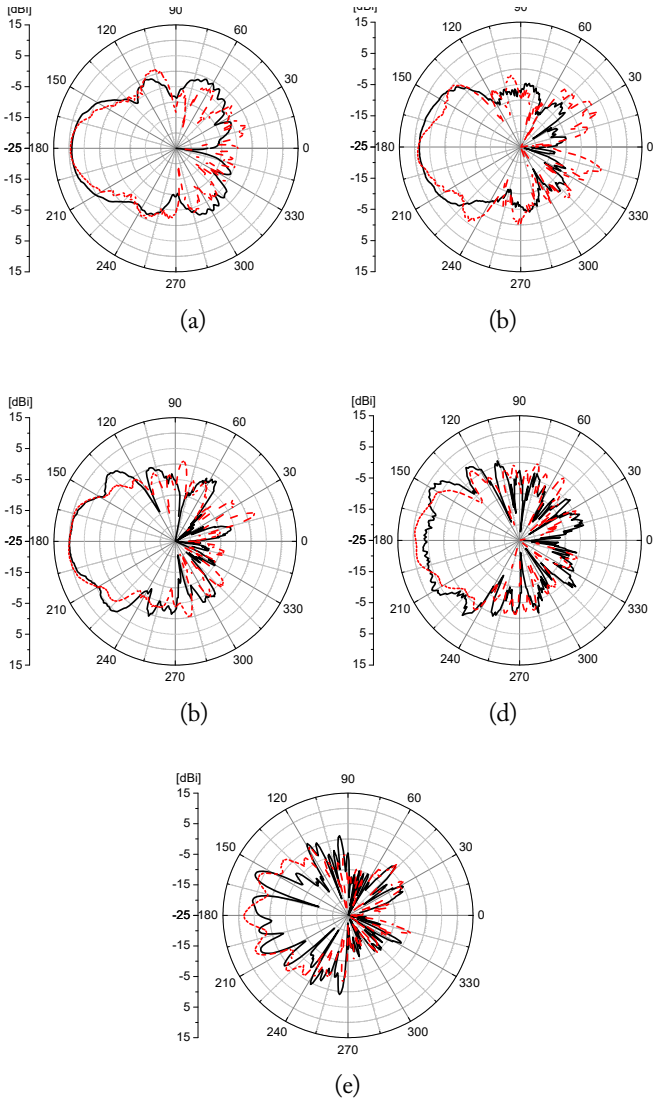


Fig. 8. Measured E-plane radiation patterns of the referenced DETASA [8] (solid line) and the proposed DETASA (dashed line) at (a) 6 GHz, (b) 10 GHz, (c) 12 GHz, (d) 18 GHz, and (e) 22 GHz.

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Sensitivity Enhancement of a Vertical-Type CMOS Hall Device for a Magnetic Sensor

Sein Oh · Byung-Jun Jang · Hyungil Chae*

Abstract

This study presents a vertical-type CMOS Hall device with improved sensitivity to detect a 3D magnetic field in various types of sensors or communication devices. To improve sensitivity, trenches are implanted next to the current input terminal, so that the Hall current becomes maximum. The effect of the dimension and location of trenches on sensitivity is simulated in the COMSOL simulator. A vertical-type Hall device with a width of 16 μm and a height of 2 μm is optimized for maximum sensitivity. The simulation result shows that it has a 23% better result than a conventional vertical-type CMOS Hall device without a trench.

Key Words: Hall Sensor, Magnetic Sensor, Sensitivity, Trench, Vertical-Type Hall Device.

I. INTRODUCTION

Magnetic sensors are commonly used in various applications, such as the contactless measurement of current [1], magnetic resonance imaging [2], or biosensors [3], which require the accurate measurement of the magnetic field. A Hall sensor is a type of magnetic sensor that is widely used because of its capability to integrate into the CMOS process. The typical architecture of a Hall sensor consists of a Hall device, a chopper-stabilized amplifier, and a modulator, among others [4–6]. Although a Hall sensor includes circuits that can reduce the DC offset and amplify the signal for a high signal-to-noise ratio, a Hall device is still the most important because it is the front-end block that turns the external magnetic field into an electrical signal. Two types of Hall devices are implemented in the CMOS process: a horizontal-type Hall device (HHD) and a vertical-type Hall device (VHD). An HHD is designed to sense an external magnetic field, which is perpendicular to the wafer surface. However, in some applications such as magnetic communication, the measurement of a magnetic field in 2D or 3D is necessary for better accuracy in a space, which mandates the sensing of a magnetic field that is parallel to the wafer surface. A VHD depends on the vertical current flow in N-well and can be used in combination with HHDs for the 2D or 3D sensing of a magnetic field. However, VHDs suffer from low sensitivity and a large offset when compared with HHDs [7]. Sensitivity is critical for a CMOS VHD because a magnetic input signal is usually small, and the DC offset and 1/f noise are not negligible at a low-frequency region [8]. One of the most popular methods to reduce the offset of a VHD is a four-folded and three-contact (4F-3C) structure, but its sensitivity is still limited [9, 10]. This structure consists of four basic VHDs, and the sensitivity of each VHD contributes to the overall sensitivity. Therefore, the sensitivity of a single VHD needs to be improved regardless of the offset reduction feature. We intend to introduce a technique to achieve high sensitivity. The current-mode operation of a VHD is advantageous because its temperature dependency is low, and the ohmic voltage drop in the intercon-

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nection barely affects the performance [11]. It also provides better sensitivity [12]. For these reasons, current-mode VHDs are used throughout this work.

This paper is organized as follows. Section II analyzes a conventional VHD and explains its operation. A method to improve sensitivity is then proposed. Section III shows the optimization process of a VHD and some of the simulation results obtain from the COMSOL simulator.

II. Vertical-Type Hall Device

1. Operation and Analysis of the Current-Mode VHD

In a VHD, the current does not flow straight as in an HHD. Fig. 1 shows the current-mode VHD model implemented in the CMOS process with a bias current of I_{bias}. The bias current goes into the VHD through the input terminal C2 in the middle, and it propagates as illustrated in Fig. 2. As the two current output terminals are on the sides of the input terminal, most of the current flows along the wafer surface, and only a small portion of the current flows near the bottom, making a long detour due to Ohm's law. Therefore, the horizontal current density is higher than the vertical current density near the input terminal. The direction of the current density significantly affects the Hall output current and thus sensitivity.

The current density in Fig. 2 is expressed by Eq. (1), the charge velocity by Eq. (2), and Lorentz force by Eq. (3).

$$\overrightarrow{J_\theta} = -q\mu_n n\overrightarrow{E_e}, \tag{1}$$

$$\vec{v} = \mu_n\overrightarrow{E_e} = -\frac{1}{q*n} * \overrightarrow{J_\theta}, \tag{2}$$

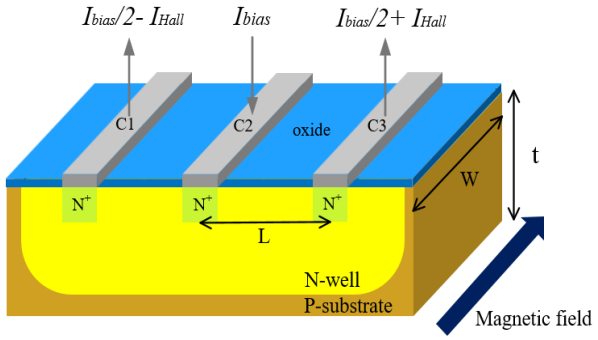


Fig. 1. Current-mode VHD implemented by N-well in the CMOS process.

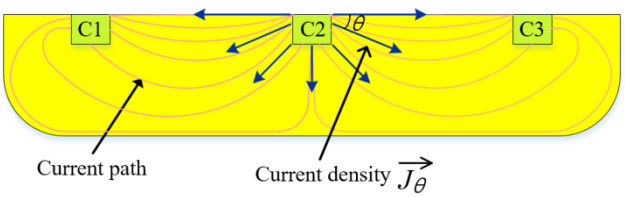


Fig. 2. Current density distribution and current path in the 2D VHD model.

$$\vec{F} = -q\left(\overrightarrow{E_{Hall}} + \vec{v}\ X\ \vec{B}\right), \tag{3}$$

where μ_n is the Hall mobility, n is the carrier concentration, and $\overrightarrow{E_e}$ is the electric field caused by the potential difference between terminals C2 and C3. When the external magnetic field is applied to the bias current flowing through a VHD, the moving electrons with a velocity of $\vec{v}$ subject to a magnetic field are biased in a direction perpendicular both to the current and to the magnetic field by magnetic force $\overrightarrow{F_m}$ expressed by Eq. (4).

$$\overrightarrow{F_m} = -q\left(\vec{v}\ X\ \vec{B}\right). \tag{4}$$

By combining the expressions in Eqs. (2) and (4), we can obtain Eq. (5), which shows the relationship between the magnetic force and the current density.

$$\overrightarrow{F_{m\theta}} = -q * R_H\left(\overrightarrow{J_\theta}\ X\ \vec{B}\right), \tag{5}$$

where R_H is the Hall coefficient and is equal to $-\frac{1}{qn}$. The sensitivity of a VHD depends on the direction of the current density near the input terminal C2 in Fig. 2. The current density near C2 has a distribution of $\overrightarrow{J_\theta}$, where θ is the angle between the current density direction and the Hall device surface. Owing to the bilateral symmetry of a VHD, we can consider only one side of a VHD for simplicity as shown in Fig. 3. The current density distribution near C2 can be represented by a vector $\vec{J}$, which is derived by the integration of $\overrightarrow{J_\theta}$ as shown in Eq. (6).

$$\vec{J} = \int_0^{\pi/2} \overrightarrow{J_\theta}\ d\vec{\theta}. \tag{6}$$

The total magnetic force on the current toward C3 can be derived from $\vec{J}$. It is expressed by Eq. (7) with the assumption that the magnetic field is constant and exists only in the z-direction.

$$\overrightarrow{F_m} = -q\overrightarrow{E_m} = -qB_zR_HJ\left(\sin\theta * \overrightarrow{a_x} + \cos\theta * \overrightarrow{a_y}\right). \tag{7}$$

In steady state, charges are accumulated on one side of the Hall device because of $\overrightarrow{F_m}$. The Hall electric field $\overrightarrow{E_{Hall}}$ is generated, and the electrons are forced back by $\overrightarrow{F_{Hall}}$. Therefore,

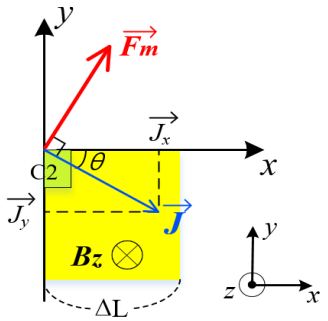


Fig. 3. Direction of magnetic force vs. current density.

the Lorentz force in Eq. (3) is zero. The Hall electric field is expressed by Eq. (8):

$$\overrightarrow{E_{Hall}} = -B_z R_H \left(J_y * \overrightarrow{a_x} + J_x * \overrightarrow{a_y}\right). \quad (8)$$

As mentioned above, sensitivity is determined around the terminal C2 within the distance of ΔL. The Hall voltage is an integral of the Hall electric field as expressed in Eq. (9).

$$V_{Hall} = -\int_{\Delta L} \overrightarrow{E_{Hall}} \cdot d\vec{L} = B_z R_H \int_{\Delta L} \left(J_y \overrightarrow{a_x} + J_x \overrightarrow{a_y}\right) \cdot d\vec{L}. \quad (9)$$

As $d\vec{L}$ is in the x-direction, Eq. (9) can be simplified as Eq. (10).

$$V_{Hall} = B_z R_H J_y \Delta L. \quad (10)$$

As the VHD operates in the current mode, Eq. (10) needs to be converted to a current equation. By applying Ohm's law and using R_{in}, which is defined in Eq. (11) [12], the Hall current is expressed by Eq. (12):

$$R_{in} = \frac{1}{q\mu_n n}\frac{\Delta L}{Wt} \quad (11)$$

$$I_{Hall} = \frac{V_{Hall}}{R_{in}} = \frac{B_z R_H J_y \Delta L}{R_{in}} = -\mu_n B_z \mathrm{W} t J_y \quad (12)$$

where W is the width, L is the length, and t is the thickness of the VHD shown in Fig. 1. Eq. (12) shows that the current density in the x-direction does not contribute to the Hall current. The current along the y-axis is expressed by Eq. (13). We obtain Eq. (14) by substituting Eq. (13) into Eq. (12).

$$\alpha I_y = -J_y W \quad (\alpha: \text{constant } [\mathrm{m}^{-1}]) \quad (13)$$

$$I_{Hall} = -\alpha \mu_n B_z t I_y \quad (14)$$

The sensitivity of a VHD in the current mode is expressed in Eq. (15). From Eqs. (14) and (15), we obtain Eq. (16).

$$S_I = \left|\frac{I_{Hall}}{I_{Bias} B_z}\right| \ [T^{-1}] \quad (15)$$

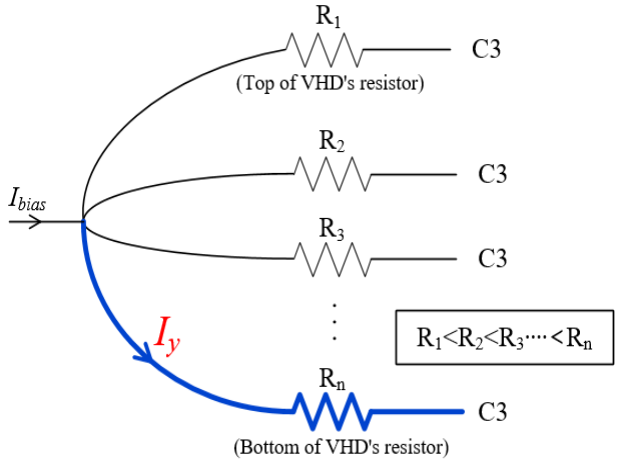


Fig. 4. Resistive model of the current path from C2 to C3.

$$S_I = \left|\alpha \mu_n t \frac{I_y}{I_{Bias}}\right| \ [T^{-1}] \quad (16)$$

In Eq. (16), μ_n and t are usually determined by the process. Therefore, I_y/I_{Bias} needs to be increased for high sensitivity. The increase in I_y/I_{Bias} indicates that more current needs to flow through the longest path in a VHD. However, the resistance increases as the current path becomes longer. The current path in a VHD can be modeled as the parallel resistors shown in Fig. 4. To increase I_y/I_{Bias}, more current needs to flow through the high-resistance path in Fig. 4. This action violates Ohm's law, but the implantation of a trench in N-well causes change in the resistance distribution and can lead the current flow as intended.

2. Sensitivity Enhancement by Trench Implantation

According to the analysis on a conventional VHD in Section II-1, the sensitivity of a VHD will increase if more current flows perpendicular to the surface around the terminal C2. For this change of current flow, a current-blocking trench can be implanted next to C2, as shown in Fig. 5. The trench does not need to be dielectric, but it can be formed simply by p-doping on N-well. The trench affects the resistive distribution in Fig. 4, and the resistance of the longest path becomes the smallest. It guides the bias current to flow vertically and helps to increase the sensitivity of a VHD. However, as the trench becomes deeper or wider, the input resistance of a VHD, R_{in} in Eq. (12), increases significantly, and sensitivity starts to decrease. Therefore, there exists an optimized dimension of a trench and its location for maximum sensitivity.

The implantation of trenches on N-well is advantageous to sensitivity, but it may contribute to the DC offset because of the variation in the CMOS process. However, this offset can be easily suppressed by the 4F-3C structure using the current spinning [9, 10] and is not significant in most cases.

III. Simulation Results

1. Conventional VHD

A Hall device is usually modelled and simulated in 2D. First, a conventional VHD in the current mode is designed and simulated in the COMSOL simulator. The VHD is sized to have a width of 16 μm and a height of 2 μm. The width of the input

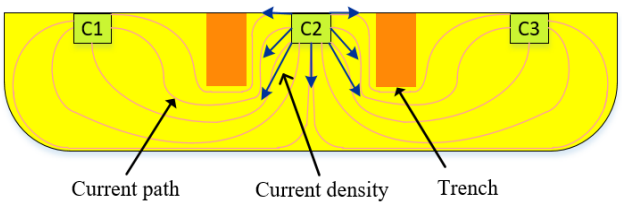


Fig. 5. Current density distribution and current path in the 2D trench implanted in a VHD model.

terminal is 1 μm, and the output terminals have a width of 0.5 μm as half of the input current flows in each output terminal. The height of all terminals is 0.1 μm. In this work, a fixed voltage is used at the input terminal instead of a fixed current to generate bias current throughout the simulations. In the CMOS process, the use of a fixed current may cause overvoltage if the resistance of a VHD increases. Fig. 6 shows the simulated electric potential and the current density distribution in a VHD when the input terminal is biased by 100 μA and the magnetic field is 1 T. The sensitivity of the reference VHD is 0.0095 (T^{-1}). Table 1 shows the model parameters used for the COMSOL simulation.

2. Proposed VHD with Trench Implantation

Fig. 7 shows the simulation model of a VHD with implanted trenches. Both the dimension of the trenches and their location affect the current density distribution, thus contributing to the sensitivity of a VHD. To evaluate the effect of a trench on sensitivity, we choose three variables, namely, height, width, and distance, between the input terminal and the trench. We can optimize the sensitivity of a VHD by analyzing the effect of each variable.

2.1 Distance

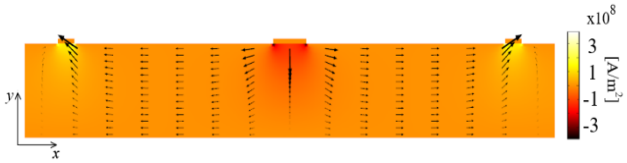


Fig. 6. Current density distribution of a conventional current-mode VHD. Arrows represent the current density, and color means the current density in the y-axis.

Table 1. Hall device model parameters

Parameter	Value
Carrier concentration, n (/cm^3)	7.38e16
Mobility, μ (cm^2/V/sec)	1,058
Silicon conductivity, σ (μ/cm)	$q \cdot \mu \cdot n$
Charge, q (C)	1.602e-19
Magnetic field, B (T)	1
Bias current (μA)	100

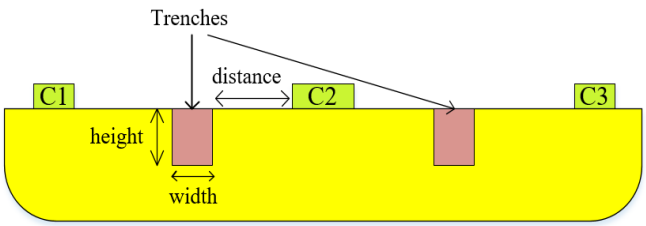


Fig. 7. Trench implantation in the VHD.

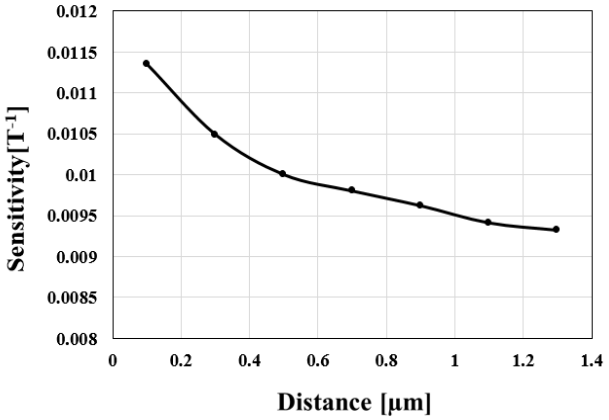


Fig. 8. Simulated sensitivity of the VHD vs. trench distance.

We vary the distance between the input terminal and the trench from 0.1 μm to 1.3 μm by a 0.1 μm step, and the height and the width are fixed at 0.5 μm. Fig. 8 shows the simulated sensitivity versus distance, with sensitivity being inversely proportional to distance. The reason for this is that the bias input current flows in parallel with the surface, and the Hall current decreases as the distance increases. Therefore, the trench needs to be as close to the input terminal as possible.

2.2 Width

Trench width can also affect the sensitivity of a VHD because it changes the current density distribution. The width is varied from 0.1 μm to 1.7 μm by a 0.2 μm step, and the height changes from 0.5 μm to 0.9 μm by a 0.1 μm step. The distance is fixed at the minimum of 0.1 μm to maximize sensitivity. Fig. 9 shows the simulated sensitivity versus the trench width. Sensitivity decreases as the width increases for all values of height. This inverse proportion comes from the resistance change of the VHD. If the trench width increases, the resistance between the terminals (R_{in}) becomes large. The increased R_{in} reduces the ΔI_{Hall} as indicated in Eq. (12). Therefore, sensitivity is inversely proportional to the trench width.

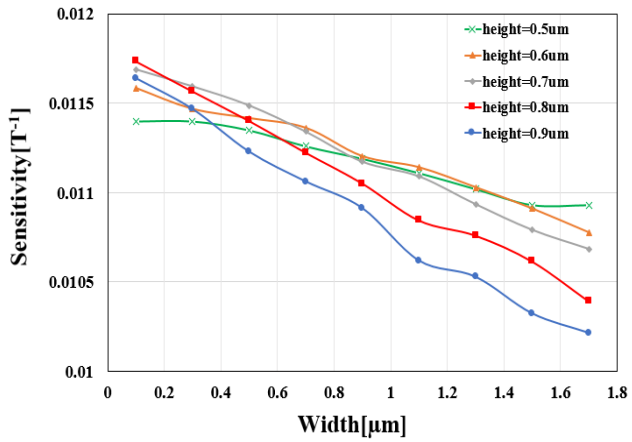


Fig. 9. Simulated sensitivity of the VHD vs. trench width.

2.3 Height

The trench height is directly related to the direction of the current density around the input terminal. The trench height is varied from 0.1 μm to 1.2 μm by a 0.1 μm step for a width of 0.1 μm and 0.2 μm, and the distance is fixed at 0.1 μm. As shown in Fig. 10, sensitivity becomes maximum at a certain value of height. When the height is small, sensitivity increases as the height increases. The reason for this is that more current will flow in the vertical direction if the trench is higher. However, the sensitivity of a VHD begins to decrease when the height is over 0.8 μm. Although the directivity of the current density is improved, the resistance between the terminals increases, thus reducing the sensitivity. As the height of N-well is 2 μm, the sensitivity has an optimal value when the trench is sized to be approximately 40% of the height of N-well.

3. VHD Simulation with an Optimized Trench

From the result above, we optimize a VHD with implanted trenches for maximum sensitivity. The trench is sized to have a width of 0.1 μm and a height of 0.8 μm. The distance between the trench and the input terminal is 0.1 μm. Fig. 11 illustrates the simulated current density distribution. As expected, the current density is guided in the vertical direction around the input terminal unlike the conventional one, as shown in Fig. 12. Consequently, ΔI_{Hall} increases (Fig. 13), and sensitivity is calculated as 0.011734 (T^{-1}), which is 23% better than that of the conventional one.

IV. Conclusion

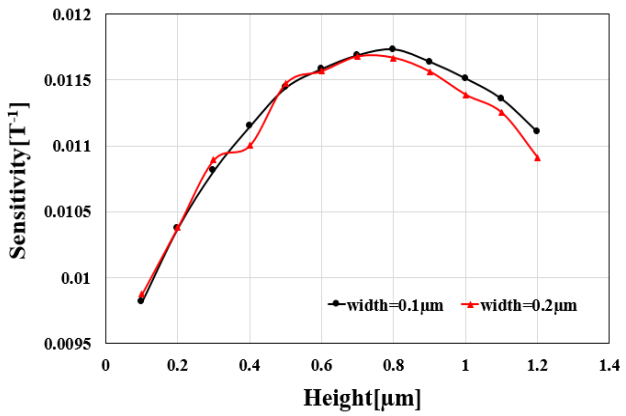


Fig. 10. Simulated sensitivity of VHD vs. trench height.

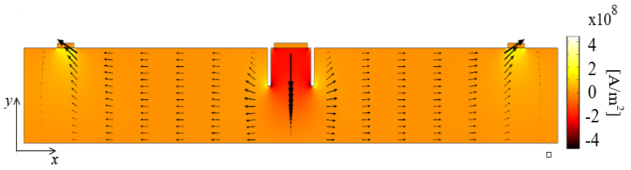


Fig. 11. Current density of the proposed VHD with optimized sensitivity. Arrows represent the current density, and color means the current density in the y-axis.

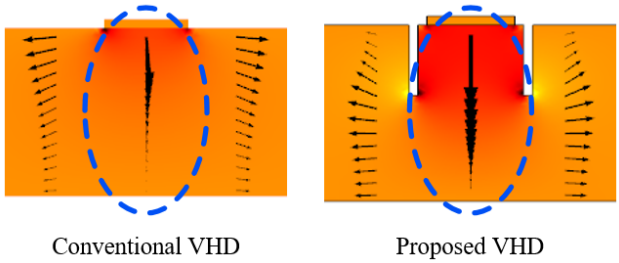


Fig. 12. Current direction comparison of the conventional VHD and the proposed VHD near the input terminal. Arrows represent the current density, and color means the current density in the y-axis.

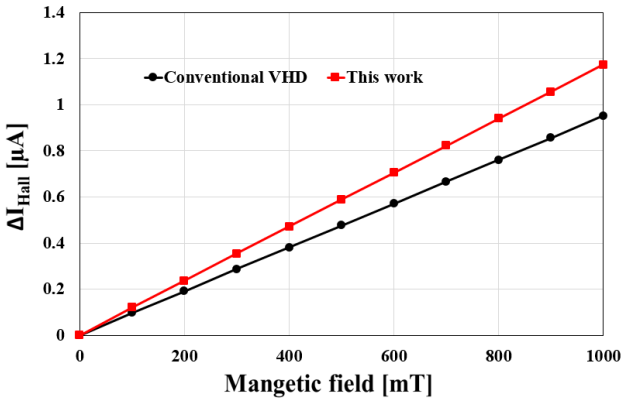


Fig. 13. Hall current of the conventional VHD vs. the proposed VHD.

This study analyzes a conventional VHD and the reason for its low sensitivity. Then, a new method of trench implantation is proposed to increase the sensitivity. The implanted trench next to the input terminal changes the direction of the current flow vertically. Therefore, the resultant Hall current is generated maximally. Three types of dimensional factors of the trench affect sensitivity. In the simulation results, the trench width and the distance between the trench and the input terminal are inversely proportional to sensitivity. Conversely, the trench height produces a curve, and the sensitivity is optimized by a certain height. When the three factors of the trenches are optimized simultaneously, the result shows 23% higher sensitivity than that of the conventional one without trenches. The trench can be implanted on N-well without any post-processing of the wafer. Therefore, we can achieve the higher sensitivity of a VHD, which is useful in a 4F-3C structure using the current spinning.

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A Coupled Line Impedance Transformer for High Termination Impedance with a Bandpass Filtering Response

Phirun Kim · Yongchae Jeong*

Abstract

In this study, a short-ended coupled line with a short-circuit stub transmission line bandpass filtering impedance transformer is presented. The general designed equations are derived on the basis of circuit theory. The design curves are provided to examine the characteristic of the proposed impedance transformer. The proposed circuit is suitable for high termination impedance. To validate the design formulas, a 400–50 Ω impedance transformer is designed and fabricated at the operating center frequency (f_0) of 2.6 GHz. The measured results show a good agreement with the simulation. The measured insertion and return losses are 0.6 dB and 22.5 dB at f_0, respectively. The measured return loss is higher than 20 dB within the passband frequency of 2.51–2.7 GHz. Moreover, the stopband attenuation is higher than 25 dB from DC to 1.64 GHz of the lower stopband and from 3.12 GHz to 6.4 GHz of the higher stopband.

Key Words: Bandpass Response, Coupled Line, Impedance Transformer, Transmission Pole.

I. INTRODUCTION

The bandpass filtering impedance transformer (IT) is rapidly developing in modern communication systems. This IT is increasingly becoming necessary in many applications, such as power dividers [1], antenna feeding lines [2], and amplifier design [3]. Indeed, a conventional quarter-wavelength transformer is a well-known IT [4], but this network has some limitations, such as difficulty in the realization of a high impedance transforming ratio and poor out-of-band suppression. In recent years, various structures of filtering IT have been proposed [5–11]. A mixed lumped and distributed IT with low-pass filtering response is designed in [5]. However, the shunt capacitors can produce a self-resonance frequency, which introduces additional parasitic effects at high frequency. Wideband ITs are proposed using an open-circuit parallel-coupled line with an interconnecting transmission line (TL) [6] and a multiple-section stepped-impedance TL with input/output couplings [7]. However, their impedance transforming ratio is low. In [8], an open-circuit stub coupled-line IT with a bandpass filtering response is designed to transform 20 Ω to 50 Ω. In [9], a bandpass response coupled-line IT with an ultra-high impedance transforming ratio is presented by using two-section open-ended coupled lines. This IT is applicable to low termination impedance. Therefore, a short-ended coupled-line IT with an ultra-high impedance transforming ratio is proposed [10] for high load termination impedance. High selectivity is obtained by increasing the number of stages, but the insertion loss is degraded. Moreover, the stopband characteristic is limited. In [11], a bandpass filtering IT is designed with a section of open-ended coupled lines and an open-circuit stub TL.

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The transformer is valid for low termination impedances at the source and/or load terminals.

In this study, an IT with a short-circuit stub parallel-coupled line and a TL with a bandpass filtering response is presented. The proposed network can be applied to high load termination impedance. A short-circuit stub TL can produce several transmission zeros in the stopband. Moreover, a bandpass filtering response with high selectivity can be obtained.

II. Design Equation

Fig. 1(a) shows the schematic of the proposed IT, which is composed of a short-end parallel-coupled line and a short-circuit stub TL with a characteristic admittance of Y_1 connected at coupling port 2 of the coupled line. The electrical lengths of the parallel-coupled line (θ) and the short-circuit stub TL (2θ) are given as a quarter-wavelength ($\lambda/4$) and a half-wavelength ($\lambda/2$) at f_0, respectively. A short-circuit stub TL is used to create two transmission poles in the passband and three transmission zeros in the stopband [11], where the transmission zeros are located on the lower and upper sides of the passband, respectively. The proposed circuit is efficient at high termination impedance. Fig. 1(b) shows the schematic of the impedance transformer in [11] with an open-end parallel-coupled line and an open-circuit stub TL. The network is used only for low termination impedance.

The reflection and transmission coefficients of the proposed IT network, where input port 1 and output port 3 are terminated with the admittances Y_S and Y_L, respectively, can be derived as (1).

$$S_{11}=\frac{\left(Y_{11}-Y_S\right)\left(Y_{22}+rY_S\right)-Y_{12}Y_{21}}{\left(Y_{11}+Y_S\right)\left(Y_{22}+rY_S\right)-Y_{12}Y_{21}} \tag{1a}$$

$$S_{21}=\frac{2Y_{21}\sqrt{Y_SY_L}}{\left(Y_{11}+Y_S\right)\left(Y_{22}+rY_S\right)-Y_{12}Y_{21}}, \tag{1b}$$

where

$$Y_{11}=j\frac{\cot\theta}{2}\left[\frac{\left(Y_{0o}-Y_{0e}\right)^2\cot\theta}{2Y_1\cot 2\theta+\left(Y_{0e}+Y_{0o}\right)\cot\theta}-\left(Y_{0e}+Y_{0o}\right)\right] \tag{2a}$$

$$Y_{12}=Y_{21}=j\frac{Y_{0o}-Y_{0e}}{2}\csc\theta\left[\frac{\left(Y_{0e}+Y_{0o}\right)\cot\theta}{2Y_1\cot 2\theta+\left(Y_{0e}+Y_{0o}\right)\cot\theta}-1\right] \tag{2b}$$

$$Y_{22}=j\frac{Y_{0e}+Y_{0o}}{2}\left[\frac{\left(Y_{0e}+Y_{0o}\right)\csc^2\theta}{2Y_1\cot 2\theta+\left(Y_{0e}+Y_{0o}\right)\cot\theta}-\cot\theta\right] \tag{2c}$$

$$\theta=\frac{\pi}{2}\frac{f}{f_0}. \tag{2d}$$

Moreover, $Y_S=1/Z_S$, $Y_L=1/Z_L$, $Y_{0e}=1/Z_{0e}$, and $Y_{0o}=1/Z_{0o}$. By solving (1a), the even-mode characteristic admittance Y_{0e} of the coupled line with the specified S_{11}, Y_S, and r at f_0 can be derived as (3).

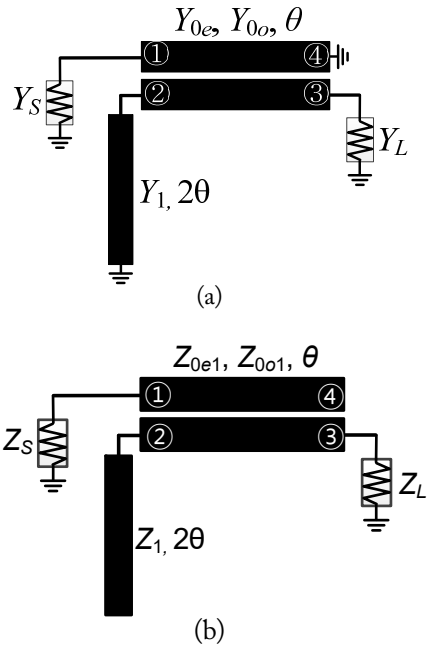


Fig. 1. Schematic of the impedance transformer. (a) Proposed network and (b) network in [11].

$$Y_{0e}=Y_{0o}-2Y_S\sqrt{\frac{r\left(1+S_{11}\big|_{f=f_0}\right)}{\left(1-S_{11}\big|_{f=f_0}\right)}}, \tag{3}$$

where

$$r=\frac{Y_L}{Y_S}=\frac{Z_S}{Z_L}. \tag{4}$$

r is an impedance transforming ratio of Z_S to Z_L. S_{11} and Y_{0o} are the predefined magnitude of return loss and odd-mode admittance of the coupled line at f_0, respectively. By giving (1a) equal to zero, the characteristic admittance of Y_1 can be derived as (5).

$$Y_1=\frac{Y_{0e}+Y_{0o}}{r-1}. \tag{5}$$

Using (5), two transmission poles can be obtained in the passband [11].

From the above design equations, the relationship of Z_{0e} and Z_{0o} with r is plotted in Fig. 2. Z_{0e} dramatically decreases as r increases. With the same r, Z_{0e} increases as Z_{0o} increases. Therefore, a higher r requires a relatively looser coupling than a lower r, where the coupling coefficient is given by $C=20\log[(Z_{0e}-Z_{0o})/(Z_{0e}+Z_{0o})]$ [dB]. The calculation is conducted by choosing $Z_L=50\ \Omega$, $S_{11}=-20$ dB at f_0, $Z_{0o}=27\ \Omega$, 30 Ω, 35 Ω, and r varies from 2 to 12.

Similarly, Fig. 3 shows the variations of Z_1 according to r and Z_{0o}. Z_1 increases as r increases. Moreover, Z_1 increases more with a higher Z_{0o}. As shown in Figs. 2 and 3, a low r prevents

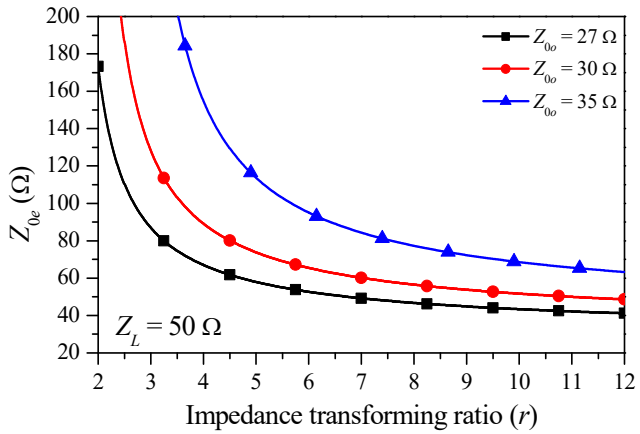


Fig. 2. Variations of Z_{0e} according to r and Z_{0o}.

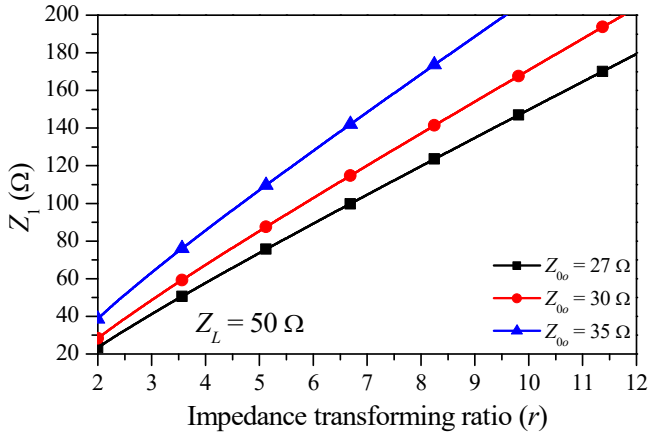


Fig. 3. Variations of Z_1 according to r and Z_{0o}.

the realization of the coupled line, whereas a high r hinders the realization of a short-circuit stub TL. Therefore, a trade-off between r and the characteristic impedances of Z_1 and Z_{0e} is required.

Fig. 4 shows the S-parameter characteristics of the proposed IT with different r. The stopband attenuation is improved with a high r; but the bandwidth of the passband becomes narrow. Moreover, two transmission poles occur in the passband.

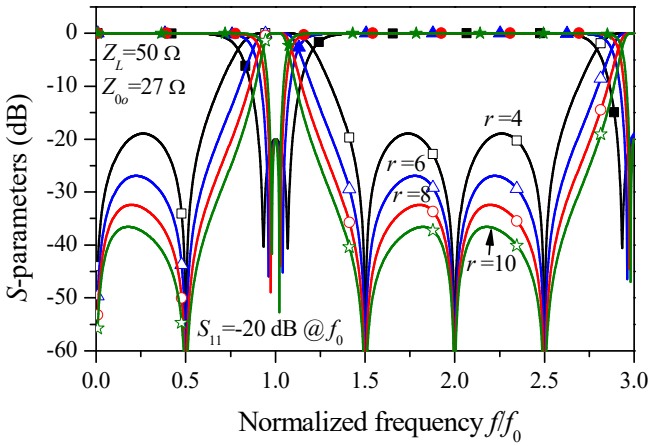


Fig. 4. S-parameter characteristics with different r.

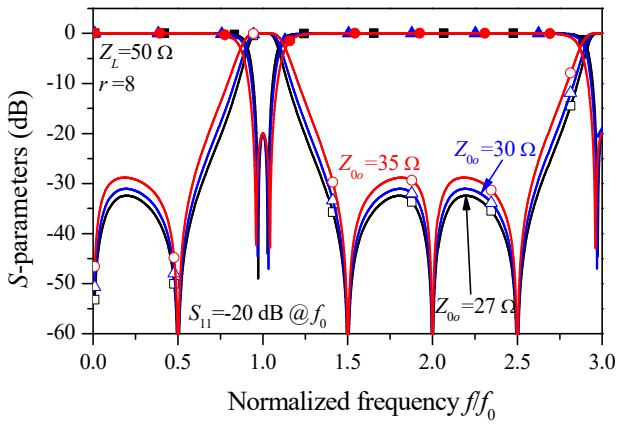


Fig. 5. S-parameter characteristics with different Z_{0o}.

Table 1. Calculated variables of the proposed impedance transformer

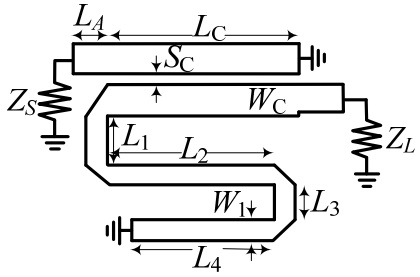
	Z_L = 50 Ω and S_{11} = –20 dB at f_0		
	Z_{0o} (Ω)	Z_{0e} (Ω)	Z_1 (Ω)
r = 4	27	66.99	57.73
r = 6	27	52.68	89.25
r = 8	27	46.72	119.78
	30	56.5	137.17
	35	77.3	168.64
r = 10	27	43.38	149.77

Transmission zeros are produced in the stopband, and a wide stopband characteristic can be obtained. Transmission zeros at 0.5f_0, 1.5f_0, and 2.5f_0 are produced by a short-circuit stub TL. Moreover, a transmission zero at 2f_0 is produced by a coupled line. The simulation is conducted by fixing Z_{0o} = 27 Ω and varying r = 4, 6, 8, and 10. Fig. 5 illustrates the S-parameter characteristics of the proposed IT with different Z_{0o}. The bandwidth of a 20 dB return loss is improved with a high Z_{0o}, but the stopband attenuation is degraded. The simulation is conducted by fixing r = 8 and varying Z_{0o} = 27 Ω, 30 Ω, and 35 Ω. The calculated values of all variables of the simulation in Figs. 4 and 5 are shown in Table 1.

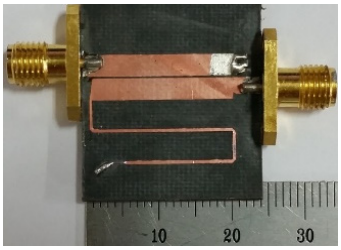
III. Simulation and Measurement Results

An experimental validation was conducted by designing the IT and fabricating it on an RT 5880 substrate, with ε_r = 2.2 and h = 0.787 mm. The proposed IT network with a 400–50 Ω (r = 8) of termination impedance, Z_{0o} = 27 Ω, and reflection coefficient of 20 dB at f_0 = 2.6 GHz was designed. All the calculated variables are listed in Table 1. An electromagnetic (EM) simulation was performed using the Ansoft HFSS v15.

Fig. 6 shows the EM simulation layout and photograph of the fabricated IT network. The physical dimensions of the fabricated circuit are listed in Table 2. To minimize the circuit size, a half-wavelength short-circuit stub TL is designed as a mean-



(a)



(b)

Fig. 6. The fabricated IT: (a) EM simulation layout and (b) photograph.

Table 2. Physical dimensions of the fabricated IT (unit: mm)

Parameter	Value	Parameter	Value
W_c	3.1	$L_1 = L_3$	4
W_1	0.4	L_2	19
S_c	0.1	L_4	15.7
L_c	19.2	L_A	1.5

der structure. The overall circuit size of the proposed IT is 24 mm × 16 mm.

Fig. 7 illustrates the EM simulation and measurement results. The measured results are in good agreement with the simulations. The insertion and return losses are obtained 0.6 dB and 22.5 dB at f_0, respectively. Similarly, the bandwidth of the 20 dB return loss is 0.19 GHz, which extends from 2.51 GHz to 2.7 GHz. The insertion loss of the passband is better than 0.8 dB. Moreover, one transmission zero on the lower side and three transmission zeros on the upper side of the passband are obtained, and these transmission zeros provide a wide stopband characteristic. The lower- and upper-side stopband attenuations of 25 dB are obtained from dc to 1.64 GHz and from 3.12 GHz to 6.4 GHz, respectively.

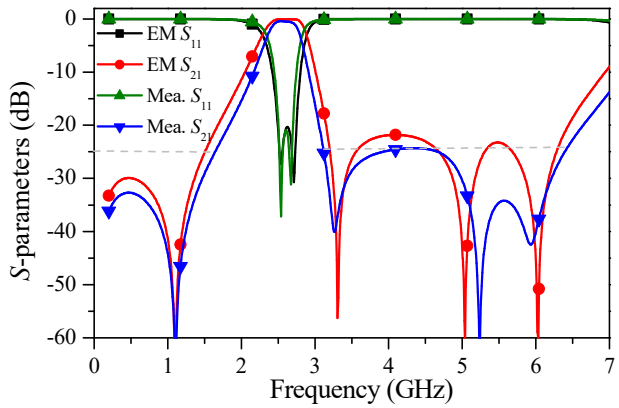


Fig. 7. EM simulation and measurement results of the proposed IT.

IV. Conclusion

An IT with a bandpass filtering response for high termination impedance is proposed and demonstrated in this study. The designed equations are derived with a predefined return loss. To show the validity of the proposed analysis, an IT is designed, fabricated, and measured. The simulated and measured results agree well with the analysis. The proposed IT with high selectivity, multi-transmission zeros, and bandpass response can be obtained simultaneously. The proposed IT is simple to design and is expected to be applied in various applications, such as baluns, power dividers, and antenna feeding lines.

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Design of a Short/Open-Ended Slot Antenna with Capacitive Coupling Feed Strips for Hepta-Band Mobile Application

Kyoseung Keum · Haiyan Piao · Jaehoon Choi*

Abstract

In this paper, a planar printed hybrid short/open-ended slot antenna with capacitive coupling feed strips is proposed for hepta-band mobile applications. The proposed antenna is comprised of a slotted ground plane on the top plane and two capacitive coupling feed strips with a chip inductor on the bottom plane. At the low frequency band, the short-ended long slot fed by strip 1 generates its half-wavelength resonance mode, whereas the T-shaped open ended slot fed by strip 2 generates its quarter-wavelength resonance mode for the high frequency band. The antenna provides a wide bandwidth covering GSM850/GSM900/DCS/PCS/UMTS/LTE2300/LTE2500 operation bands. Moreover, the antenna occupies a small volume of 15 mm × 50 mm × 1 mm. The operating principle of the proposed antenna and the simulation/measurement results are presented and discussed.

Key Words: Capacitive Feed, Chip Inductor, Hepta-Band Mobile Antenna, Slot Antenna.

I. INTRODUCTION

With the rapid development of mobile communication technology and diversified demand of mobile users, multi-band antennas become essential for mobile communication systems [1]. In some mobile applications, it is desirable to integrate hepta-band systems, such as GSM850/GSM900 bands in the lower band and DCS/PCS/UMTS/LTE2300/LTE2500 bands in the upper band, into a single mobile device. However, it is a big challenge for a single mobile device to accommodate a hepta-band antenna due to the space limitation of the printed circuit board (PCB). For this reason, a mobile antenna should be designed in low profile planar structure and in compact size without deteriorating the performance. Slot antenna, with the advantages of low-profile, simple structure, and easy integration with other devices, is a good candidate for the design of multi-band antennas [2, 3]. However, the slot antenna has difficulty covering the low-frequency bands by a single resonance [4]. To overcome this problem, various researches using multi-resonance at the lower frequency band have been studied [5–9]. In these studies, multi-resonance characteristic was achieved by using capacitive coupling feed. However, these antennas have narrow bandwidth at high-frequency bands.

In this paper, we propose a compact slot antenna covering GSM850 (824–894 MHz), GSM900 (880–960 MHz), DCS (1,710–1,880 MHz), PCS (1,850–1,990 MHz), UMTS (1,920–2,170 MHz), LTE2300 (2,305–2,400 MHz), and LTE2500 (2,500–2,690 MHz) bands. The proposed antenna consists of two capacitive coupling feed strips, a chip inductor, a short-ended slot, and an inverted T-shaped open-ended slot. The

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proposed antenna is simulated by ANSYS High-Frequency Structure Simulator (HFSS; ANSYS Inc., Canonsburg, PA, USA).

II. Antenna Design

The configuration of the proposed antenna for hepta-band mobile application is shown in Fig. 1. The antenna is designed on a FR-4 (ε_r=4.4, tanδ=0.02) substrate with a thickness of 1 mm. The dimension of FR-4 substrates is 50 mm × 115 mm × 1 mm, and the antenna with a dimension of 15 mm × 50 mm × 1 mm is located at the upper side of the substrate. The short-ended slot forms a loop along the edge of the antenna and the T-shaped slot is placed at the upper edge of the ground plane. Coupling feed strip is comprised of strip 1 and strip 2. The chip inductor is connected in series with the strip 1 and strip 2. The short-ended slot and T-shaped slot, are excited by strip 1 and strip 2, respectively. By electromagnetic coupling between the short-ended slot and strip 1, dual resonances are generated to cover GSM850/900 bands. A wideband characteristic at the high-frequency band is achieved by the combination of the harmonic resonance of the short-ended slot and the resonances due to the T-shaped slot and the strip 2. Table 1 shows

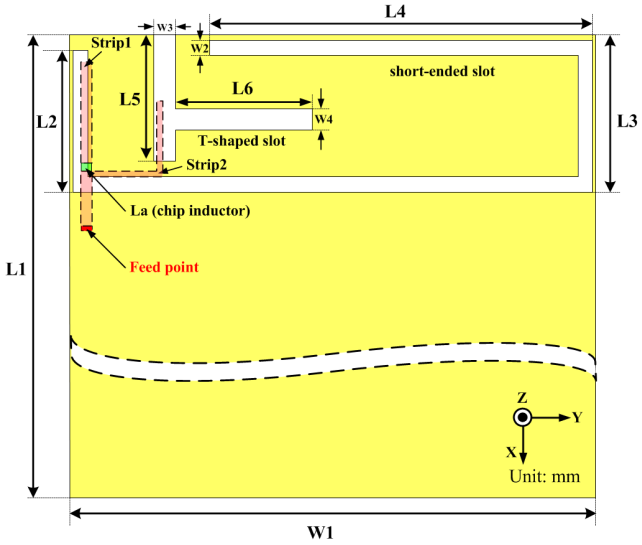
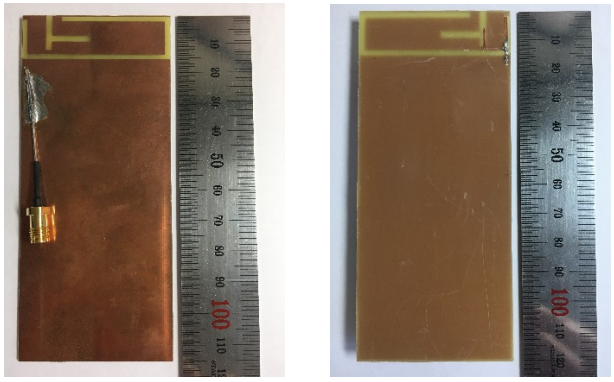


Fig. 1. The geometry of the proposed antenna.

Table 1. Design parameters of the proposed antenna

Parameter	Value	Parameter	Value
L_1	115 mm	W_1	50 mm
L_2	13.5 mm	W_2	1.5 mm
L_3	15 mm	W_3	2 mm
L_4	36.5 mm	W_4	2 mm
L_5	12 mm	L_a	22 nH
L_6	13 mm	-	-



(a) (b)

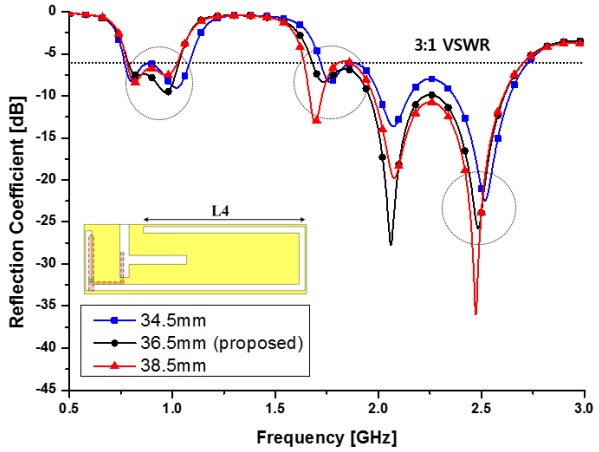
Fig. 2. Prototyped antenna: (a) top view and (b) bottom view.

the design parameters of the proposed antenna and the fabricated antenna is shown in Fig. 2. Fig. 2(a) shows the slots of the ground plane and the feed cable and Fig. 2(b) shows the strips and chip inductor.

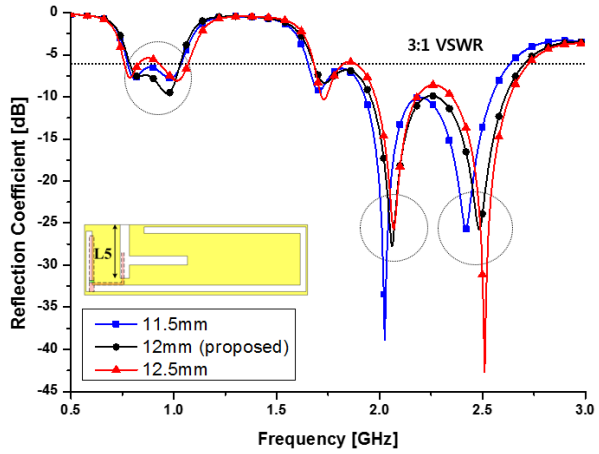
III. Antenna Analysis

Fig. 1 shows the various design parameters mostly affecting the antenna performances. In order to analyze and verify the operation principle of the proposed antenna, the parametric study has been conducted.

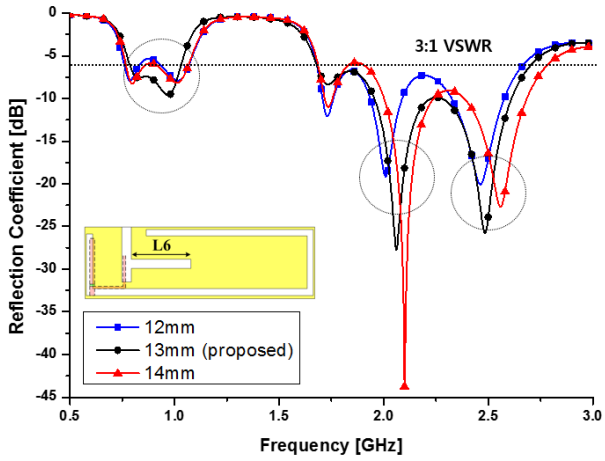
Fig. 3(a) shows the simulated reflection coefficient when the length of the top slot (L_4) varies from 34.5 mm to 38.5 mm. It can be seen that as the length of L_4 increases the whole frequency band is shifted to the lower frequency side. In particular, L_4 largely affects the reflection coefficient characteristics at GSM900, DCS and LTE bands. These frequencies correspond to the half-wave resonance of the short-ended slot and its second and third harmonic resonances, respectively. The optimal length of L_4 is chosen as 36.5 mm. Fig. 3(b) shows the simulated reflection coefficient when the length of the T-shaped slot (L_5) changes from 11.5 mm to 12.5 mm. It is observed that the upper frequency band is shifted to lower frequency as the length of L_5 increases. In particular, L_5 largely affects the performance at GSM900, UMTS, LTE bands. The optimal length of L_5, satisfying the –6 dB reflection coefficient is 12 mm. Fig. 3(c) shows the simulated reflection coefficient when the length of the T-shaped slot (L_6) is changing from 12 mm to 14 mm. It is shown that as the length of L_6 increases, the upper frequency band is shifted to the low frequency side. In particular, the GSM900, UMTS, LTE bands are affected critically by the change of L_6. When L_6 becomes 14 mm, the antenna has wider –6 dB reflection coefficient bandwidth than that of 13 mm in the high frequency band. However, since the –6 dB reflection coefficient in the GSM and PCS bands is satisfied marginally, the length of L_6 is set to 13 mm. Fig. 4 shows the simulated



(a)



(b)



(c)

Fig. 3. The simulated reflection coefficient with different values of L_4 (a), L_5 (b), and L_6 (c).

reflection coefficient when the values of the chip inductor L_a are 18 nH, 22 nH, and 27 nH. The change of the inductor value affects the reflection coefficient characteristic in the lower frequency GSM band more critically than in the upper frequency bands. It can be seen that as the inductor value increases, the resonance occurs at the low side of the lower frequency band,

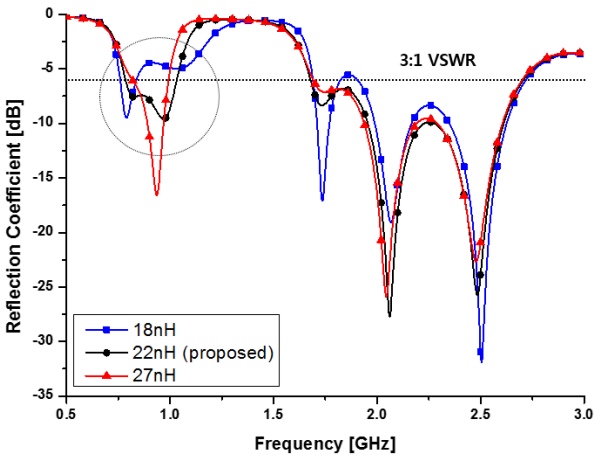


Fig. 4. Simulated reflection coefficient with different chip inductor values of L_a.

and when the inductor value decreases, the resonance occurs at the high side of the lower band. The –6 dB reflection coefficient at the GSM band is satisfied when the inductor value is 22 nH.

Fig. 5 shows the surface current distributions on the proposed antenna. At lower bands, the surface current around the short-ended slot is strongly excited by the capacitive coupling due to feed strip 1. This current distribution shows that the short-ended slot with the strip 1 generates the half-wavelength resonance [10, 11] and its harmonic modes at the vicinity of 0.85, 1.73, and 2.50 GHz. At the upper bands, the intensity of the surface current distribution around the T-shaped slot becomes relatively stronger than that at the lower-frequency band. The current distribution shows that the T-shaped open-ended slot with the strip 2 generates the quarter-wavelength resonance mode [10] at the vicinity of 2.06 GHz and 2.50 GHz.

IV. Simulation and Measurement Results

Fig. 6 shows the simulated and measured reflection coeffi-

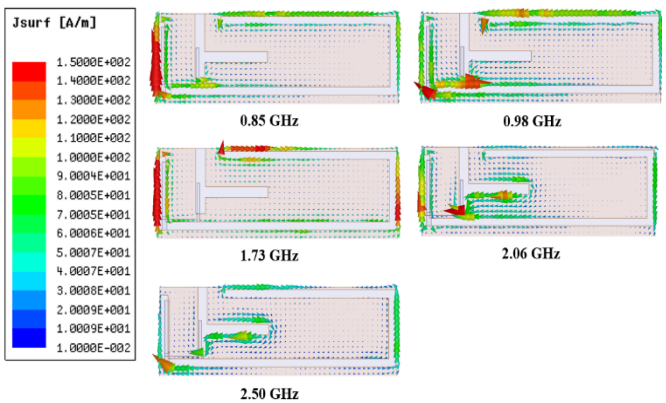


Fig. 5. The simulated surface current distributions on the proposed antenna.

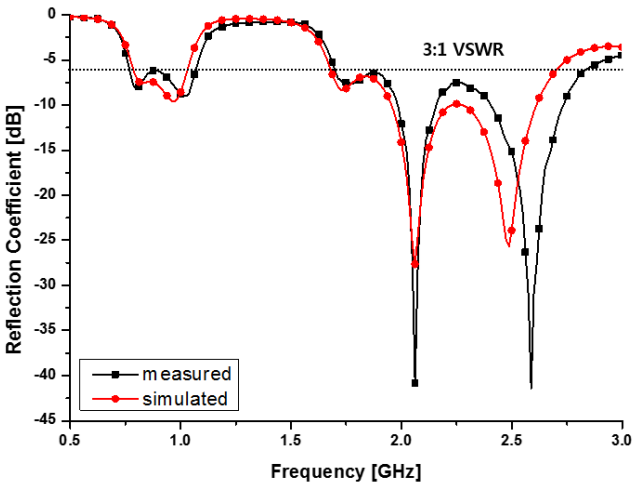


Fig. 6. Simulated and measured reflection coefficient of the proposed antenna.

cients of the proposed antenna. The –6 dB reflection coefficient bandwidths are 260 MHz (0.77–1.03 GHz) at the lower band and 1,040 MHz (1.66–2.70 GHz) at the upper band. The bandwidth of the proposed antenna is wide enough to cover the desired hepta-band. The measured 2D radiation patterns of the proposed antenna are shown in Fig. 7. The 2D radiation patterns were measured at the center frequency of each band. At low frequencies, with the center frequencies of the GSM850 and GSM900 bands at 0.86 GHz and 0.92 GHz, respectively, dipole-like radiation patterns can be observed in the x-z plane. In the x-y plane, the radiation patterns are omni-directional. At high frequencies, with the center frequencies of the DCS, PCS, UMTS, LTE2300, and LTE2500 bands at 1.80, 1.92, 2.05, 2.35, and 2.60 GHz, respectively, the proposed antenna has quasi-omnidirectional radiation patterns. Fig. 8 illustrates the

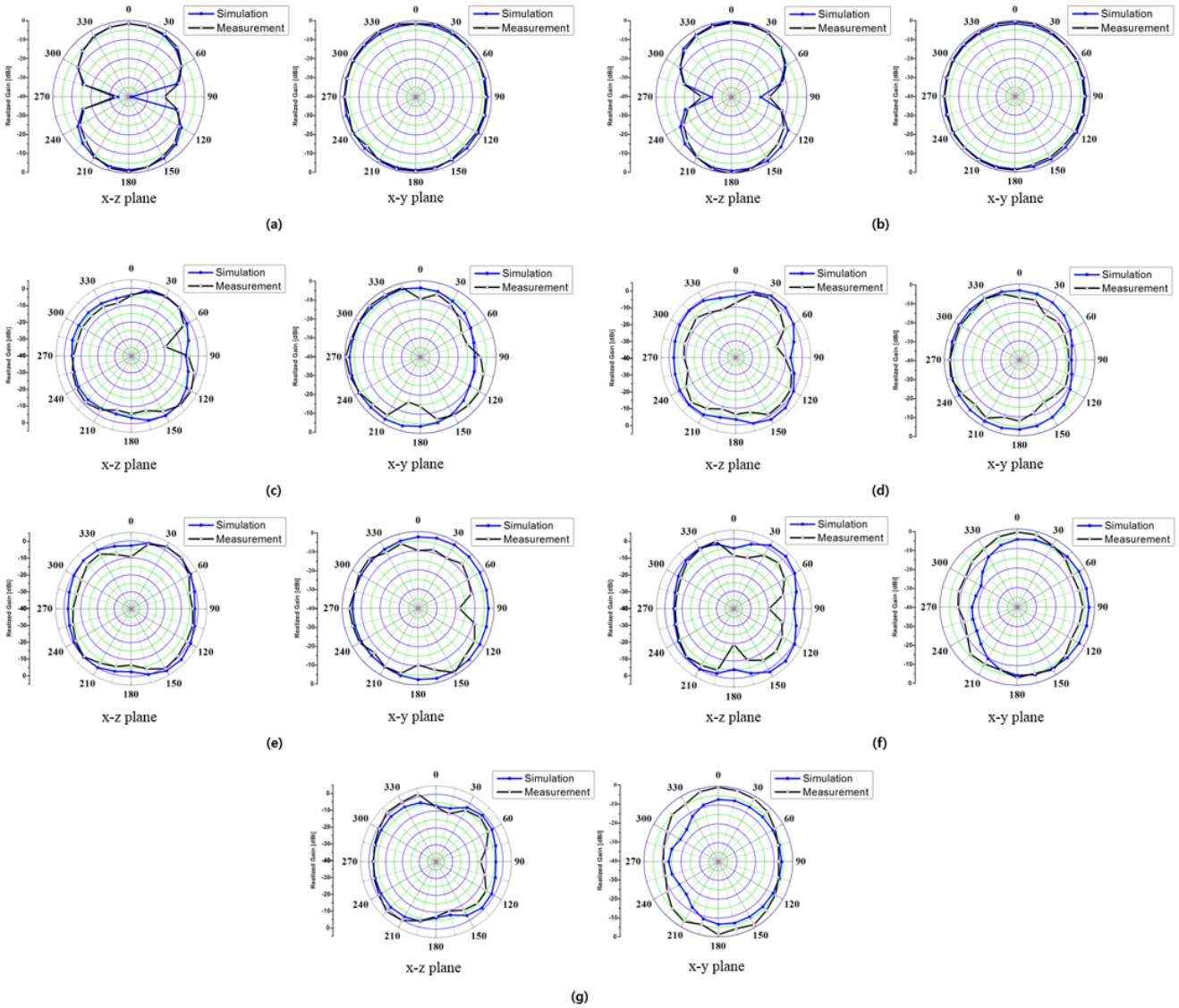


Fig. 7. Simulated and measured radiation patterns of the proposed antenna at (a) 0.86 GHz, (b) 0.92 GHz, (c) 1.80 GHz, (d) 1.92 GHz, (e) 2.05 GHz, (f) 2.35 GHz, and (g) 2.60 GHz.

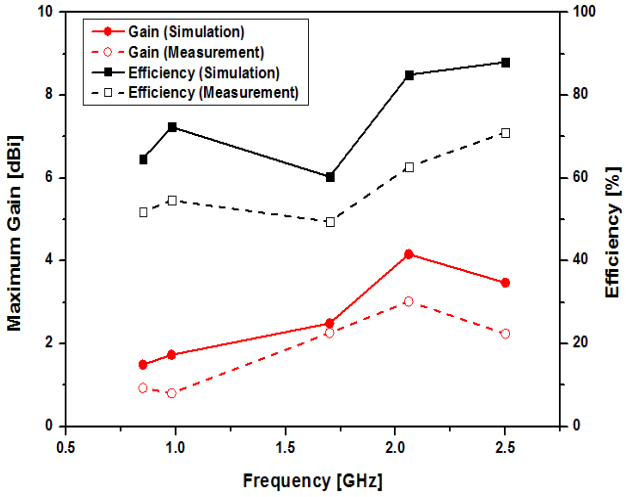


Fig. 8. Simulated and measured efficiencies and maximum gains of the proposed antenna.

simulated and measured efficiencies and maximum gains of the proposed antenna. The measured efficiencies and maximum gains are 51.8% and 0.93 dBi, 54.6% and 0.8 dBi, 49.5% and 2.26 dBi, 62.7% and 3.02 dBi, 71.1% and 2.24 dBi, at 0.85, 0.98, 1.73, 2.06, and 2.50 GHz, respectively. As shown in Fig. 8, the measured efficiencies and maximum gains are substantially lower than the simulated results. This is mainly due to the metal surface roughness of the FR4 substrate which is not accounted for in the simulation [12]. Moreover, the percent error of the chip inductor and fabrication errors of capacitive coupling strips lead to the loss of gain.

V. Conclusion

In this paper, a hepta-band planar printed hybrid short/open-ended slot antenna with capacitive coupling feed strips is proposed. By properly choosing the lengths of feed strip 1 and strip 2 and the value of chip inductor, the proposed antenna has a wide bandwidth of 260 MHz covering GSM850 and GSM900 bands in the lower frequency band and 1,040 MHz covering DCS, PCS, UMTS, LTE2300, and LTE2500 bands in the upper frequency band. The geometrical parameters of the short-ended slot and the feeding strip 1 are mainly affecting the antenna performances at GSM900, DCS, and LTE bands and those of open-ended T-shaped slot and the feeding strip 2 determines the antenna characteristics at GSM900, UMTS, and LTE bands. Along with the wide bandwidth, the antenna has good radiation properties desired for mobile communication applications.

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Numerical Investigation of Scattering from a Surface Dielectric Barrier Discharge Actuator under Atmospheric Pressure

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Abstract

Surface dielectric barrier discharge (SDBD), which is widely used to control turbulence in aerodynamics, has a significant effect on the radar cross-section (RCS). A four-way linearly synthesized SDBD air plasma actuator is designed to bolster the plasma effects on electromagnetic waves. The diffraction angle is calculated to predict the RCS because of the periodic structure of staggered electrodes. The simplified plasma modeling is utilized to calculate the inhomogeneous surface plasma distribution. Monostatic RCS shows the diffraction in the plane perpendicular to the electrode array and the notable distortion by plasma. In comparison, the overall pattern is maintained in the parallel plane with minor plasma effects. The trends also appear in the bistatic RCS, which has a significant difference in the observation plane perpendicular to the electrodes. The peaks by Bragg's diffraction are shown, and the RCS is reduced by 10 dB in a certain range by the plasma effect. The diffraction caused by the actuator and the inhomogeneous air plasma should be considered in designing an SDBD actuator for a wide range of application.

Key Words: Actuator, Diffraction, Plasma, Radar Cross-Section, Surface Dielectric Barrier Discharge.

I. Introduction

A surface dielectric barrier discharge (SDBD) actuator has been examined as a flow controller in aerodynamics [1]. It has several advantages, such as switchable plasma property, absence of mechanical movement, and simple structure, among others. Several efforts have been made to utilize the actuator in flying vehicles. In some studies, SDBD is attached to turbine blades [2, 3], and it presents an outstanding performance in the control of flow separation. Other studies use the actuator to change the air flow around the airfoil [4]. The plasma effect as the change in the angle of flow attack is demonstrated in [5]. At the other end of the spectrum, researchers have attempted to enhance the plasma effect by enlarging the plasma area. Linearly synthesized SDBD has been proposed [6], and it generates a large plasma area. The sliding plasma discharge is examined by designing the electrodes properly [7].

Accordingly, the plasma effects on electromagnetics have gained the interest of researchers because the scattered waves can be influenced by the staggered electrode structures. In addition, the incident waves and plasma may possibly interact. Therefore, the radar cross-section (RCS) of flying vehicles,

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which is one of the crucial factors for the survival rate of aircrafts, may change. Unexpected effects on RCS may occur because of the SDBD actuators installed for the purpose of controlling flow. To demonstrate the effect on electromagnetic waves, experiments in a certain observation plane have been made [8]. However, various observation planes are required because the SDBD structure has staggered electrodes that can cause reflection and diffraction. Another study analyzes the diffraction based on simulation under low pressure. It reveals that the RCS increases at a specific angle [9]. Experiments under atmospheric pressure are necessary for flying vehicles. The study [10] based on 2D simulation should be extended to 3D to enable analysis in various angles.

The current study focuses on the change in scattering by the SDBD air plasma actuator under atmospheric pressure. Monostatic and bistatic RCS in different observation planes is simulated to demonstrate the effect of a plasma actuator on survivability.

II. Structure of a Plasma Actuator

The linearly synthesized SDBD actuator consists of four unit actuators to enhance the plasma effect. The sufficient spacing between the actuators reduces the interference between the adjacent units. Each unit has an exposed and buried electrode with a dielectric barrier. The plasma is generated above the buried electrode (Fig. 1). The plasma actuator is backed by a metal plate to define the surface of the target. The targetis square, with each side measuring 120 mm. The configuration of the designed actuator is illustrated in Fig. 2. The size of the electrode strip is 100 mm × 12 mm. The spacing between two units is 8 mm. A substrate with a permittivity of 2.3 is used as the dielectric barrier.

Fig. 3 shows the air plasma generation of the designed actuator. Clearly, the visible plasma is generated on top of the buried electrodes. In addition, the interaction between the unit SDBD actuators is limited. The color of the plasma indicates that the air plasma is generated.

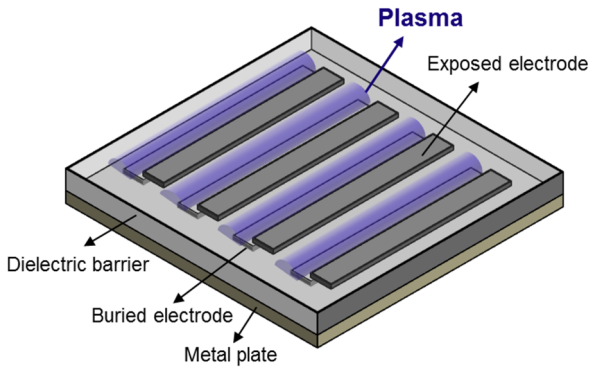
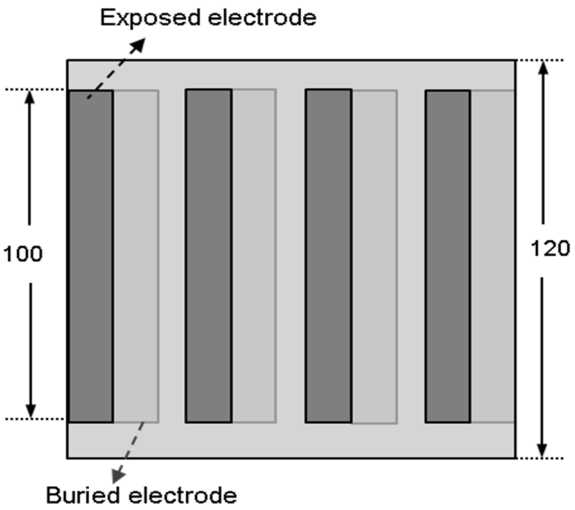
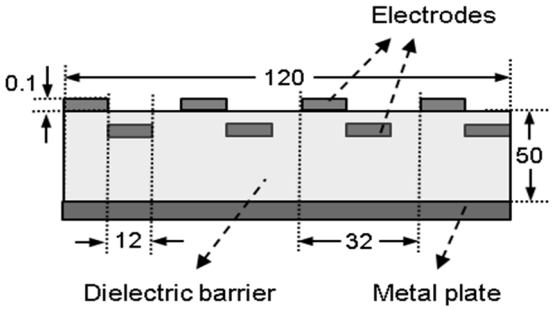


Fig. 1. Structure of the SDBD plasma actuator.



(a) Configuration



(b) Cross section of the generator

Fig. 2. Configuration of the plasma actuator (unit in mm). (a) Configuration. (b) Cross section of the generator.

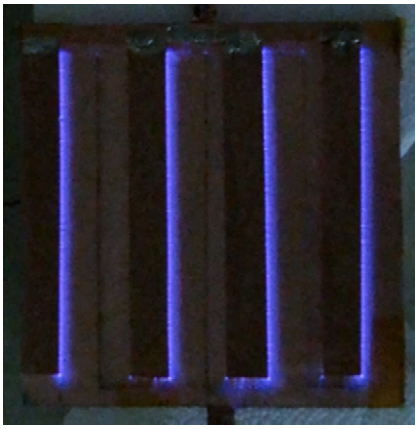


Fig. 3. Generated air plasma from the SDBD discharge actuator.

III. Effect on Electromagnetic Waves

1. Diffraction from the Plasma Actuator

The SDBD actuator has a periodic structure, which causes a grating diffraction with the conductor strips acting as a crystal [11]. In Eq. (1), the relation between the incident wave and the diffraction wave is predicted when the grating spacing is determined.

$$\sin\theta_d - \sin\theta_i = m\frac{\lambda}{\Lambda}, \tag{1}$$

where θ_d is the diffraction angle, θ_i is the incident angle, θ_i is the grating spacing between two elements, and m is the integer number. With the incident angle of 0° and Λ of 32 mm, the diffraction angle is calculated as 38.7° when the frequency of the incident wave is 15 GHz. Therefore, the maximum occurs at the angle with the bistatic RCS simulation.

The monostatic RCS can be analyzed as the Bragg diffraction:

$$2\sin\theta_d = m\frac{\lambda}{\Lambda}. \tag{2}$$

The calculation result is 18.2° when m is 1. As a result, monostatic RCS has a maximum diffraction at the angle.

2. Interaction with the Plasma Array

The interaction between cold plasma and incident electromagnetics is determined by dispersion relation,

$$k = k' - jk'' = \sqrt{\frac{\omega^2}{c^2} - \frac{{\omega_p}^2}{c^2(1 - j\frac{\nu_c}{\omega})}} \tag{3}$$

where ω_p is the plasma frequency, ν_c is the collision frequency, and ω_c is the operating frequency of the incident wave [12]. Plasma frequency and collision frequency are highly dependent on the plasma generator structure and the ambient pressure and temperature. The collision frequency is a function of the type of gas, neutral gas density, and electron temperature [13]. Measuring all of these variables is difficult. The value of collision frequency is 2.1 THz, which is obtained from the measurement of helium glow discharge under atmospheric pressure [14]. The plasma frequency is calculated from the equation,

$$\omega_p = \sqrt{\frac{n_e e^2}{m\varepsilon_0}} \tag{4}$$

where n_e is the electron density, e is the electron charge, and m is the mass of the electron. The range of electron density of the DBD actuator is 10^{17}–10^{21} m^{-3} [15]. The electron density largely varies as the structure and the environment change. It is the key to analyzing the interaction between plasma and electromagnetic waves. The electron density and its distribution should be derived. As shown in the following section, plasma modeling for the given actuator provides the electron density distribution.

IV. Plasma Modeling for the Actuator

To derive plasma modeling for the actuator, a simplified modeling for charged particles in plasma is utilized [15]. The relation between the charged particle and the electric potential is defined as Eq. (5).

$$\nabla\cdot(-\nabla\Phi) = \frac{\rho_c}{\varepsilon}, \tag{5}$$

where Φ is the electric potential in the conductive fluid, and ρ_c is the charge density. In Eq. (6), the electric potential can be separated into electric potential due to electrode ϕ and electric potential due to charged particles in plasma φ_c. Under the assumption that the Debye thickness is small and the charge on the wall is not large, the potential caused by the electric charge is the dominant factor in the decision of the charge distribution. The effect of the external electric field is relatively small [16]. Then, Eq. (5) can be converted to Eq. (7).

$$\Phi = \phi + \varphi_c, \tag{6}$$

$$\nabla\cdot(\varepsilon_r\nabla\rho_c) = -\frac{\rho_c}{\lambda_d^2}, \tag{7}$$

where ε_r is the relative permittivity, and λ_d is the Debye thickness. Based on a time-independent system, the density of the charged particle can be expressed as a normalized from

$$\rho_c^{\,*} = \frac{\rho_c}{\rho_c^{\,\max} f(t)}, \tag{8}$$

where ρ_c^{max} is the maximum charged particle density and $f(t)$ is the time-dependent factor. On the surface of the dielectric, the charged particle follows a half Gaussian distribution [17]. Then, the electron density can be derived as

$$n_e = \rho_c \exp(\frac{e\Phi}{k_B T_e}) \tag{9}$$

where k_B is the Boltzmann constant, and T_e is the electron temperature. The calculated normalized electron density is shown in Fig. 4 by using multi-physics software (COMSOL Inc., Palo Alto, CA, USA). The maximum occurs on top of the buried electrode, and the density weakens as the distance from the surface increases. The maximum electron density is set to 1×10^{21} m^{-3}, which corresponds to the electron range in a previous work

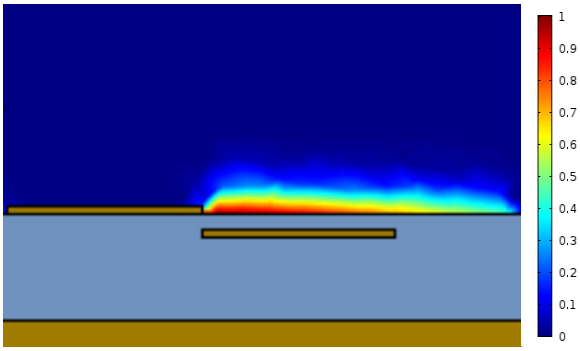


Fig. 4. Normalized charge density distribution, ρ_c^*.

V. RCS OF THE TARGET

Fig. 5 shows the coordinates for the DBD plasma actuator in the *x*-*y* plane. Monostatic and bistatic RCSs are calculated as shown in Figs. 6–9 using COMSOL multi-physics software.

1. Monostatic RCS

When the linearly polarized electromagnetic wave along the y-axis is incident toward the DBD with a frequency of 15 GHz, the monostatic RCS is observed in the *x*-*z* plane as shown in Fig. 6. The copper plate has a prominent peak at 0°. Most of the field hits the plate and reflects back toward it. If the radar is in the different direction, backward reflection will decrease because of specular reflection. For the plasma-off state, the peak around 0° is explained similarly. At a different angle, the incident fields are scattered by the electrodes, which increase the level of monostatic RCS. Accordingly, the RCS for the plasma-off state is higher than that for the copper plate. In addition, the simulated result with the plasma-off is compared with the measured result reported by Wolf and Arjomandi [8]. A good agreement was found in the simulated and measured results in the range of –20° to 20°. The main beam at 0° is observed because of a specular reflection from the copper plate. The experimental result indicated a distorted pattern in 30°–40°. The level of measurement result with the plasma-off state is higher than that of the simulation. This outcome may be due to the undesired effects from background noise. However, the simulated monostatic RCS of DBD and the analytic solution of the square copper plate show symmetry. When the plasma is generated, the graph shows no significant change. The generated plasma may have a minor effect on the monostatic RCS in the φ = 0° plane.

However, the trend is completely changed in other observation planes. When φ=90°, the monostatic RCS has an asym-

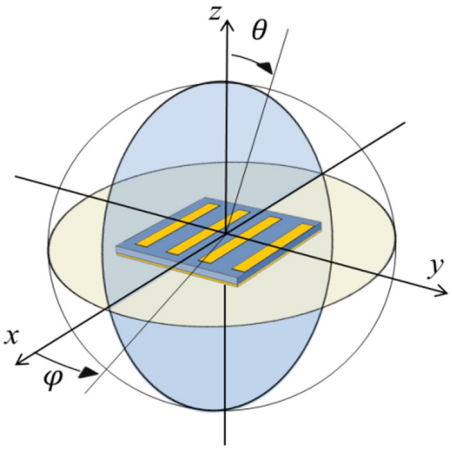


Fig. 5. Coordinates for the dielectric barrier discharge actuator.

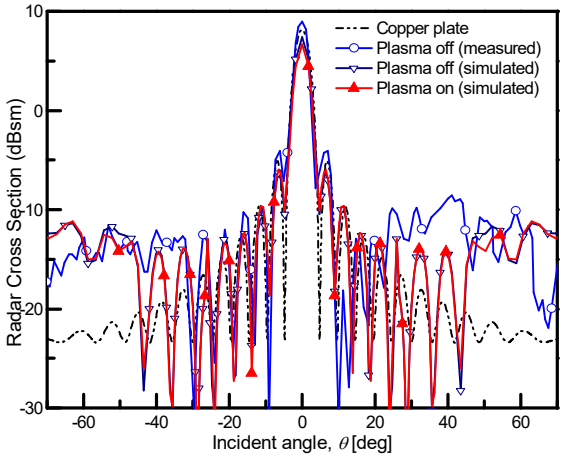


Fig. 6. Monostatic RCS for the incident wave of a 15-GHz frequency when φ=0°.

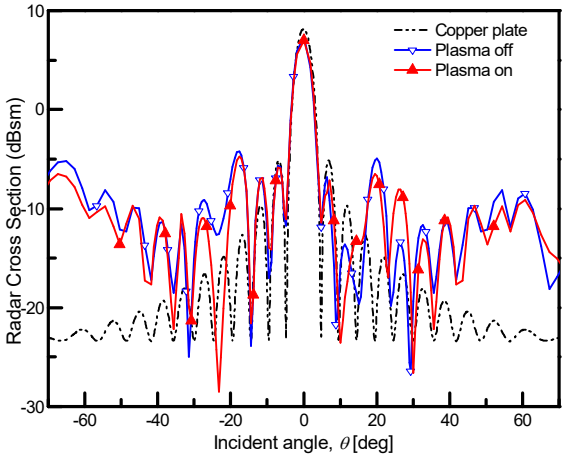


Fig. 7. Monostatic RCS for the incident wave of a 15-GHz frequency when φ=90°.

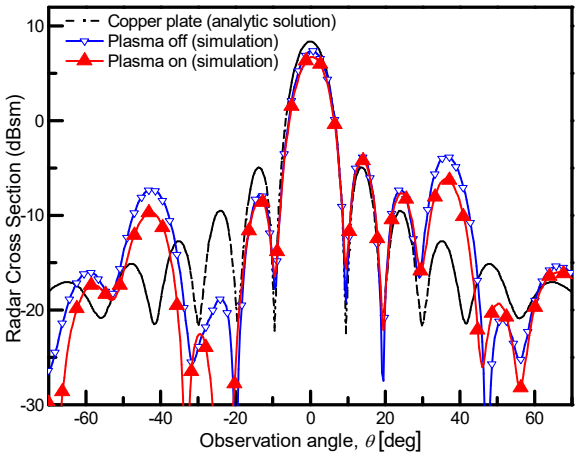


Fig. 8. Bistatic RCS for the incident wave of a 15-GHz frequency with the observation in the *x*-*z* plane.

metric pattern in the *y*-*z* plane because of the staggered electrodes in the actuator. Compared with the copper plate, sudden

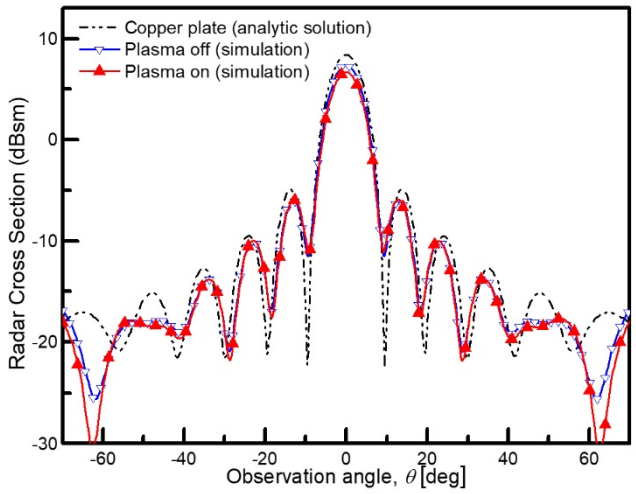


Fig. 9. Bistatic RCS for incident wave of 15 GHz frequency with the observation in *y*-*z* plane.

increases occur near −20° and 20° that are caused by the Bragg diffraction, the calculated incident angle of which is 18.2° in the previous section. When the plasma is generated, a decrease of 10°–20° occurs, and a sudden increase is observed in 30°. The reason for this outcome is that the inhomogeneity of the plasma distribution changes the direction of the scattering waves.

2. Bistatic RCS

Fig. 8 shows the bistatic RCS when the incident wave is linearly polarized in the *x*-*z* plane. The plasma-off state RCS is comparable with that in the case of the square copper plate. A peak at 0° is observed because of the specular reflection. In addition, the plasma effect does not appear in the graph. As shown in Fig. 8, the distortion due to the plasma actuator is notable in Fig. 9. When the incident angle is 0°, the diffraction angle is 38.7°. Two peaks due to diffraction can be observed at the diffraction angles. The peaks are at least 5 dB higher than that in the case of the copper plate. As the plasma is generated, the overall level decreases by 2–3 dB. The RCS near −25° has notable effects. The attenuation by plasma decreases the RCS by 10 dB at the frequency range. It is related to the distribution of plasma, which is periodically located near exposed electrodes. The reduction is caused by both the staggered electrode alignment of the generator and the plasma distribution.

VI. Conclusion

The SDBD plasma has a significant effect on RCS. Peaks by diffraction are observed in both plasma-off and -on states because of the periodicity of staggered electrodes. In the plasma-on state, the inhomogeneous plasma distribution causes both the reduction of the level and the change in scattering angles. These effects become different as the RCS observation plane changes.

If the operating frequency or spacing between electrodes is changed, the location of peaks due to diffraction can be controlled. Therefore, moving the peaks to the unimportant angles is possible, thus reducing the level of RCS in the desired range of angles. However, the changes in the structure affect the process of plasma generation. They lead to different reflections and absorption properties by plasma. Therefore, such relations should be considered in the DBD actuator design. Further research should include the effect of a complex electrode structure on plasma generation.

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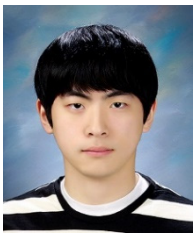
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SDR-Based Frequency Interference Emulator in the Space-Time Domain and Its Application

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Abstract

In this study, we propose a software-defined radio-based frequency interference emulator in the space-time domain. This emulator can easily model actual interference environments because of the versatile programming capability of the universal software radio peripheral and LabVIEW. As an example of an interfering network using the contention-based multiple access scheme in the time domain, we emulate a coordinated Wi-Fi network that consists of one access point and two Wi-Fi nodes. Results show that our emulator can successfully model multiple interfering signals in the Wi-Fi network and easily adjust various space–time domain parameters.

Key Words: Emulator, Interference, Software-Defined Radio (SDR), Wi-Fi

I. INTRODUCTION

A world without wireless communication is difficult to imagine. For various wireless services, wireless devices such as smart phones, laptops, and sensors are rapidly increasing, thus causing the use of the frequency spectrum to increase sharply. However, only a limited amount of frequency spectrum is available, and this limitation may cause wireless devices to experience adverse interference from nearby wireless devices that simultaneously transmit in the same or adjacent frequency band. Therefore, a quantitative understanding and analysis of this interference are important for future wireless services [1].

Various experimental methods to analyze interference problems are reported in [2–5]. Some studies use real wireless hardware to emulate interference environments [2–4]. However, when real hardware is used, changing or obtaining physical (PHY) and/or media access control (MAC) layer parameters is difficult because they are commonly embedded in the modem chip. A typical parameter in interference analysis, namely, bit error rate, can hardly be measured at the PHY layer of real hardware without modifying its software. One study generally used signal generators to emulate interfering signals [5].

Although a signal generator is highly controllable, the number of interferers that it can emulate remains unclear. Therefore, a necessary task is to implement an interference emulator that can reflect a realistic interference environment and easily adjust various parameters of multiple interferers in the space, frequency, and/or time domain.

This study aims to develop a versatile emulator for imitating various frequency interference environments consisting of multiple interferers. We propose a software-defined radio (SDR)-based frequency interference emulator in the space–time domain. In our emulator, space and frequency domain parameters in the PHY layer are adjusted by a Universal Software Radio

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Peripheral (USRP) and the LabVIEW program. Time domain parameters used in the MAC layer are controlled by a discrete Markov model implemented in the LabVIEW program. Given the versatile programming capability of USRP and LabVIEW, we can adjust various space and frequency domain parameters (separation distance, path loss according to the separation distance, center frequency, bandwidth, spectrum mask, transmit power, etc.) and time domain parameters (packet size, duty cycle, back-off time, etc.), respectively. Accordingly, we analyze the effects of the victim's performance degradation due to realistic multiple interferers in a given interference scenario.

II. INTERFERER MODEL

To show an explanatory emulation of a multi-interferer environment in detail, we assume an interference scenario with Zigbee as the victim and the Wi-Fi network as the multiple interferers. We take these assumptions because the 2.4 GHz unlicensed frequency band is used worldwide, and previous studies showed that the Wi-Fi network is the most significant interference source of Zigbee [2]. Fig. 1 depicts a typical interference scenario with Wi-Fi and Zigbee devices in the space and time domain, respectively. In the space domain, the probability of interference increases as the separation distance between the victim's wireless device (Zigbee) and the interferer (Wi-Fi) decreases. Moreover, the interference may occur as the operating frequency band between wireless devices overlaps. As shown in Fig. 1(a), the Zigbee channel may overlap with the Wi-Fi channel. For example, if the Wi-Fi network operates on channel 6 (CH6) and Zigbee operates on channel 17 (CH17), then the Zigbee channel will be completely overlapped by the Wi-Fi channel. This interference is probably the most severe because the spectrum overlapping factor is almost 1. To extend this finding to a case with K Wi-Fi interferers, the received interference signal power at the victim's antenna input port can be defined as

$$P_{Ik} = P_{wifi} \cdot \alpha_k(d) \cdot \gamma_k, \text{for } k = 0,1,\cdots,K-1, \qquad (1)$$

where P_{wifi} is the transmit power of the WiFi node, and $\alpha_k(d)$ is the path loss at the distance, d, from the kth node to the victim receiver, γ_k and is the spectrum overlapping factor of the kth Wi-Fi interferer; the typical values of γ_k and α_k are specified in [6]. An important detail is that even though K Wi-Fi interferers exist, only one node including the access point (AP) transmits data at any instant, as shown in Fig. 1(b).

We then consider the time domain shown in Fig. 1(b). Both standards for Zigbee (IEEE 802.15.4) and Wi-Fi (IEEE 802.11b) specify three clear channel assessment (CCA) methods to determine the channel occupancy. The CCA default mode of Wi-Fi and Zigbee devices operate in the carrier sensing mode, in which a Wi-Fi node will consider the channel free if no other Wi-Fi device is detected. If we assume that both Wi-Fi and Zigbee devices operate in the carrier sensing mode, then they will essentially be blind to each other's transmissions[7-8]. This assumption provides the worst-case performance of an environment in which Wi-Fi and Zigbee coexist [9]. Therefore, the interfering Wi-Fi network can be emulated independently regardless of the Zigbee victim nodes. Moreover, only one Wi-Fi node may transmit at any instant, and it interferes with the Zigbee nodes regardless of the number of coordinated Wi-Fi nodes. Fig. 1(b) shows an example in which the Wi-Fi network has full traffic, where t_{DIFS} and t_{SIFS} are the periods of time for the distributed interframe space (DIFS) and the short interframe space (SIFS), respectively. If the channel is idle for a period of time that is equal to t_{DIFS}, node #1 will

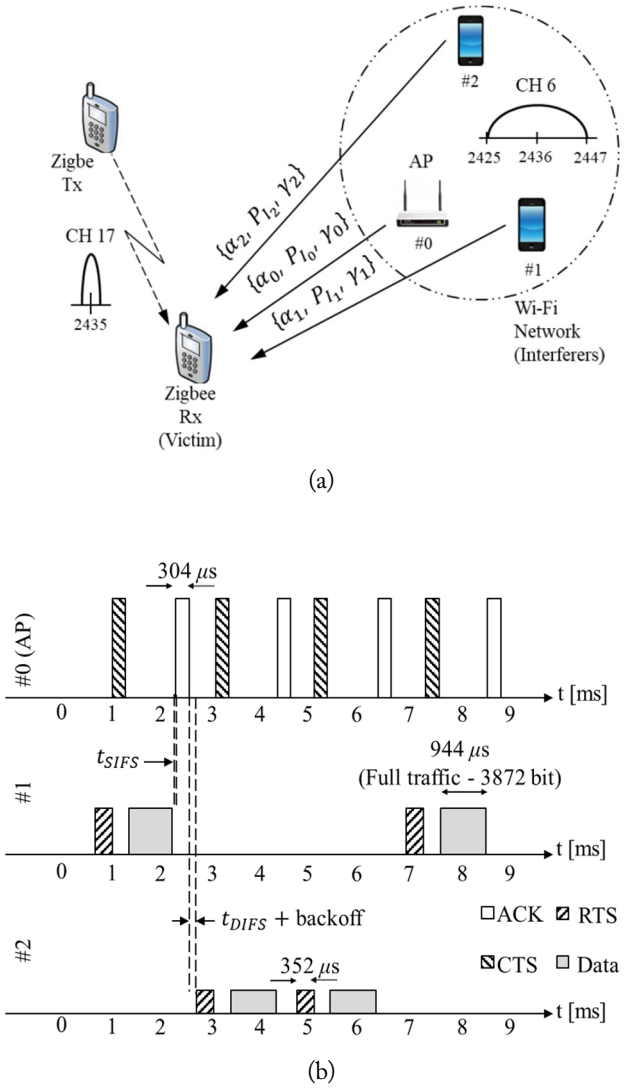


Fig. 1. Interference scenario that consists of a Zigbee pair as the victim and three Wi-Fi interfering nodes. (a) Space domain and (b) time domain.

transmit the Request-to-Send (RTS) message. If node #1 receives the Clear-to-Send (CTS) message from AP, then node #1 will transmit its data packet. In case of full traffic, the packet size is 3872 bit (about 944 μs). It the AP successfully receives the packet from node #1, the AP transmits the acknowledgement (ACK) message to node #1 after the t_{SIFS} second. This process is repeated until there is no data packet to be transmitted.

III. Implementation of Interference Emulator

We designed an SDR-based frequency interference emulator that consists of one Wi-Fi AP and two Wi-Fi nodes, as shown in Fig. 2(a). We connected the host PC to the USRP with gigabit Ethernet cables for a fast interface, and we used an RF switch that can selectively connect three antennas with one USRP. The Wi-Fi transmitter output power is set to 30 mW in accordance with the standard. Our emulator can change the individual path loss α_k according to the distribution of the nodes. Moreover, the proposed emulator can calculate path loss in the software by using a specific path loss model without the actual spatial distribution of the interferers. That is, as shown in Fig. 1(a), if we determine the spatial distribution of interferers and calculate path loss in the software, then we can easily implement our emulator using only one USRP. For this reason, the suggested emulator is advantageous for emulating multi-interferer environments.

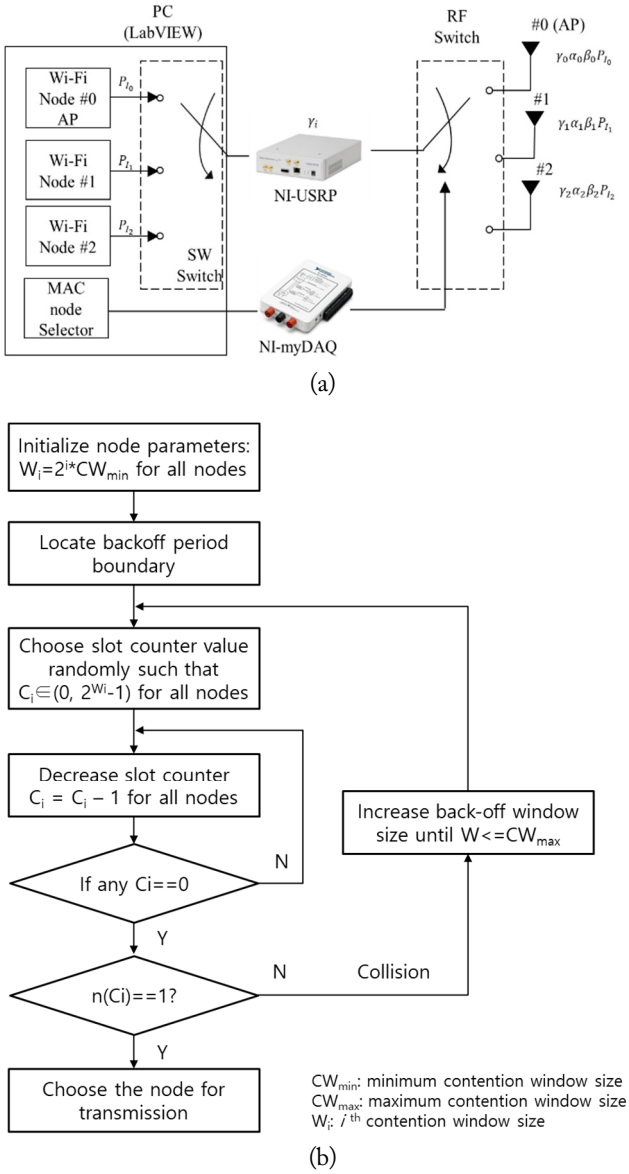


Fig. 2. Block diagram of the interference emulator. (a) Block diagram and (b) transmitting node selection algorithm.

In the time domain, a transmitting node selection algorithm is implemented using LabVIEW. According to the channel access scheme of the Wi-Fi network, if more than two nodes transmit data at a given instant, then a collision will occur, and the collided nodes will have to wait during the slot counter value, C_i, where the subscript i denotes the backoff stage, $i \in \{0,1,\cdots,m\}$, and m is the maximum backoff stage. When only one Wi-Fi node transmits data at a given time, the Wi-Fi node transmits immediately without collision. Otherwise, each unsuccessful transmission doubles the contention window size, W_i, up to a maximum value, CW_{max}. This process is depicted in Fig. 2(b) and is based on [10].

To support the validation of our emulator, we generate interfering data packets with the LabVIEW program using the transmitting node selection algorithm depicted in Fig. 2(b). Then, we configure the emulator as shown in Fig. 3(a). In the configuration illustrated in Fig. 3(a), we expect the AP signal to have the highest amplitude at the victim receiver and the #2 node signal to have the smallest amplitude at the victim receiver. Verification of the measured waveform at the victim receiver using an oscilloscope in Fig. 3(b) confirms that our emulator can successfully generate time domain packets of multiple Wi-Fi interferers. All parameters, except for center frequency, are consistent with the Wi-Fi standard, but center frequency is set to 400 MHz because of the limitation of the oscilloscope. Fig. 3(b) shows the interference waveform of the Wi-Fi interferers when the Wi-Fi network has a full traffic case. The time domain waveform in Fig. 3(b) is almost identical to that in Fig. 1(b). To test its usability, we emulate an interference waveform when the Wi-Fi network has no traffic and the Wi-Fi AP transmits only a beacon every 100 ms, as shown in Fig. 3(c). These results show that our emulator can successfully emulate interfering signals in the Wi-Fi network and easily adjust various space-time domain parameters.

IV. Conclusion

We have proposed an SDR-based frequency interference emulator in the space-time domain. Actual interference envi-

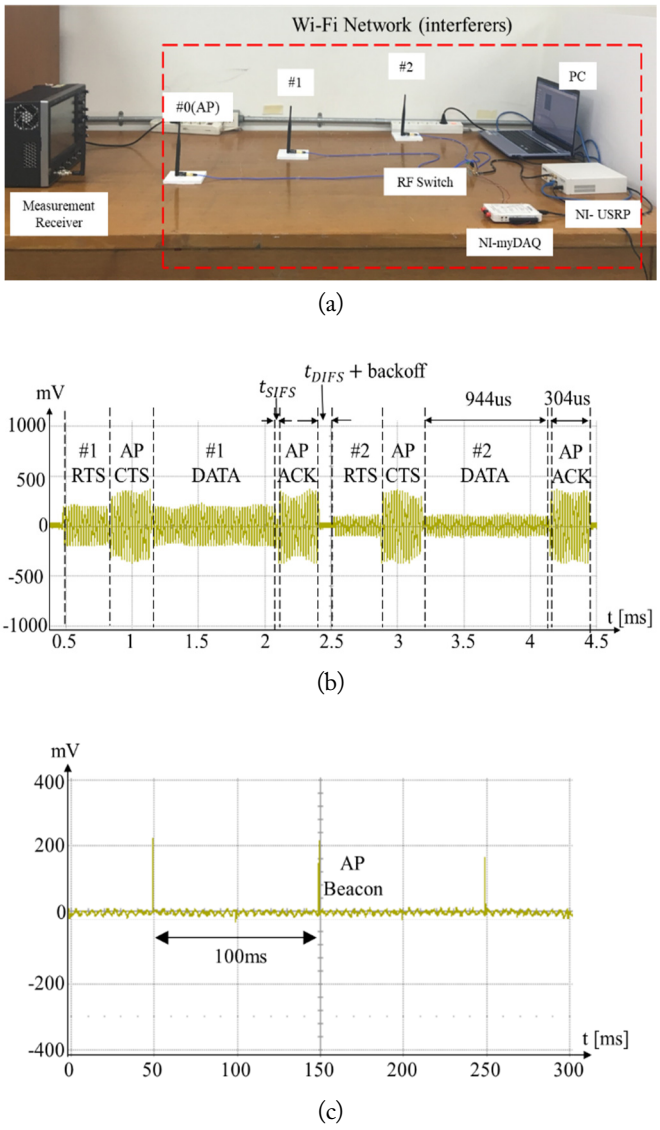


Fig. 3. Interference emulation of the Wi-Fi network at the input of the victim receiver. (a) Photograph of the implementation, (b) full traffic case, and (c) no traffic case.

ronments can be modeled easily because of the versatile programming capability of the USRP and LabVIEW. Our interference emulator can easily model a large number of simultaneous interferers that are correlated. The emulator can also help with the requirements of the extensive and controlled testing of the frequency interference mitigation algorithm prior to its use within a system.

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Attenuation Effects of Plasma on Ka-Band Wave Propagation in Various Gas and Pressure Environments

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Abstract

This work demonstrates attenuation effects of plasma on waves propagating in the 26.5–40 GHz range. The effect is investigated via experiments measuring the transmission between two Ka-band horn antennas set 30 cm apart. A dielectric-barrier-discharge (DBD) plasma generator with a size of 200 mm × 100 mm × 70 mm and consisting of 20 layers of electrodes is placed between the two antennas. The DBD generator is placed in a 400 mm × 300 mm × 400 mm acrylic chamber so that the experiments can be performed for plasma generated under various conditions of gas and pressure, for instance, in air, Ar, and He environments at 0.001, 0.05, and 1 atm of pressure. Attenuation is calculated by the difference in the transmission level, with and without plasma, which is generated with a bias voltage of 20 kV in the 0.1–1.4 kHz range. Results show that the attenuation varies from 0.05 dB/m to 9.0 dB/m depending on the environment. Noble gas environments show higher levels of attenuation than air, and He is lossier than Ar. In all gas environments, attenuation increases as pressure increases. Finally, electromagnetic models of plasmas generated in various conditions are provided.

Key Words: Dielectric-Barrier-Discharge Actuator, Plasma Absorption Properties, Plasma Attenuation.

I. Introduction

Plasma is a state of matter consisting of ionized gas. A typical example of the application of plasma is utilizing its response to electromagnetic fields to etch dielectrics in the process of fabricating semiconductors. It is well known that the propagation of electromagnetic waves is attenuated when plasma is introduced in their path. The attenuation characteristics of plasma are affected by the collision frequency and the electron density, which has been deeply investigated both theoretically [1–5] and experimentally [6–13].

Research on the attenuation effects of plasma on electromagnetic waves resulted in an important application of plasma: reducing the radar cross section (RCS) of a target to improve its low observability to radars [11–13]. In [11], measurement of a 12 cm × 12 cm metallic plate plasma showed no significant effect on the RCS, which the authors speculate is due to the very limited life time of the free electrons and the small extent of the plasma sheath. However in [12], RCS reduction of as much as 3 dB was demonstrated experimentally at 7.4 GHz for a 20 cm × 20 cm rectangular metallic plate with a circular dielectric-barrier-discharge (DBD) plasma generator. Finally, [13] reports that the frequency of maximum RCS reduction for a 20 cm × 20 cm rectangular metallic plate can be tuned by varying the number of electrodes that generate plasma, that is, by the area and location of the

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generated plasma.

In this work, the attenuation effects of plasma on propagating Ka-band waves is investigated experimentally. A DBD generator consisting of 20 layers of electrodes is fabricated. Then the generator is centered between two Ka-band antennas that are set 30 cm apart. The DBD generator is placed in an acrylic chamber to investigate the difference in the attenuation effects of plasmas generated under various gas and pressure environments. Then the transmission is measured in the Ka-band with and without plasma; the difference is the electromagnetic attenuation caused by the generated plasma. Experimental results are presented along with an electromagnetic model of the plasma obtained based on the experimental results.

II. Experimental Setup

The reduction of monostatic RCS of metallic plates due to plasma as described in [12, 13] is largely due to its reflection characteristics. However, the purpose of this work is to investigate the effect of plasma on the transmission of electromagnetic waves in the Ka-band. Therefore, the plasma must be more than a large fraction of the wavelength in the direction of propagation of the plasma.

Fig. 1 shows the proposed parallel electrode DBD generator for investigation of the attenuation effects of plasma on electromagnetic waves in the Ka-band. The electrodes of the DBD generators are placed laterally in [12, 13] to generate a planar plasma, thus transforming the shape of the DBD generator electromagnetically. In contrast, we stack the electrodes vertically, as shown in in Fig. 1. When plasma is generated between the two electrodes by applying a bias, the propagating wave must pass through the plasma layer, whose extent is between the depth of the electrode and the substrate, 60 mm and 100 mm, respectively. This corresponds to 5.2–8.7 free space wavelengths at 26 GHz, the lowest frequency in the Ka-band. This is sufficient to observe the absorbing effect of plasma.

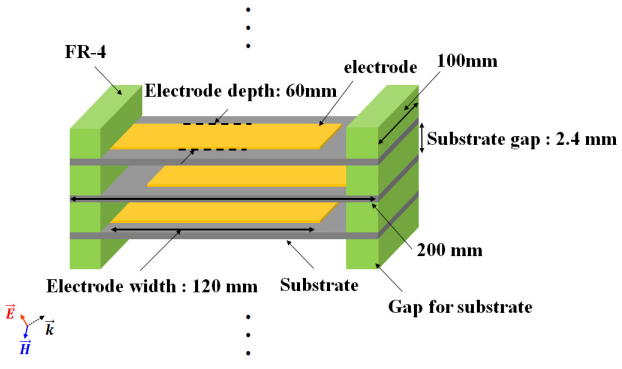
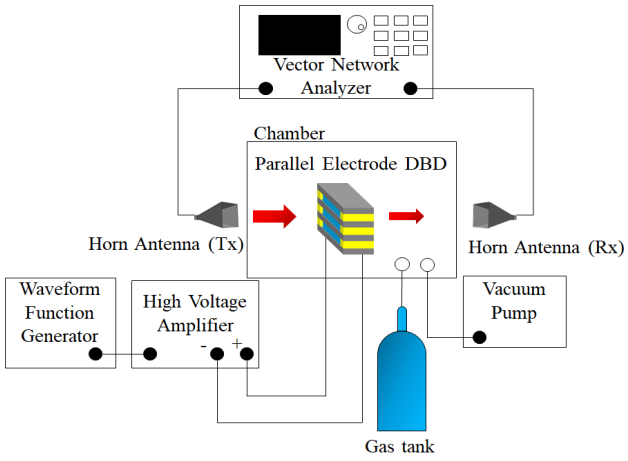
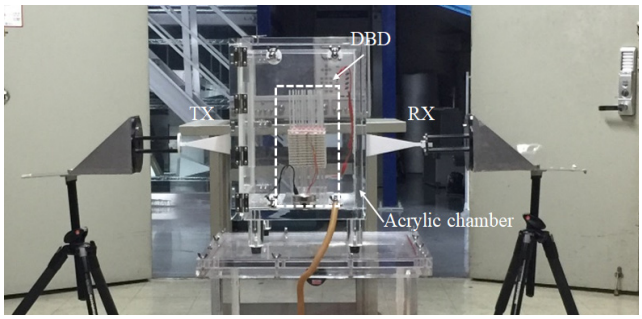


Fig. 1. Parallel electrode DBD structure.



(a)



(b)

Fig. 2. Experimental setup. (a) Complete experimental setup and (b) Photograph of experimental setup.

There are a total of 20 electrodes, each 120 mm wide, with a gap of 2.4 mm between each electrode, which is maintained using 2.4-mm-thick dielectric pieces. Thus, the total size of the electrode is 200 mm × 100 mm × 70 mm. Although it is not shown in Fig. 1, acrylic poles are used for vertical alignment of the electrodes. Alumina substrates are used to minimize structural deformation such as bending in order to prevent plasma arcs even at low bias levels.

The DBD structure is placed in a 400 mm × 300 mm × 400 mm acrylic chamber (48 L in volume). The chamber is hermetic with 3-cm-thick walls such that the pressure inside the chamber can be reduced to 1 Torr, or approximately 0.001 atm.

A diagram of the complete setup is shown in Fig. 2(a). The chamber is placed between two MTG SGH-40 Ka-band standard horn antennas. The longer the distance between the two antennas, the more vulnerable the measurement is to any misalignment between the antennas. The shorter the distance, the more difficult it becomes to remove multipath signals by gating the time-domain response. In this work, the two antennas are separated by 30 cm, which is an optimal distance.

Bias voltage is applied using a Keysight 33500B waveform function generator and amplified via a Trek Model 10/40A high-voltage power amplifier. The transmission is measured in the Ka-band using an Anritsu 37247D Vector Network Analyzer. The time-gating function and an in-house algorithm are utilized to find the exact location and width of the time window, which is critical in obtaining accurate responses. Fig. 2(b) shows a photograph of the setup.

III. Plasma Generation in Various Environment

The purpose of this work is to investigate the attenuating effects of plasma on Ka-band electromagnetic wave propagation. Another important purpose is exploring the difference in the effect for plasmas generated under different conditions. In this work, transmission is measured with and without plasma generated in three different gas environments: air, Ar, and He. For each gas, the effect is measured at three different pressure levels: 1, 0.05, and 0.001 atm.

First, the pressure inside the chamber is reduced to 0.001 atm. Then gas is introduced to the chamber until the chamber is completely filled or when the pressure recovers to 1 atm. This is repeated three times to ensure that the chamber is filled with the gas being tested. After the first measurement at 1 atm, the pressure is lowered to 0.05 atm. After the second measurement at 0.05 atm, pressure is further reduced to 0.001 atm and a final measurement is taken.

For each set of conditions, the transmission between the antennas is measured with and without the plasma.

Accurate investigation requires maintaining a stable discharge. This is possible when the applied bias voltage is sufficiently higher than the breakdown voltage, the minimum voltage at which the plasma is generated.

Table 1 summarizes the breakdown voltages for the parallel electrode DBD generator in Fig. 1 for every gas and pressure condition tested in this work. At 1 atm, the minimum bias voltage required to generate plasma is highest in air and lowest in the He environment. This is because noble gases such as Ar and He are much less reactive than air. At lower pressures, the minimum bias voltage decreases dramatically, which reaches saturation at some point.

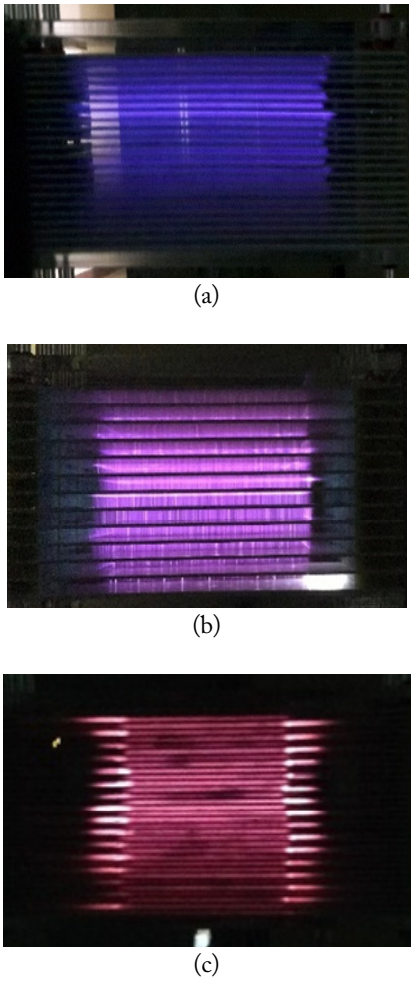
For instance, at 1 atm, the minimum voltage for air is more than twice than that for Ar. However, at 0.05 atm, the difference is within 15% and vanishes at 0.001 atm. Saturation in the bias voltage occurs at a higher pressure for He, so the minimum bias levels at 0.05 atm and 0.001 atm are the same. When the pressure decreases, the bias voltages decreases, because there are fewer collisions between electrons and gas in the lower pressure. In other words, when the pressure is low, the distance between the electrons and gas molecules is high. To generate same amount of plasma at different pressures, low bias voltage is required at low pressures.

Table 1. Measured breakdown voltage (frequency: 1 kHz)

Pressure (atm)	Breakdown voltage (kV)		
	Air	Ar	He
1	14	6	1.6
0.05	1.6	1.4	0.4
0.001	0.8	0.8	0.4

Fig. 3 shows photographs of the plasmas generated at 0.05 atm for all three gases. As can be seen, the three plasmas have different colors; the air plasma is dark blue, the Ar plasma is purple, and the He plasma is orange.

These different colors are due to the different energy levels of the plasma ions. The ionization energies of various gas ions are summarized in Table 2. Helium ions have the highest ionization energy level, while Ar, nitrogen, and oxygen ions have similar energy levels. Because air is mostly nitrogen and oxygen, the air and Ar plasmas are similar in color while He contains more red due to its substantially higher ionization energy.



(a) (b) (c)

Fig. 3. Plasmas generated at various gas environments at 0.05 atm. (a) Air, (b) argon, and (c) helium plasma.

Table 2. Ionization energy of various gas ions [14]

	He	Ar	Nitrogen	Oxygen
Ionization energy (eV)	24.7	15.8	14.6	13.7

IV. Experimental Results

Using the experimental setup shown in Fig. 2, the transmission is measured in the entire Ka-band between the two antennas with and without plasma generated under various gas and pressure conditions. The difference between the two transmissions corresponds to the attenuation due to the plasma.

Fig. 4 shows the measured attenuation. For all gas conditions, the attenuation effects of plasma become larger at higher pressure levels. This is due to the increased number of ions present at higher pressures.

When the pressure is the same, the air plasma shows the smallest attenuation effect compared to the Ar and He plasmas. This is because air is mostly nitrogen and oxygen, which are much more reactive than the noble gases, and it is therefore easier to produce other mixtures such as ozone rather than plasmas.

In the majority of cases, the attenuation remains nearly constant with respect to the frequency. However, at 1 atm, the attenuation of the Ar plasma increases with frequency, while that of the He plasma gradually decreases with frequency. Although further investigation into this difference in attenuation behavior remains as future work, it is presumed to be caused in part due to inevitable experimental errors associated with indoor experiments of unbounded waves based on time-gating of the responses, which is more noticeable because of the relatively high attenuation levels at high pressures (1 atm).

The following electromagnetic model of the plasmas is based on the measured results. The wave number of a plasma layer is given as [3]

$$jk = \mathrm{j}\omega\varepsilon_p = \sigma_p + j\omega\varepsilon_0 \tag{1}$$

where ω is the wave frequency, ε_p is the plasma permittivity, ε_0 is the vacuum permittivity, and σ_p is the plasma conductivity. In this work, the generator frequency ranges from 0.1 to 1.4 kHz. Such DC plasmas, that is, those generated by bias voltages with frequency below 100 kHz, have permittivity of one [3]. The ANSYS High Frequency Structure Simulator (HFSS) is used to obtain the electromagnetic model of the plasmas, by changing the conductivity. Fig. 4 also shows the fitted attenuation under various conditions, which is in good agreement with the measured results.

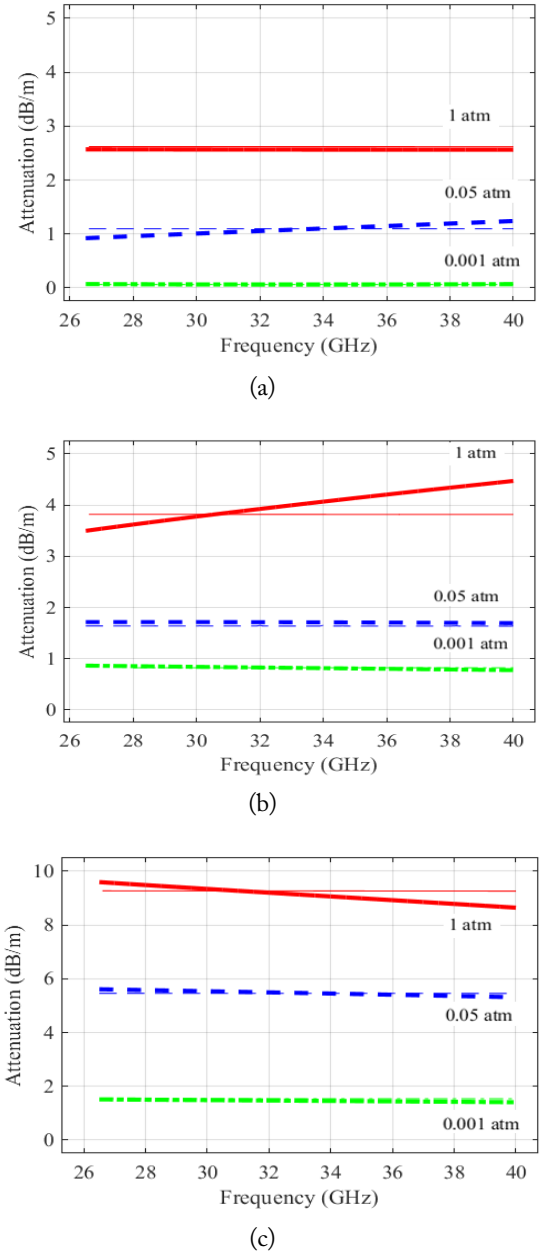


Fig. 4. Measured (thick lines) and simulated (thin lines) results of attenuation in plasma. (a) Air, (b) argon, and (c) helium plasma.

The calculated conductivity of the various gas plasmas is shown in Fig. 5. As the pressure increases, the number of ions contributing to the plasma increases, which in turn increases the conductivity. In all cases, He plasma shows the highest conductivity, which corresponds well with the highest atenuation in Fig. 4. Also, the attenuation levels remain virtually constant with respect to the frequency, which is the case in the majority of the experimental results.

Table 3 compares the conductivities obtained for the He plasma in this work and those in [9]. The conductivity in this

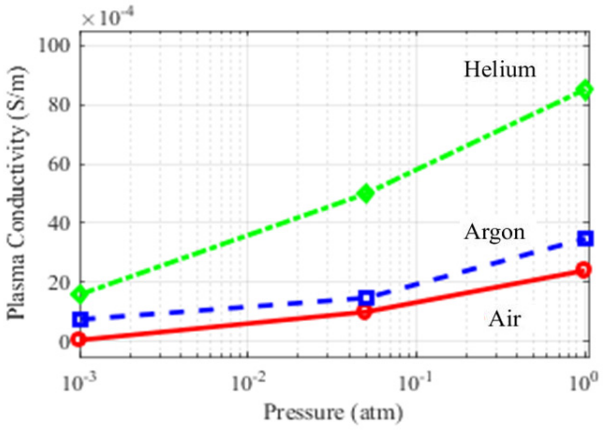


Fig. 5. Comparison of plasma conductivity according to gas composition and pressure.

Table 3. Comparison of modeled He plasma conductivity

	This work	Ref [9]
Plasma generator	DBD generator	Plasma jet
Bias voltage (kV)	20	5.26
Bias frequency (kHz)	0.1	20
Pressure (atm)	1	1.05
Plasma conductivity (S/m)	8.5×10^{-3}	5.09×10^{-2}

work is substantially lower than in [9]. Although both plasmas are generated at similar pressures, the conductivities have substantially different frequencies. The former is from the attenuation effects on Ka-band waves, while the latter is from those in the 2.7 GHz range. Most importantly, the plasma in this work is generated using a DBD generator, while that in [9] is from a jet plasma. The method of plasma discharge can result in great differences in the plasma density [3], which is presumed to be the cause of the difference in the conductivities between the two studies. Further investigation remains as future work.

V. Conclusion

In this work, the attenuation of plasma is investigated in the Ka-band. The attenuation due to plasma is measured under various gas and pressure conditions. Attenuation as high as 9.0 dB/m is obtained in the measured range. Helium plasma shows the highest attenuation effect, while air, which is composed mostly of reactive gases such as nitrogen and oxygen, is the least effective. All three tested gas environments (He, Ar, and air) showed higher attenuation at higher pressures. Based on the measured results, the plasma is electromagnetically modeled with conductivity. Results show that the He plasma has higher conductivity compared to other gases and that the difference in conductivities is maximized at 1 atm. Utilization of various gas plasmas to control RCS remains as future work.

This work has been supported by the Low Observable Technology Research Center program of the Defense Acquisition Program Administration and Agency for Defense Development.

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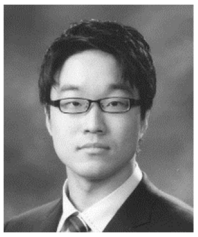
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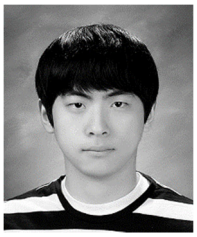
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Yuna Kim



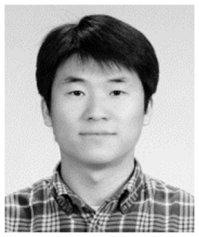
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Noninvasive Brain Stimulation Using a Modulated Microwave Signal

Taeyoon Seo[1] · Seongwoog Oh[2] · Dahee Jung[2,3,4,5] · Yeowool Huh[2,3] · Jeiwon Cho[2,3] · Youngwoo Kwon[1,*]

Abstract

We propose a microwave signal generation system for brain stimulation. The existing brain stimulation system uses a signal of several tens of kHz, and the magnetic field distribution is wide. Microwave is used to locally limit the distribution of the electromagnetic field and to change the action potential of the cell with less power. The switch modulates the microwave signal to obtain a pulse envelope. The action potential of the cell can be controlled to the excitation/inhibition state by adjusting the repetition frequency. These results are confirmed by measuring the cell potential of the mouse brain.

Key Words: Microwave Brain Stimulation, Noninvasive Brain Stimulation, Pulse Modulated Stimulation Signal.

I. Introduction

The experience of a new environment can lead to changes in the connections of the nerves throughout life [1]. The ability of the brain to reorganize itself by forming new neural connections is called neuroplasticity [2]. Neuroplasticity is the basic mechanism of learning and memory and is the result of the restoration of functions after cerebral nerve damage. Conventional rehabilitation methods do not directly change the brain but mostly improve the function of the brain through appropriate environmental changes.

Noninvasive brain stimulation is a method of neuromodulation by stimulating certain parts of the brain without surgical treatment using a magnetic field or an electric current. A transcranial magnetic stimulation (TMS) is currently being used clinically [3]. Although TMS has the advantages of noninvasiveness and inability to cause skin irritation, it has the disadvantages of requiring expensive equipment and causing noise. TMS is based on the induction current induced by a change in magnetic field. The magnetic field in the tens of kHz is widely distributed to include all areas of the brain. Localizing the stimulation area in the brain is possible because microwave has a short skin depth.

In this study, microwave is used as a method to change the action potential of the nerve. Moreover, the nerve firing rate (FR) is modulated by controlling the pulse repetition frequency (PRF).

II. Integrated Brain Stimulation System

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To demonstrate the feasibility of brain stimulation using microwave, an integrated system is implemented by combining a voltage-controlled oscillator (VCO), switches, and a power amplifier (PA). The stimulus signal is transmitted to the brain through a stimulator. The block diagram of the system is shown in Fig. 1.

1. Voltage-Controlled Oscillator

As a microwave signal source, a VCO generates a 6.5-GHz microwave signal. The schematic of the VCO is shown in Fig. 2. The VCO is designed with a cross-coupled structure, which is widely adopted in VCO design. A buffer isolates the load impedance effects and offers voltage gain. By varying the load impedance of the common-source amplifier, the buffer controls the output power of the stimulation system.

2. Power Amplifier

As shown in Fig. 3, shunt switches connected with the buffer act as a modulator. A stimulation protocol is applied to the gate of the switch transistor, so that the envelope of the microwave signal has a pulse shape.

The PA amplifies the modulated microwave signal. Given the output power of the buffer (around 0 dBm), a two-stage amplifier is designed for high power (>20 dBm). A differential structure is introduced to avoid the source degeneration effect.

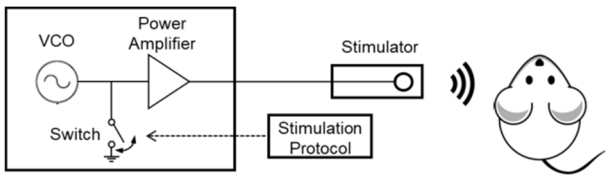


Fig. 1. Overall block diagram of the proposed brain stimulation system.

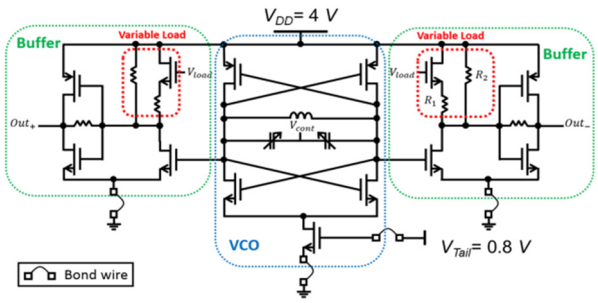


Fig. 2. A schematic of the designed VCO and variable load.

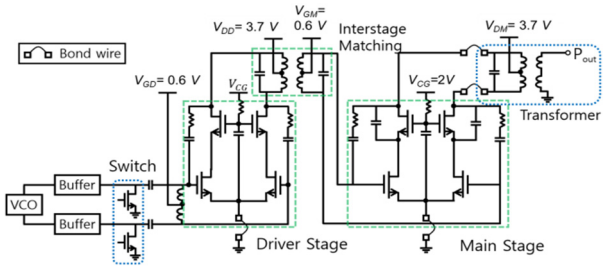


Fig. 3. A schematic of the designed switch and PA.

3. Stimulator

The stimulator is fabricated using two 0.254-mm-thick substrates, RO5880, with ε_r of 2.2. An aperture is the cross-section of the coaxial cable [4]. The structure and the photograph of the stimulator are shown in Fig. 4. Similar to the characteristics of an open-ended coaxial cable, the electric field is distributed on the surface of the aperture. Therefore, it affects only the action potential of the brain surface cell and not the deep brain.

4. Integrated Brain Stimulation System Using a Modulated Microwave Signal

The simulation system is implemented by integrating all of the circuits above. All of the active circuit components (VCO, switches, and PA) are fabricated using the 0.28 μm SOI CMOS process. The bias voltage is given in Figs. 2 and 3. Fig. 5 shows the photographs of the proposed system. In Fig. 5(b), a transformer converts the differential to a single-ended structure, which is the stimulator. To minimize combining loss, a seven-layer printed circuit board is used to fabricate the transformer.

Fig. 6(a) and (b) show the measured characteristics of the system. The frequency tuning range covers 5.7–7.12 GHz, and the output power varies at 3.2–22.4 dBm at 6.5 GHz. The measured $|S_{11}|$ of the stimulator in contact with mouse brain is 6.1 dB at 6.5 GHz as plotted in Fig. 6(c). Fig. 6(d) presents the pulse modulated stimulation signal at the output of the switch. The current consumptions of the VCO and PA are 64 mA and 180 mA, respectively.

III. Experimental Results

A mouse is used to determine the feasibility of the modulated microwave stimulation. The mouse is anesthetized, and the skin and skull are incised. Experimental setup is shown in Fig. 7(a).

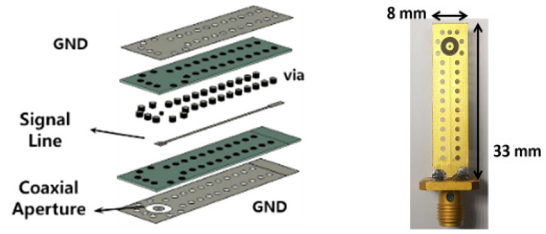


Fig. 4. Stimulator fabricated using double-layer printed circuit board technology.

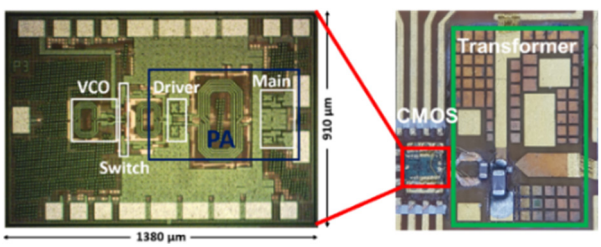
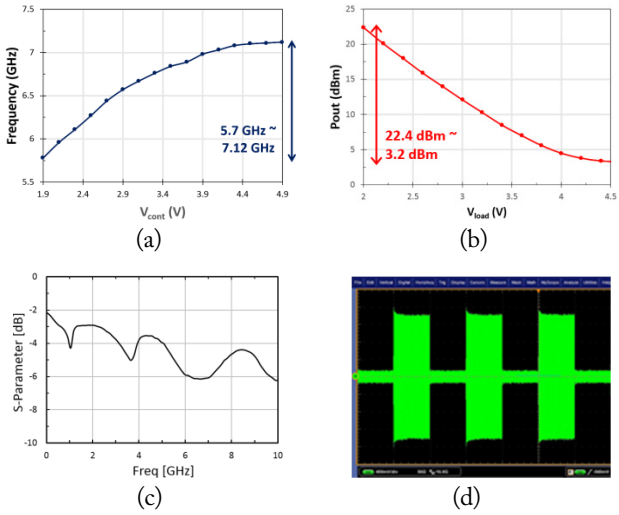
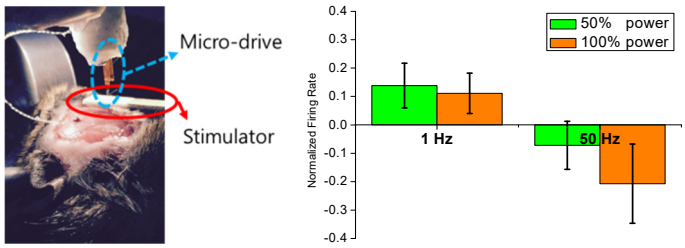


Fig. 5. Photograph of the integrated system.



(a) (b) (c) (d)

Fig. 6. Measurement results of the system: (a) frequency tuning range, (b) output power, (c) $|S_{11}|$ of the designed stimulator, and (d) pulse modulated output signal.



(a) (b)

Fig. 7. (a) Experimental setup for mouse brain stimulation. (b) Stimulation results normalized according to the baseline.

The aperture of the stimulator is brought into direct contact with the brain. Then, a micro-drive consisting of four bundles with four nichrome wires is inserted, targeting the hippocampus of the brain. The brain signals are filtered (600 Hz–6 kHz) at a sampling rate of 30,303 Hz through a data acquisition system (Digital Lynx 4SX; Neuralynx, Bozeman, MT, USA) and sorted into single-unit cell data. The effect of the modulated microwave signal on the brain stimulation is observed by comparing the FR of the single cell data before and after the stimulation. The FR of single cell data before stimulation is used as the baseline.

After the baseline measurement, the cell is stimulated with 50% power (2 mW) for 5 minutes. The pulse of the stimulation signal is 1 Hz at 1% duty cycle (10 ms width). The observation time for 5 minutes is then secured. The observation follows the same protocol as described above at 100% power (4 mW). The mouse is allowed to rest for 1 hour and is then given the same stimulus to the other cell. The 1 Hz stimulation is measured in six cells. The 50-Hz stimulation signal has a 50% duty cycle (10 ms pulse width). It is analyzed for the activity of seven cells using the same protocol as above.

For the neuronal activity analysis, the FR changes of individual neurons through stimulation are calculated using the following equation:

$$Normalized\ firing\ rate\ (\mathrm{Hz}) = \frac{After\ stimulation - Before\ stimulation}{After\ stimulation + Before\ stimulation}. \quad (1)$$

Therefore, values greater than 0 indicate increased activity compared with the baseline, and small values indicate a decreased activity. The range bars indicate the standard deviations. In the 1 Hz case, the FR is excited and inversely inhibited at 50 Hz as in Fig. 7(b).

IV. Conclusion

We have demonstrated a microwave brain stimulation using pulse modulation to enhance/inhibit the FR of a cell in the brain. The system consists of a VCO, switches, a PA, and a stimulator. This system configuration has the advantage of easily adjusting the pulse width and the PRF. The stimulation effect is confirmed by measuring the action potential of the mouse brain. The opposite cell activity is obtained according to the PRF of the modulation.

Microwave brain stimulation is advantageous in terms of energy consumption and system size compared with the conventional TMS. This study requires in-depth research of stimulation signal variations and is expected to lead to the development of neuroscience research.

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7. All figures should be of high quality meeting with the publishing requirement with legible symbols and legends. In preparing the figures, authors should consider a size reduction during the printing process to have acceptable line clarity and character sizes. Use only figures that are necessary to illustrate the meaning of the text. Figures must be black and white of high contrast. All figures should be referred to in the text as, for example, Fig. 1, Fig. 2(a), or Figs. 1-3.
8. Only those references cited in the text should be listed in the references. Authors are responsible for the accuracy and completeness of their references and the correct text citations. In the text the reference should be numbered in bracket in ascending order (e.g., [1, 3], or [4-6]; Lee [2] and Kim and Park [5]; Jang et al. [7]). In case of the paper title, only the first letter is to be capitalized. However, in case of journal and book titles, the first letter of each word should be capitalized and all of the letters should be italicized. See the example below.

Books

[1] F. Giannini and G. Leuzzi, *Nonlinear Microwave Circuit Design.* NewYork, NY: John Wiley & Sons Inc., 2004.

Journals

[2] H. Ahn and B. Kim, "Equivalent transmission-line sections for very high impedances and their application to branch-line hybrids with very weak coupling power," *Journal of Electromagnetic Engineering and Science*, vol. 9, no. 2, pp. 85-97, Jun. 2009.

Report

[3] E. E. Reber, R. L. Michell, and C. J. Carter, "Oxygen absorption in the earth's atmosphere," Aerospace Corp., Los Angeles, CA, Tech. Rep. TR-0200 (4230-46)-3, Nov. 1988.

Conference Proceedings

[4] S. P. Bingulac, "On the compatibility of adaptive controllers," in *Proceedings of the 4th Annual Allerton Conference on Circuit and System Theory*, NewYork, pp. 8-16, 1994.

Papers Presented at Conferences

[5] J. G. Kreifeldt, "An analysis of surface-detected EMG as an amplitude-modulated noise," presented at the 8th International Conference on Medical and Biological Engineering, Chicago, IL, 1969.

[6] J. Arrillaga and B. Giessner, "Limitation of short-circuit levels by means of HVDC links," presented at the IEEE Summer Power Meeting, Los Angeles, CA, Jul. 1990.

Theses (M.S.) and Dissertations (Ph.D.)

[7] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Department of Electronic Engineering, Osaka University, Osaka, Japan, 1993.

[8] J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Department of Electronic Engineering, Harvard University, Cambridge, MA, 1993.

Standards

[9] *IEEE Criteria for Class IE Electric Systems*, IEEE Standard 308, 1969.

Online Sources

[10] R. Bartle, "Early MUD History," Nov. 1990; www.ludd.luth.se/aber/mud-history.html.

9. When citing any paper published in JEES, it should be indicated the name of the journal as *Journal of Electromagnetic Engineering and Science* or *J. Electromagn. Eng. Sci*.

10. Acknowledgment, if needed, appears before the reference. Sponsorship or financial support acknowledgment should be included here.

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VII. Accepted Manuscript

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